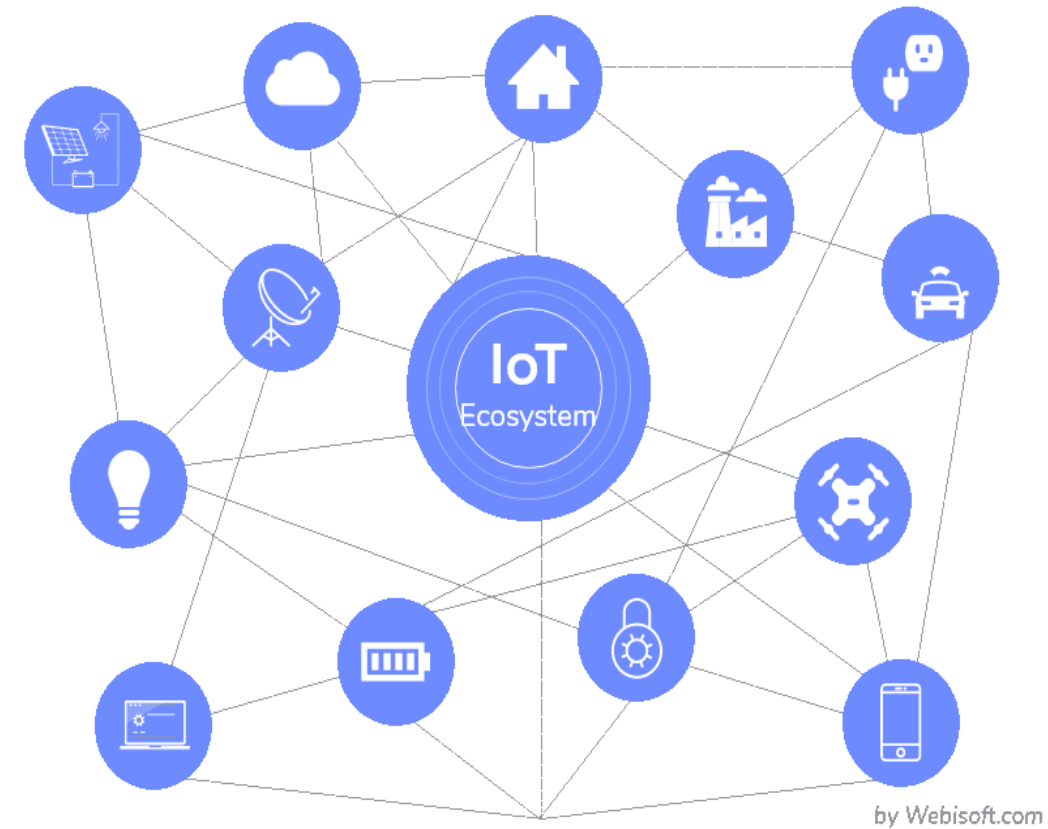


# Unit 1:

# Introduction to IoT



**Dr. Ujaval Patel**

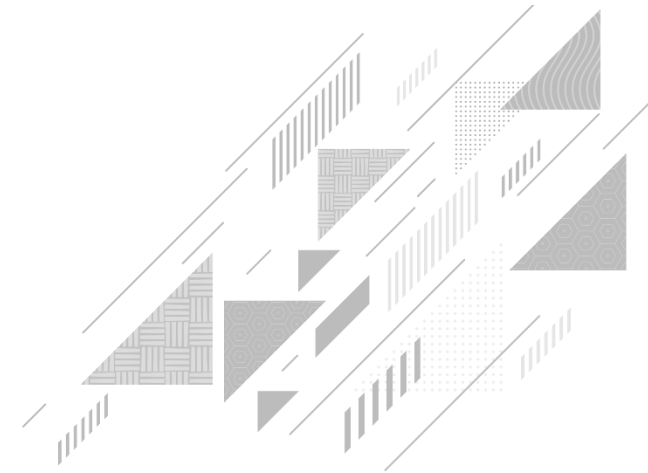
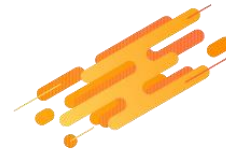
**Assistant Professor (IOT/SCADA)**

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☎ +91 987 987 9746

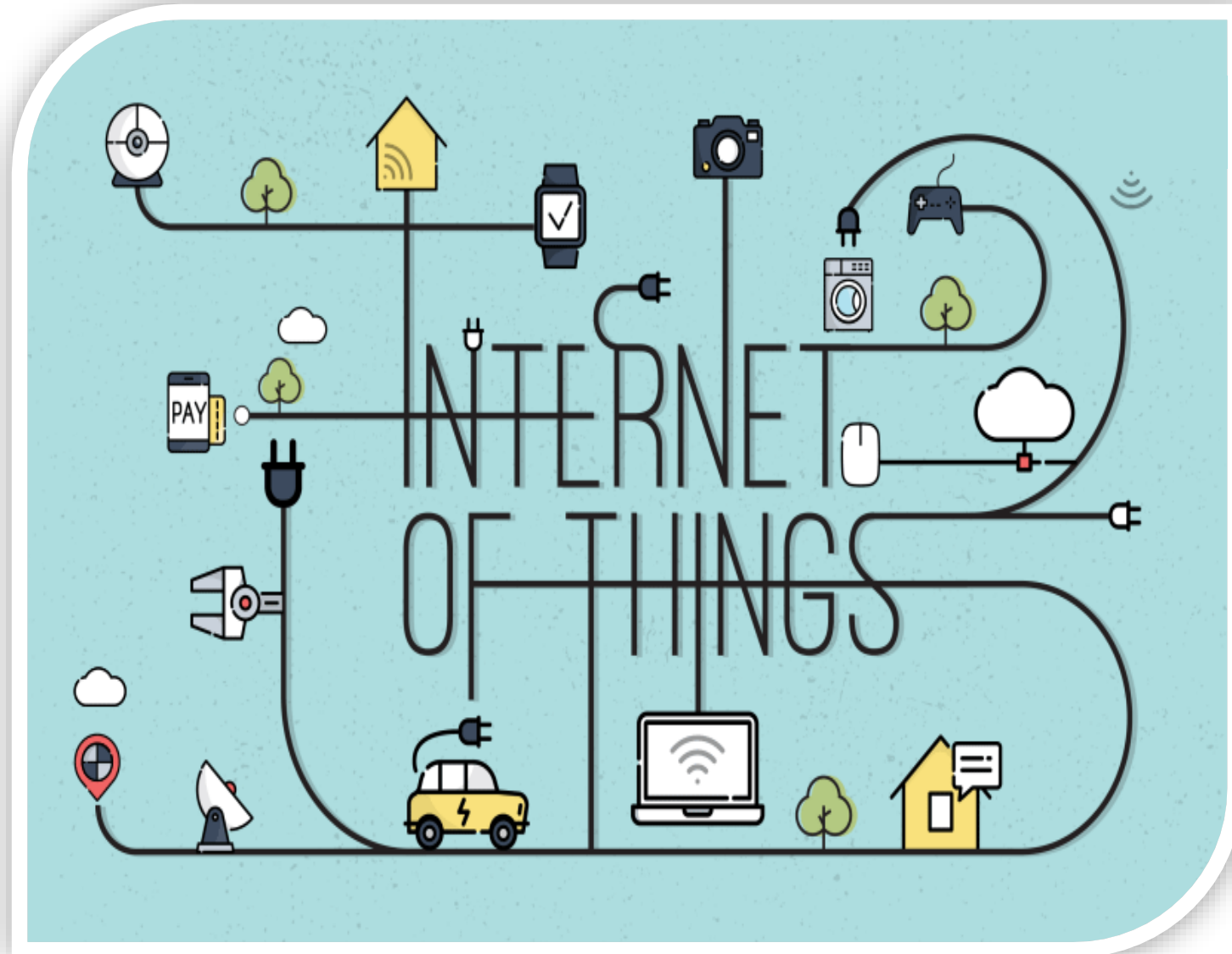


**National Forensic  
Sciences University**  
Knowledge | Wisdom | Fulfilment  
An Institution of National Importance  
(Ministry of Home Affairs, Government of India)



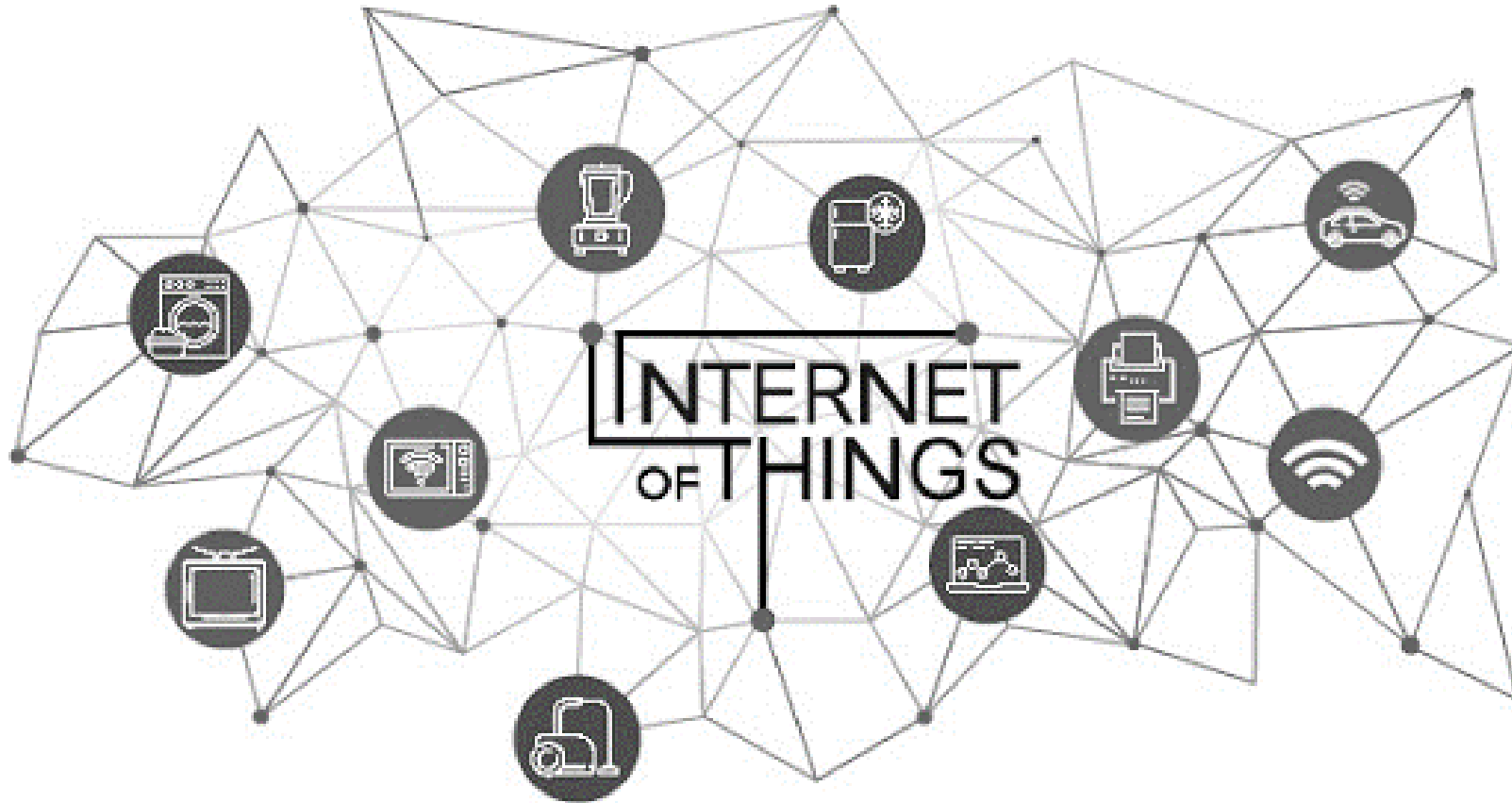
# Unit Outlines:

- ▶ Introduction to IoT
- ▶ Application areas of IoT
- ▶ Characteristics of IoT
- ▶ Things in IoT
- ▶ IoT Protocol Structures
- ▶ Enabling Technologies
- ▶ IoT Challenges
- ▶ IoT Levels
- ▶ Physical Design of IoT
- ▶ Logical Design of IoT



# Introduction to Internet of Things (IoT)

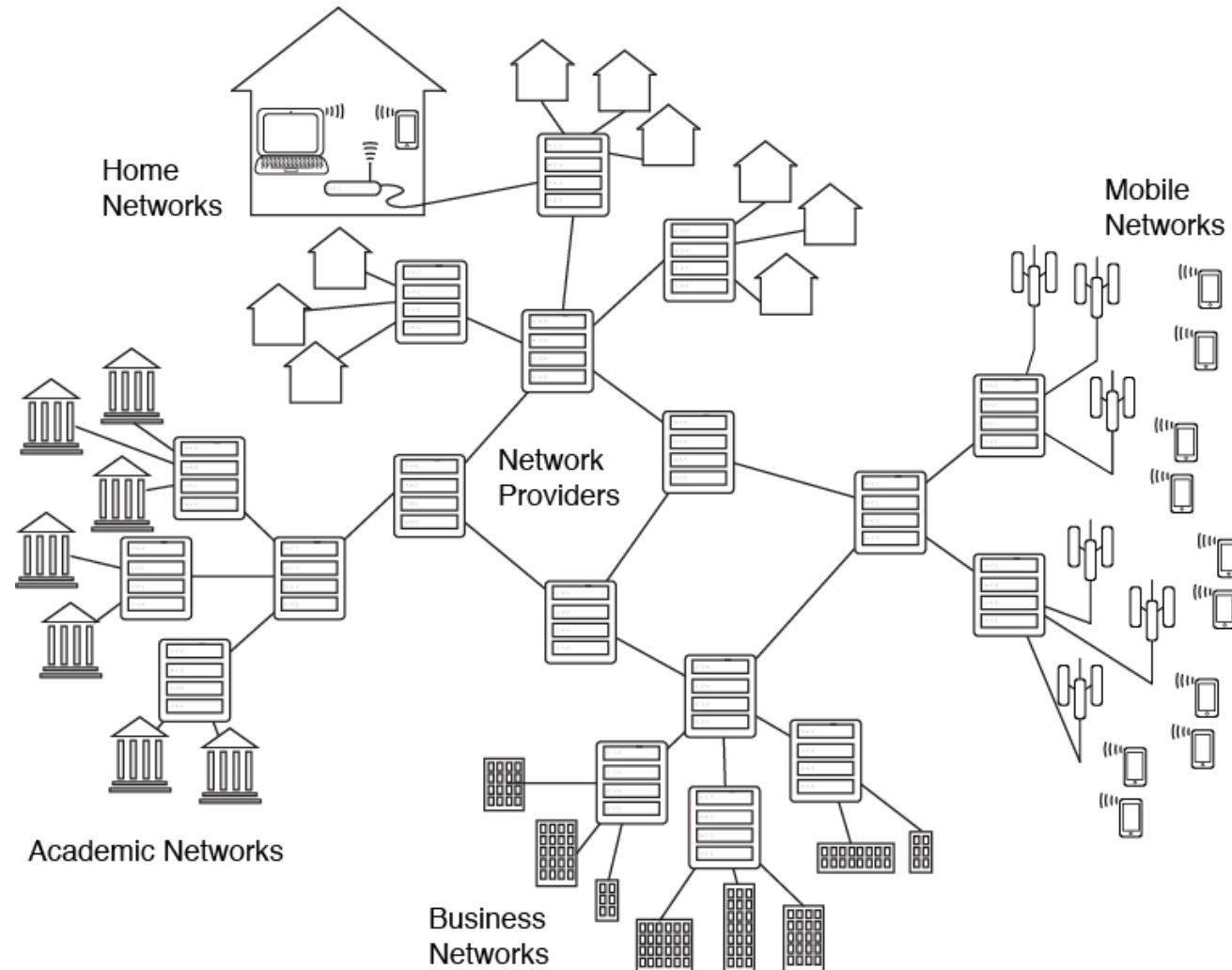
- ▶ First of all we should discuss about the name “IoT – Internet of Things” in detail.



- ▶ So we have to discuss about the first word “Internet” and then everything about the “Things”.

# Introduction to Internet of Things (IoT)

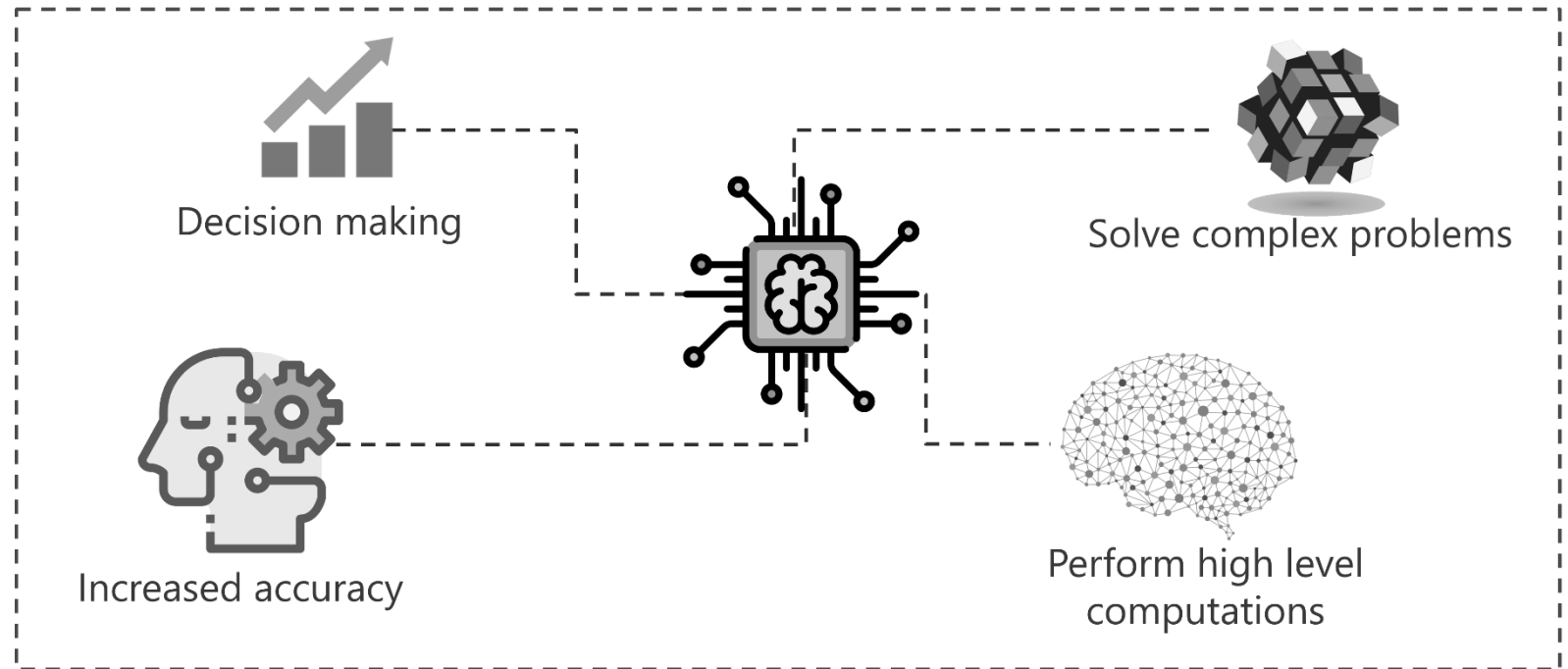
## ► What is Internet?



- In simple word, it's a **Network of Networks** or Interconnected LANs.
- So here we have to discuss about the Network, and we already learned everything about network in the last semester, Right?
- What is Network?, Requirement for networking

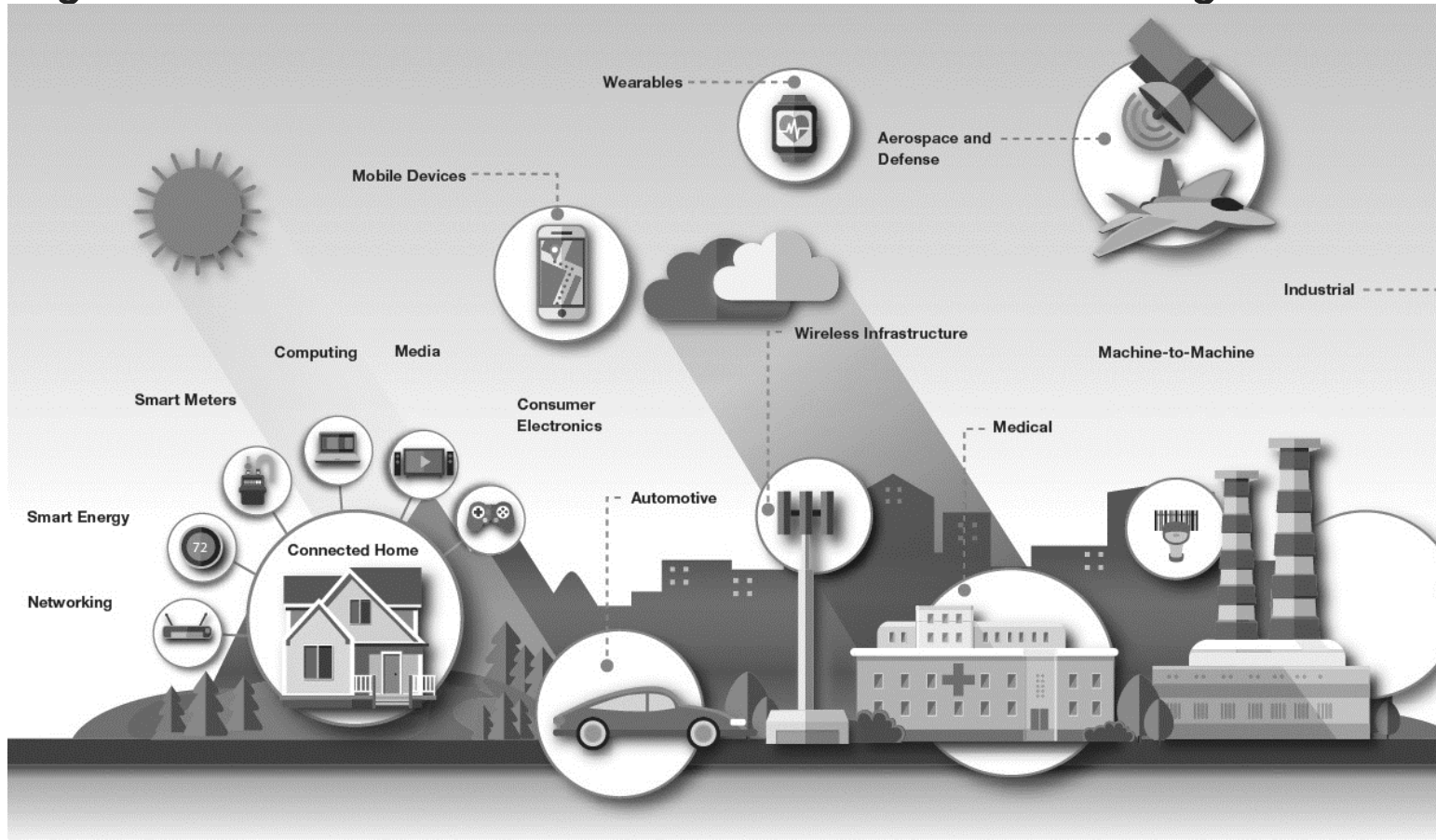
# Introduction to Internet of Things (IoT)

- ▶ Nowadays The term Internet of Things (IoT) is an emerged the popular terms.
- ▶ There are multiple ways to define IoT, but the basic of all the definitions remains the same.
- ▶ IoT is **network of interconnected computing devices** which are embedded in everyday objects, enabling them to send and receive data.
- ▶ The IoT is not just limited to the connected or networked devices, but in a broad way IoT devices exchange meaningful information from one device to another to get desire result.



# Introduction to Internet of Things (IoT)

- ▶ IoT is not a single technology, it's a combination of technologies and domain knowledge.
- ▶ As a result, engineers from different domains have to work together for building a complete IoT product.

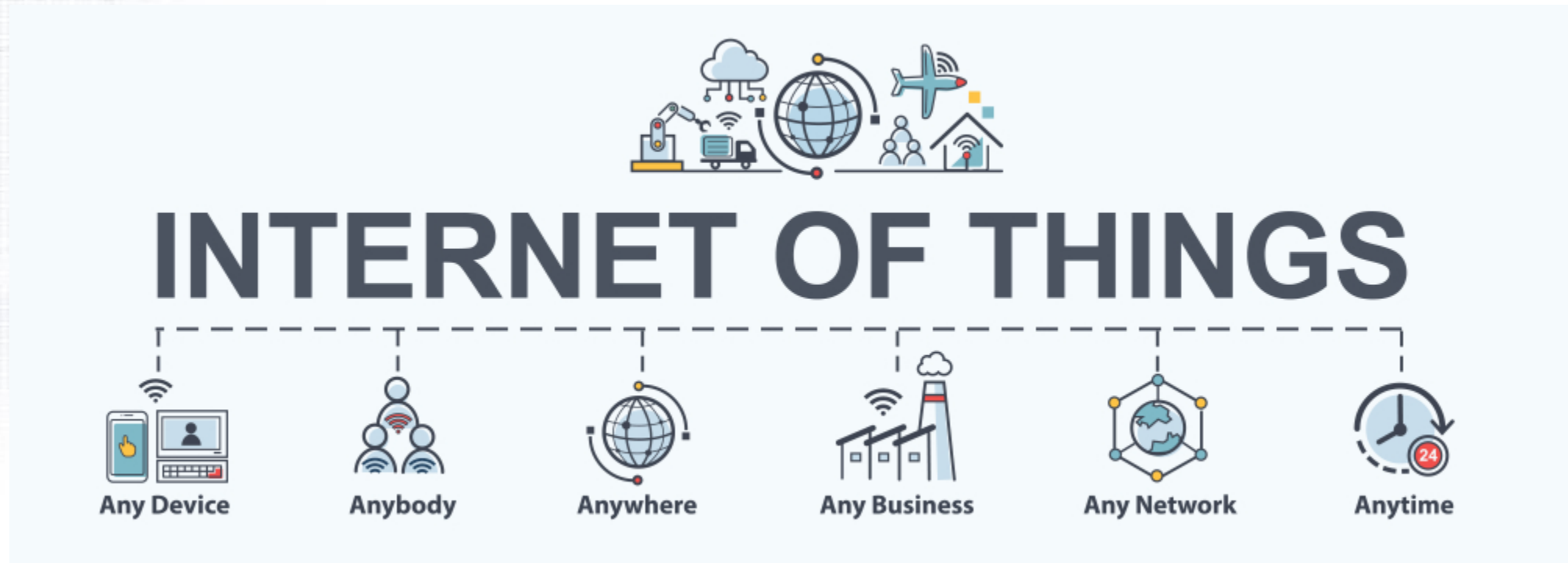


- ▶ Life would be governed entirely by Internet and IoT in the near future.



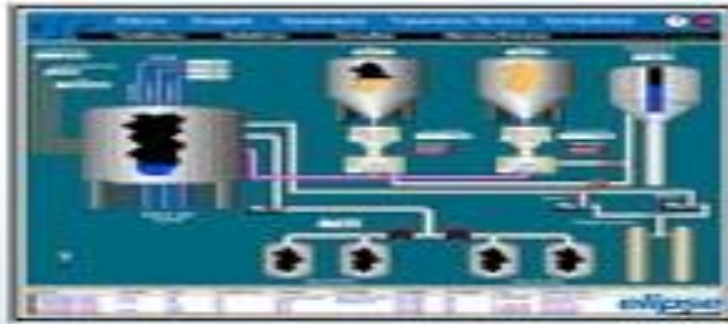
# What is IoT ?

Cyber physical embedded systems using which we can control anything from anywhere.



# What is IoT ?

## PLC & I/O Modules



SCADA



PLC



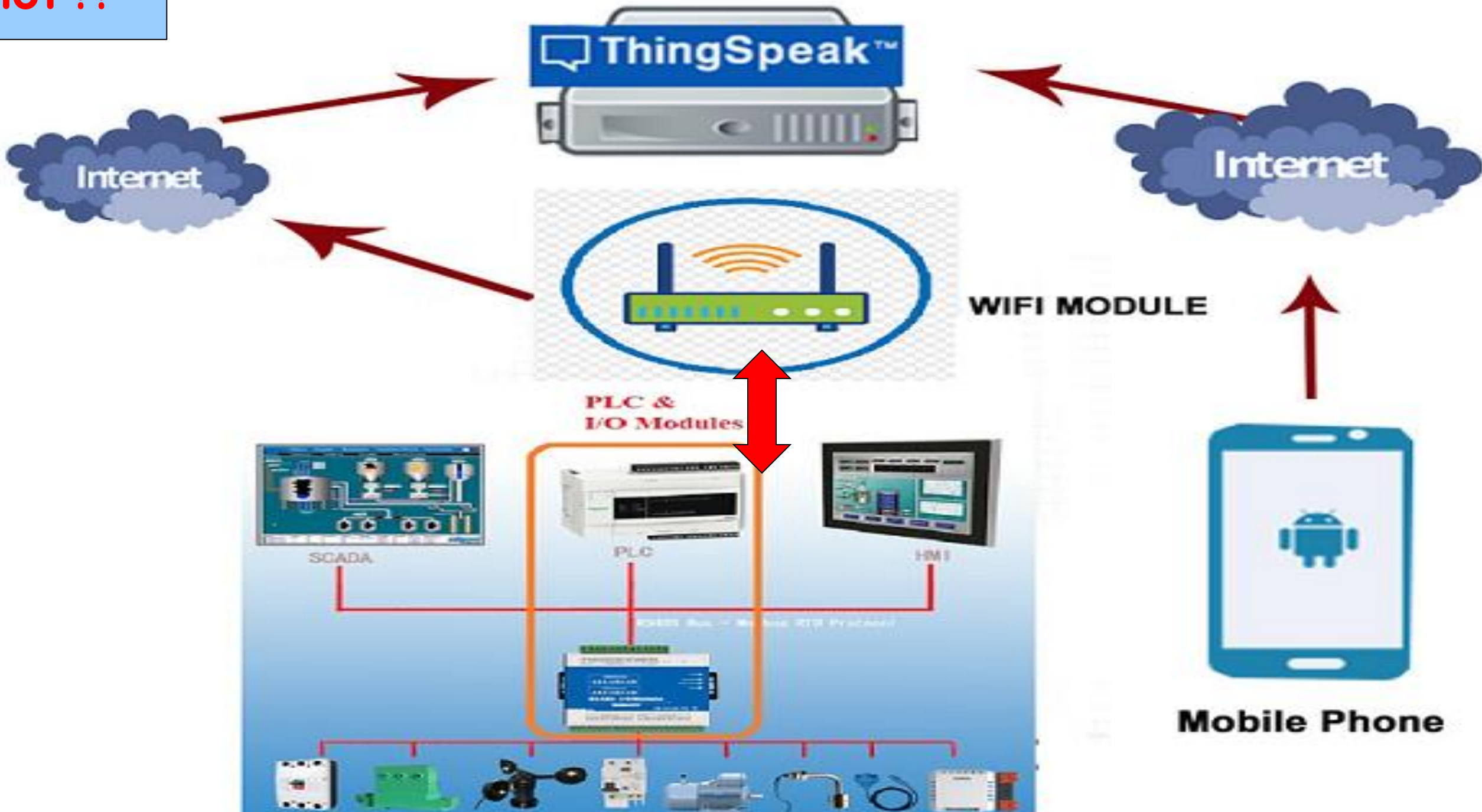
HMI

RS485 Bus - Modbus RTU Protocol



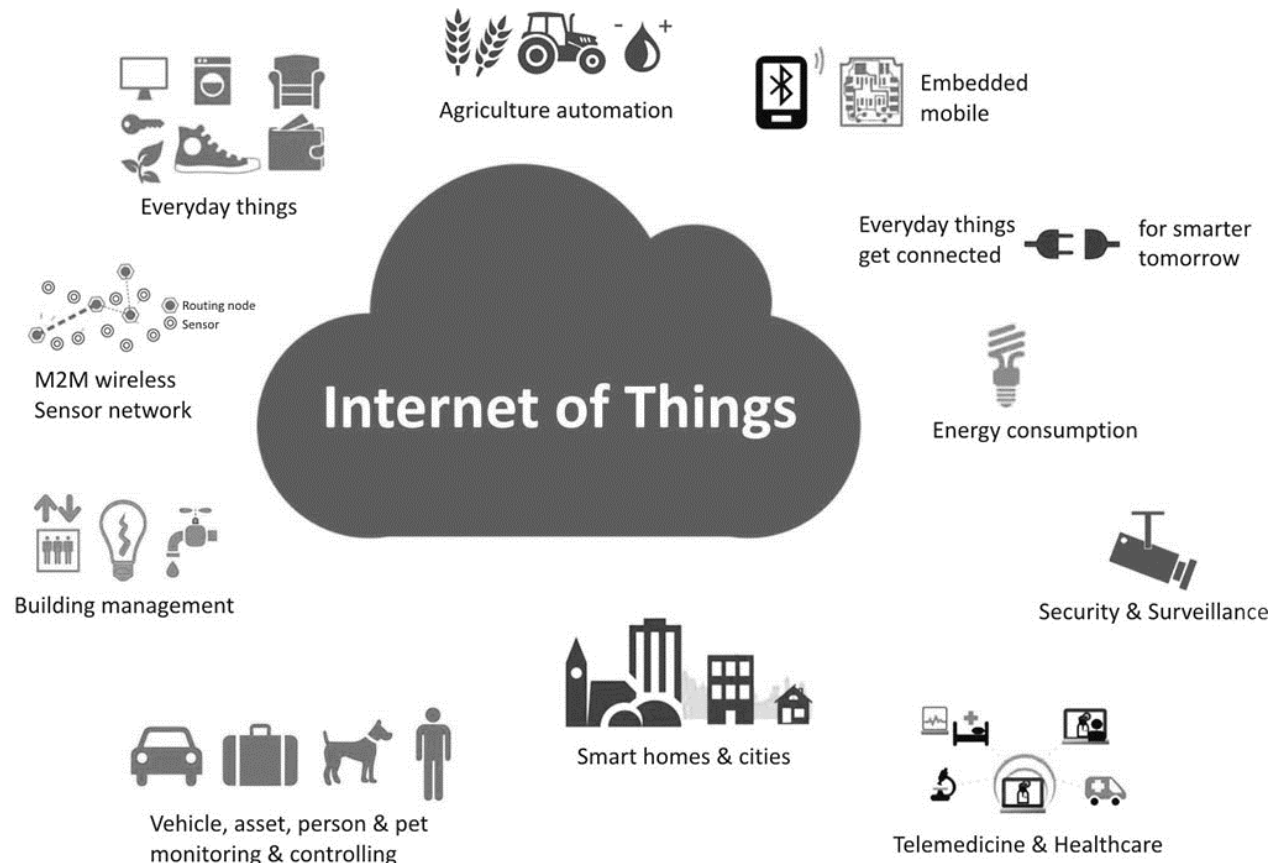
Transducers, Encode, Stepper Motor, Digital Inputs, Digital Outputs, RTD sensors, Relays etc.

IIOT ??



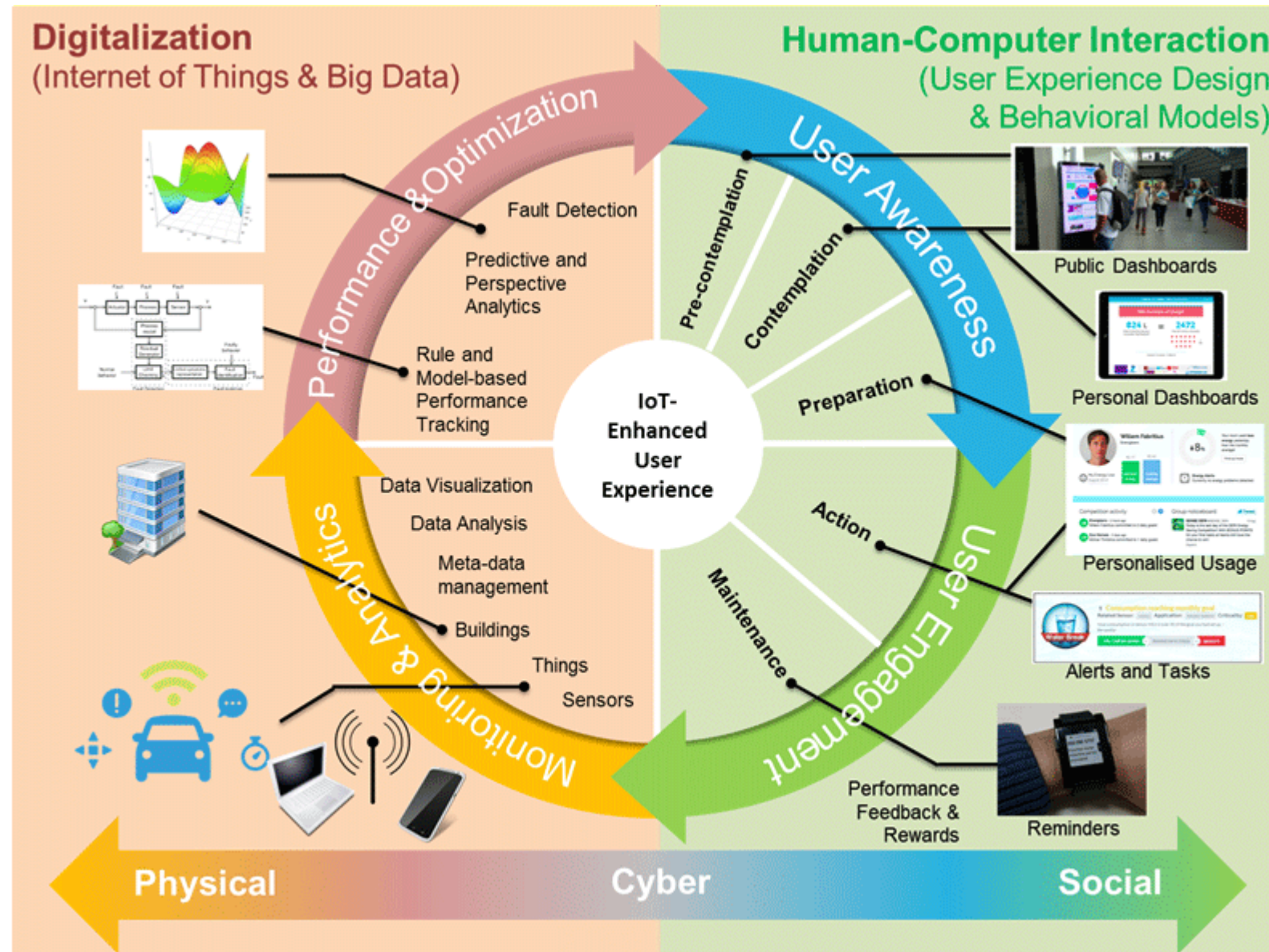
# Application areas of IoT

- ▶ The scope and application areas of IoT is very huge.
- ▶ IoT can be used to build applications for...

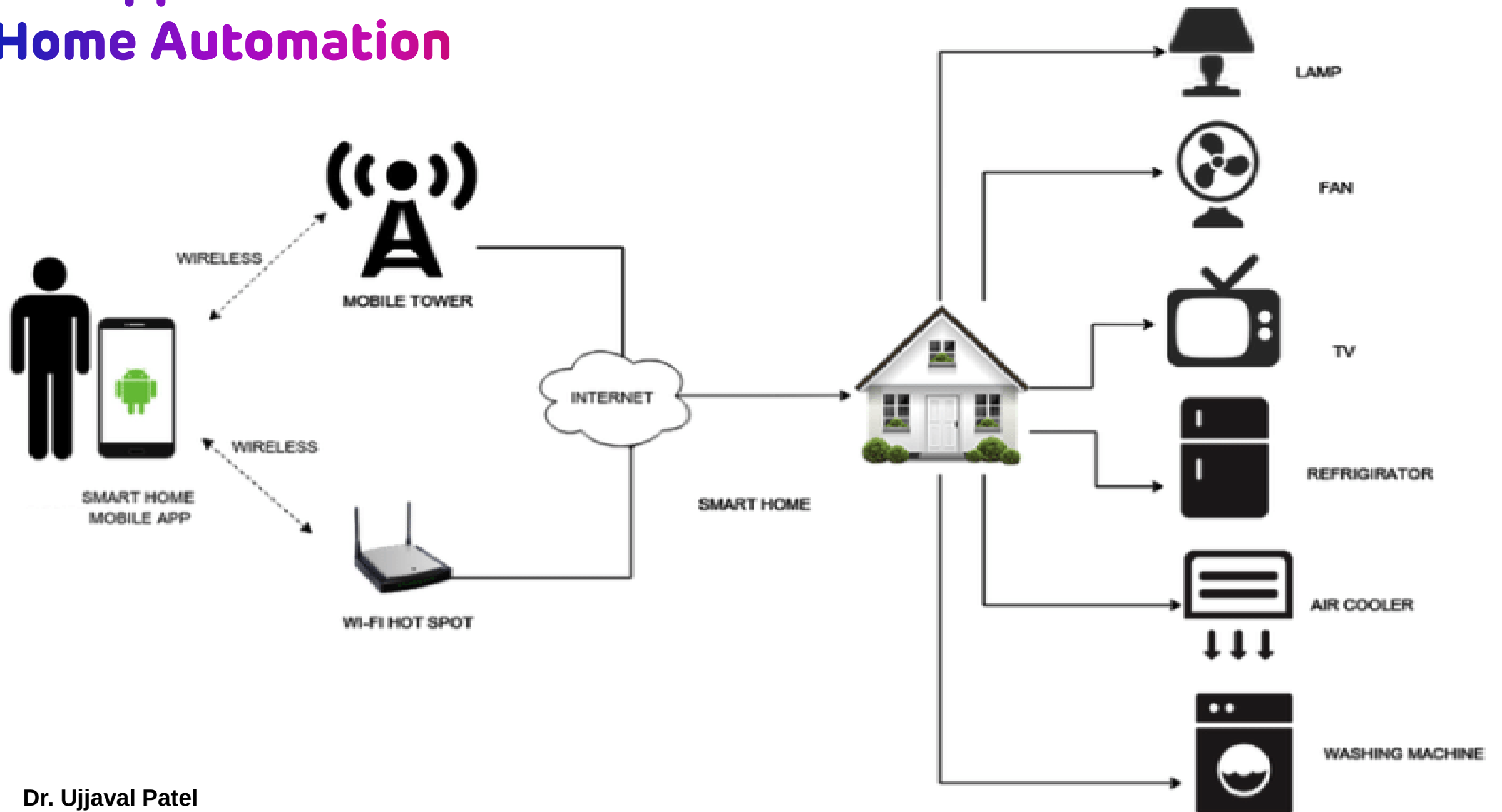


- Agriculture
- Assets Tracking
- Energy Sector
- Defense
- Embedded Applications
- Education
- Waste Management
- Healthcare Products
- Telemedicine
- Safety And Security Sector
- Smart City Applications etc.

# Application areas of IoT



# IoT Applications : Home Automation



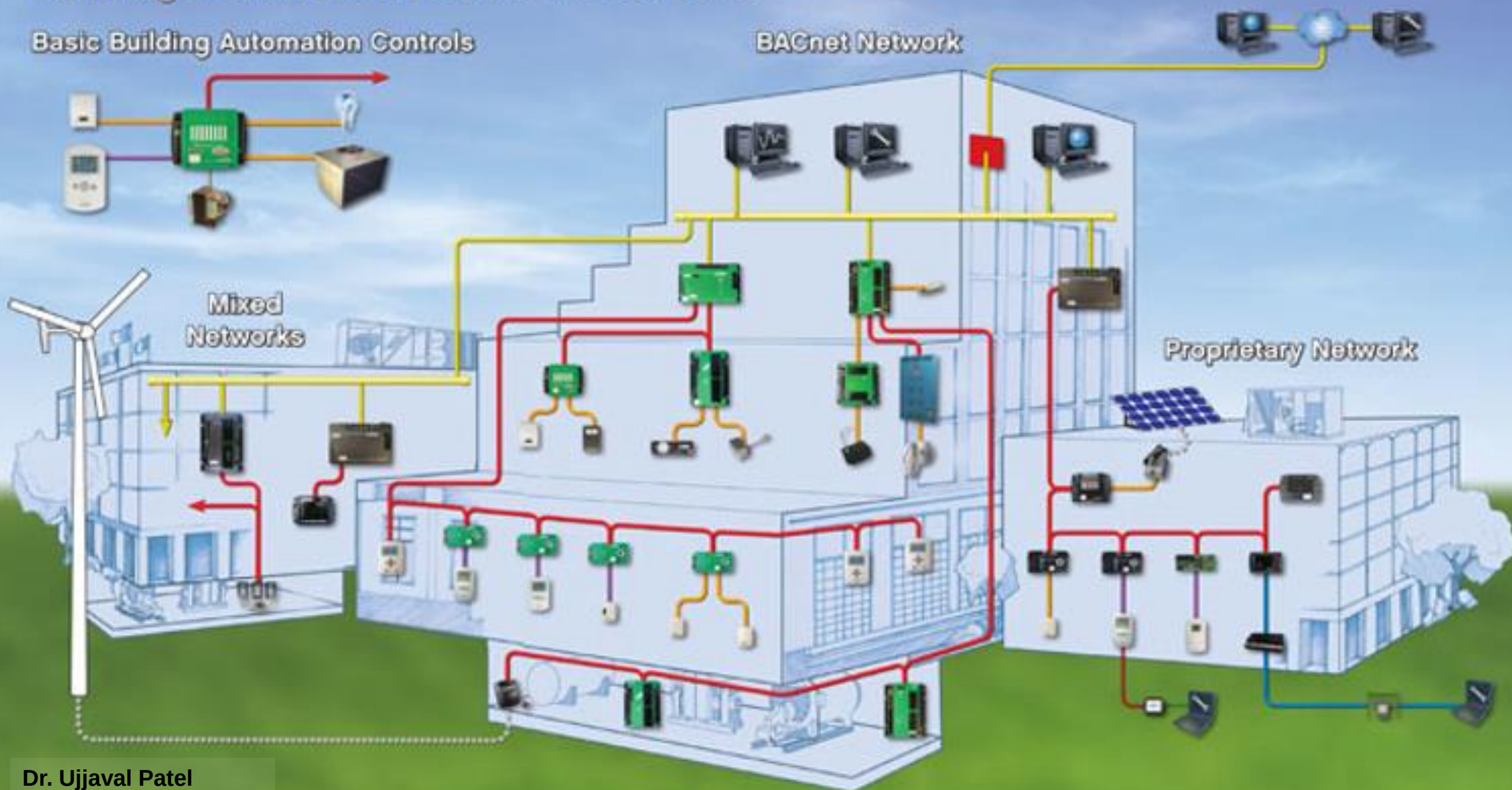
# Building Automation Behind the Scenes

Basic Building Automation Controls

BACnet Network

Mixed  
Networks

Proprietary Network



# Industrial IoT (IIoT) Systems:





# Advantages of IoT in healthcare



Lower  
expenses



Better treatment  
results



Better disease  
control



Fewer mistakes



More trust  
towards doctors



Medicines  
control



Better disease  
control

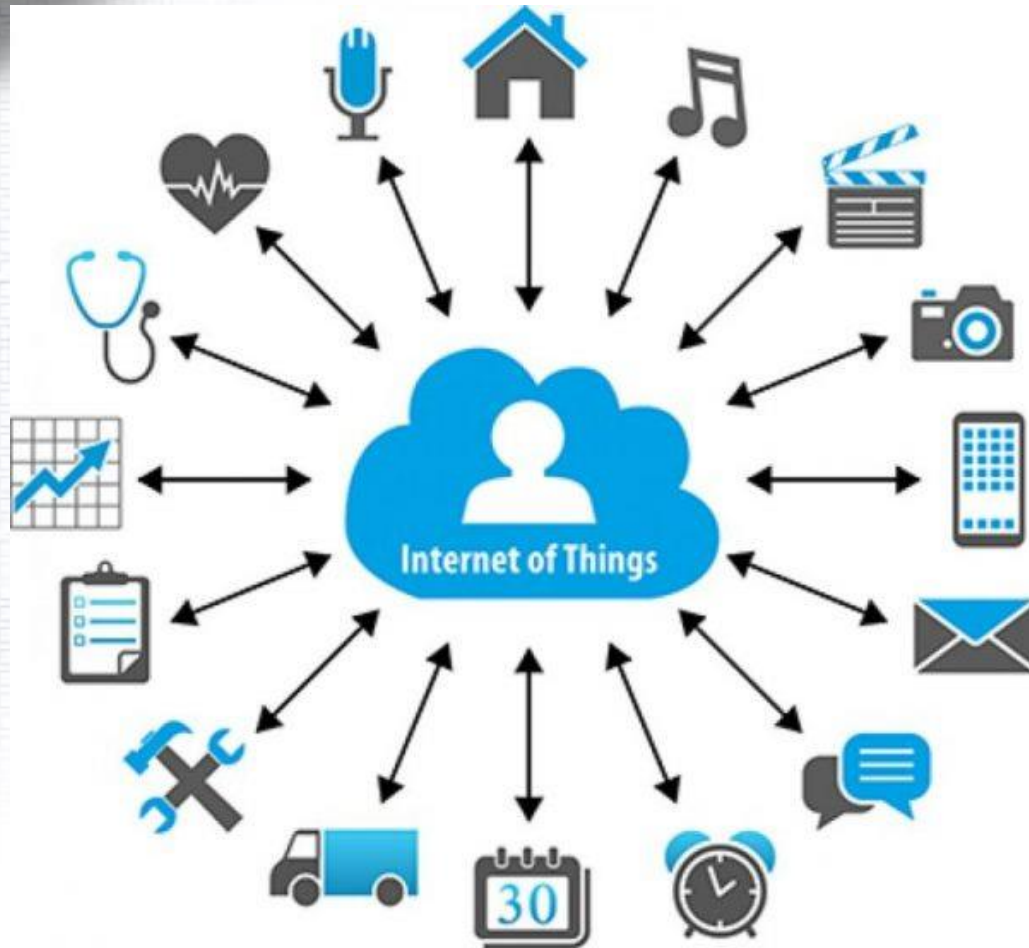


Maintenance of  
medical devices

# IoT in healthcare



# IoT Devices: Top 10

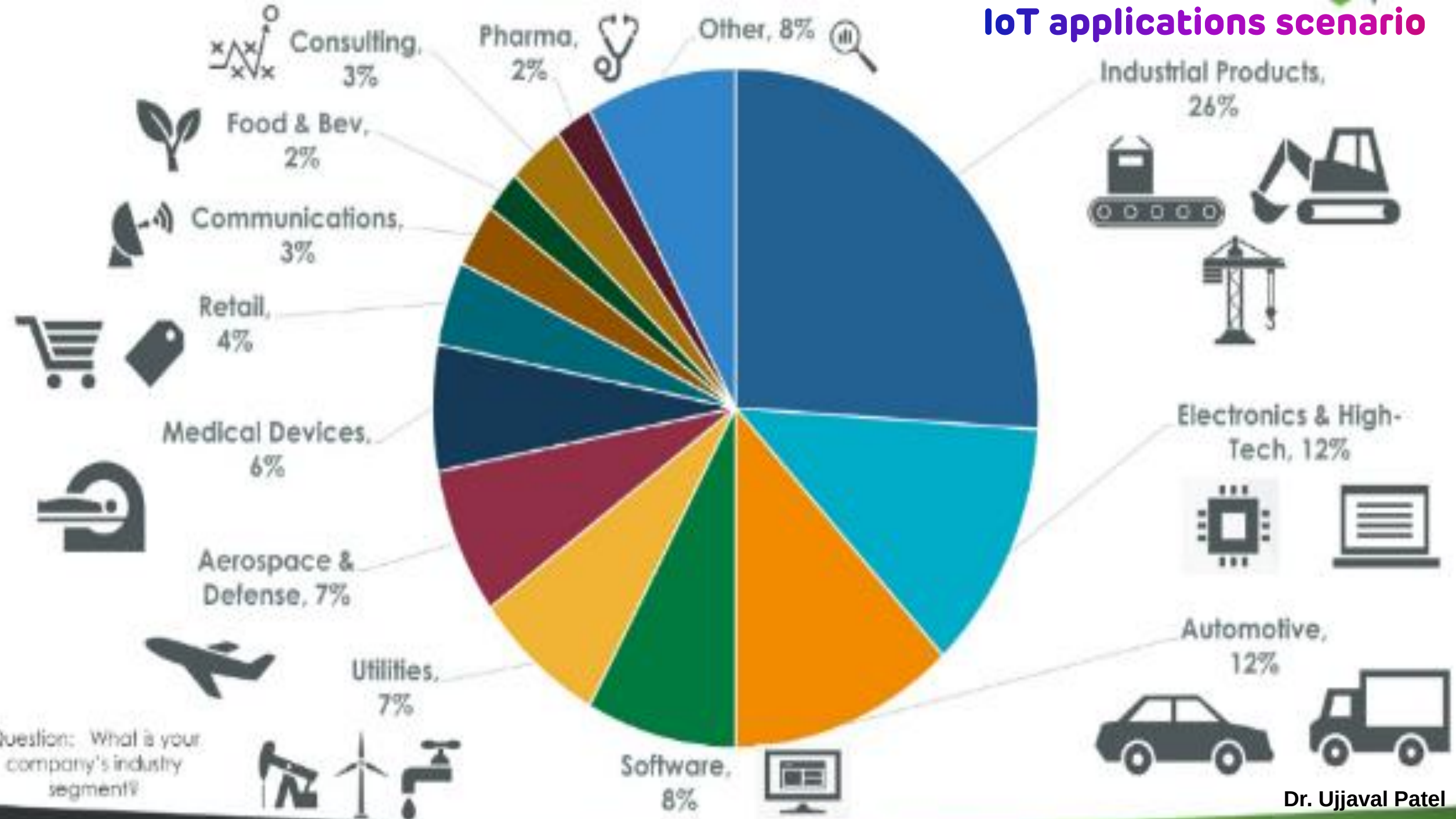


<https://www.pentasecurity.com/blog/10-smartest-iot-devices-2017/>

1. Smart Locks
2. Smart Sprinklers
3. Smart Plugs
4. Smart Baby monitors
5. Smart Cookers
6. Smart Thermostats
7. Smart Mirrors
8. Smart Cleaners
9. Smart Refrigerators
10. Smart Cars



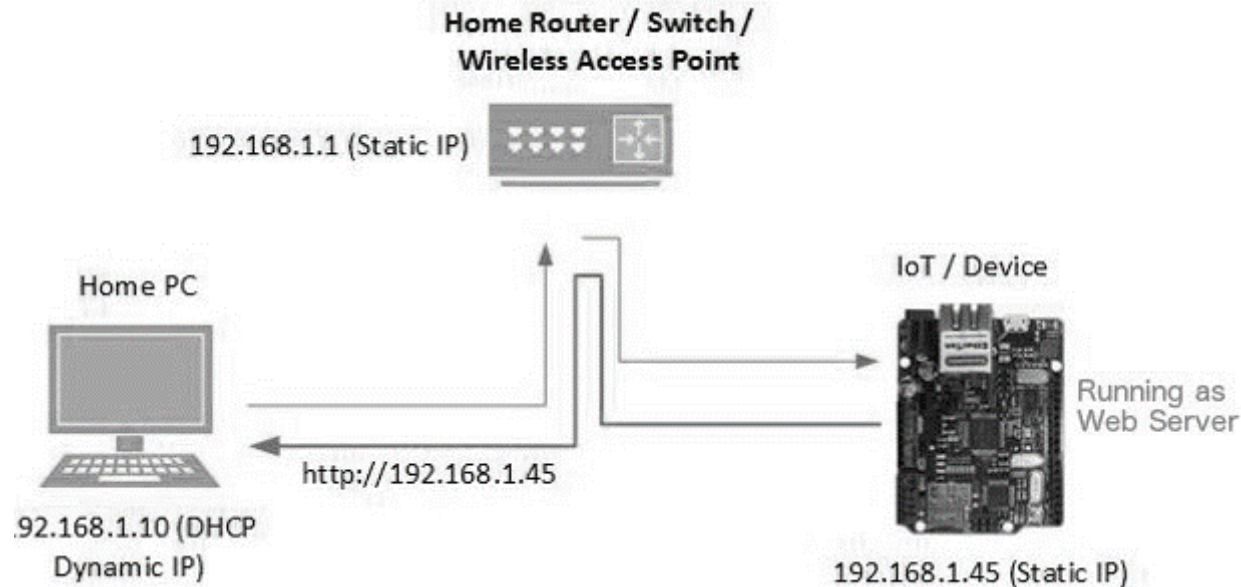
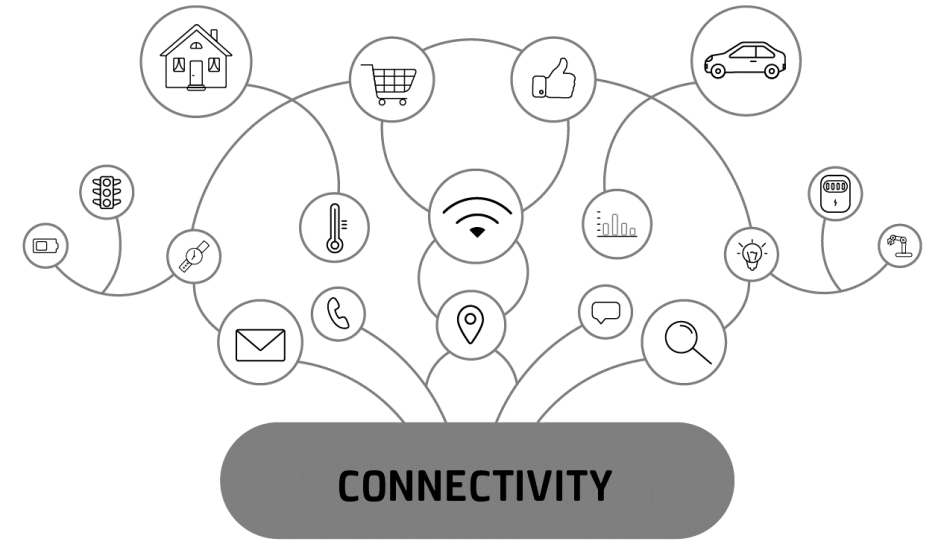
# IoT applications scenario



# Characteristics of IoT

## ► Connectivity:

- ➔ Connectivity is an important and first requirement of IoT infrastructure.
- ➔ Every Things in IoT should be connected to the IoT infrastructure.
- ➔ Connectivity should be guaranteed at anywhere and anytime.



## ► Identity:

- ➔ Each IoT device has a unique identity (e.g., an IP address).
- ➔ This identity is helpful in communication, tracking and to know status of the things.

# Characteristics of IoT

## ► Intelligence:

- ➔ Just data collection is not enough in IoT, extraction of knowledge from the generated data is very important.
- ➔ For example, sensors generate data, but that data will only be useful if it is interpreted properly.
- ➔ So intelligence is one of the key characteristics in IoT.



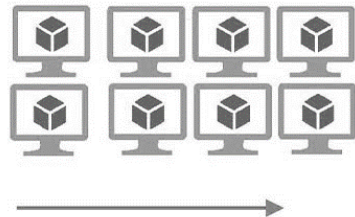
### Vertical Scaling

( Increase size of instance (RAM , CPU etc.) )



### Horizontal Scaling

( Add more instances )



## ► Scalability:

- ➔ The number of elements (devices) connected to IoT zone is increasing day by day.
- ➔ Therefore, an IoT setup should be capable of handling the expansion.
- ➔ It can be either expand capability in terms of processing power, Storage, etc as vertical scaling or horizontal scaling by multiplying with easy cloning

# Characteristics of IoT

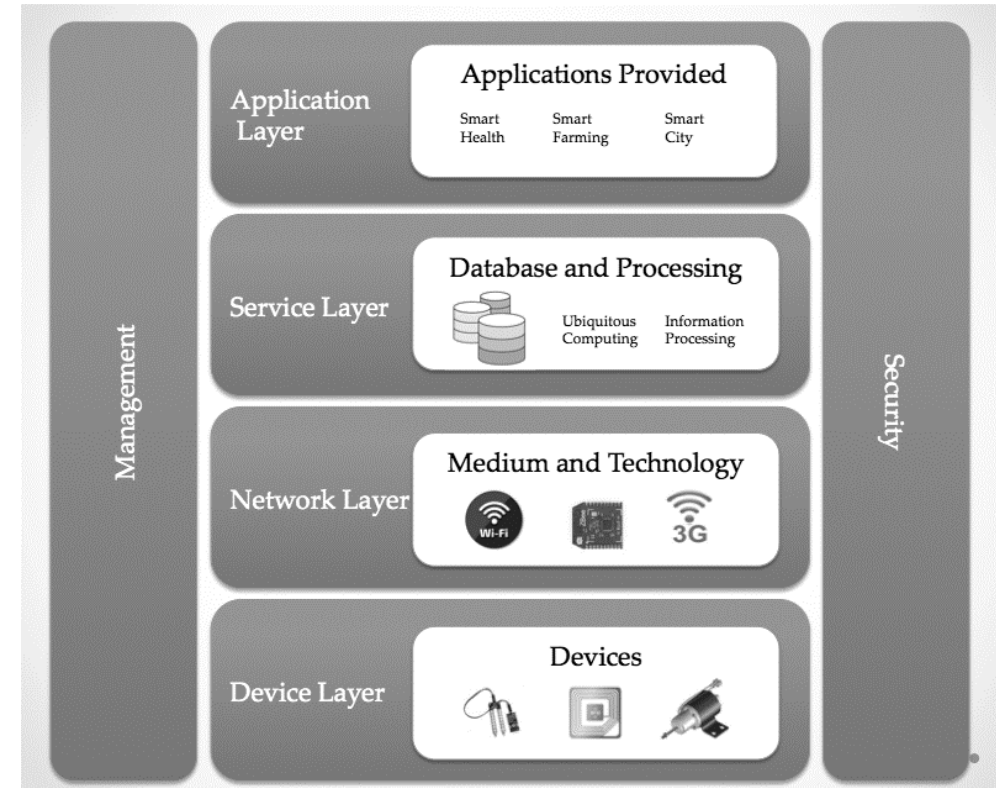


## ► Dynamic and self-adapting (complexity):

- ➔ IoT devices should dynamically adapt themselves to the changing surroundings.
- ➔ For example surveillance camera. It should be flexible to work in different weather conditions and different light situations (morning, afternoon, or night).

## ► Architecture:

- ➔ IoT architecture is yet not uniformed and standardized.
- ➔ It should be hybrid, supporting different manufacturer's products to function in the IoT network.



# Characteristics of IoT

## ► Safety:

- ➔ Sensitive personal details of a user might be compromised when the devices are connected to the Internet.
- ➔ So data security is a major challenge.
- ➔ This could cause a loss to the user.
- ➔ Equipment in the huge IoT network may also be at risk.
- ➔ Therefore, equipment safety is also critical.

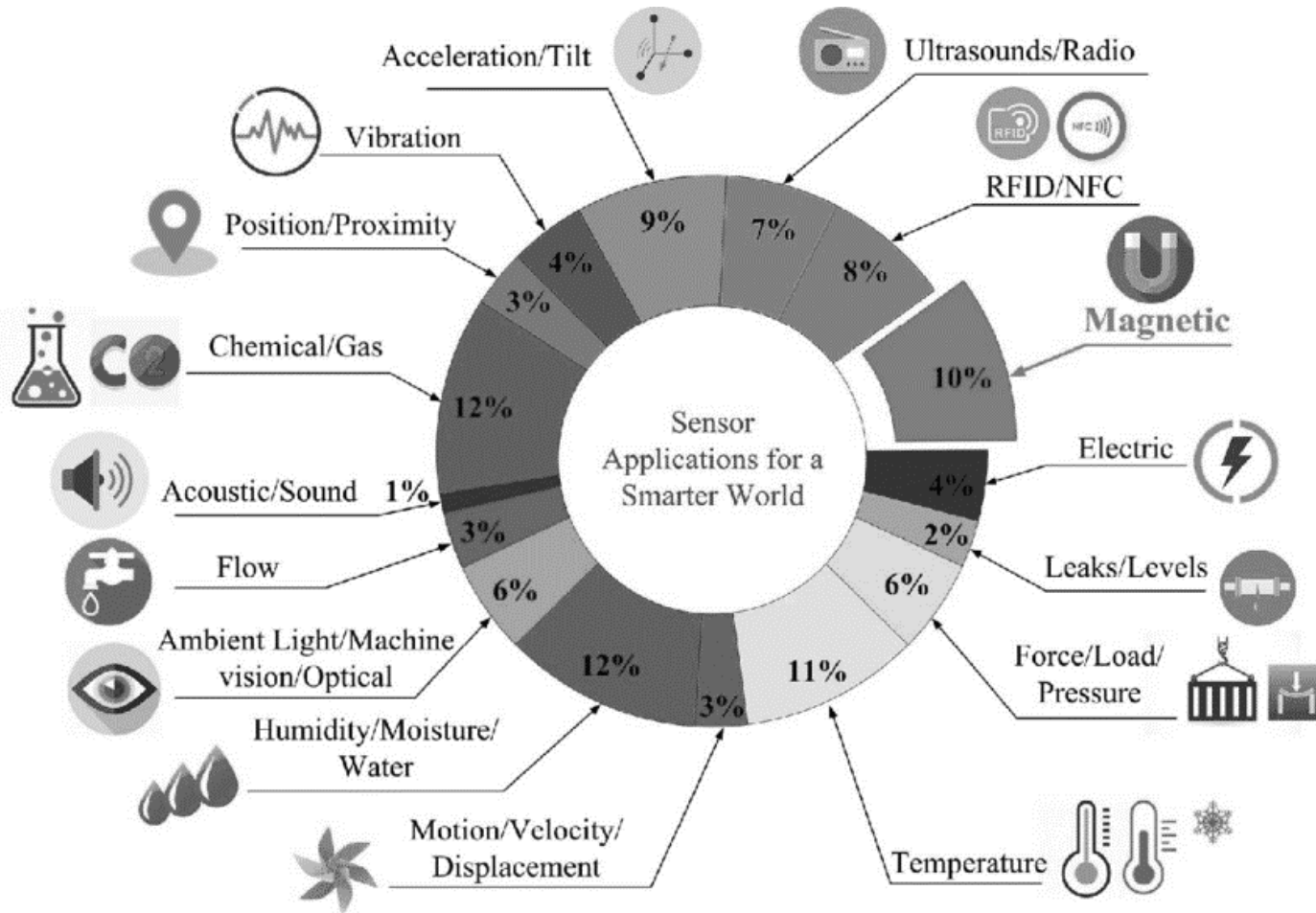


# Things in IoT

- ▶ In the IoT, things refer to a variety of devices. It can be anything even humans in it become a thing.
- ▶ For something to qualify as a “thing”, it requires identity of its existence.
- ▶ The “thing” in a network can be monitored/measured. For example, a temperature sensor could be a thing.
- ▶ Things are capable of exchanging data with other connected devices in the system.
- ▶ The data could be stored in a centralized server (or cloud), processed there and a control action could be initiated.
- ▶ The devices having all the above characteristics are known as things.



# Things in IoT



- ▶ Some of the famous “things” are temperature sensors, pressure sensors, humidity sensors, etc.
- ▶ The data from these sensors are collected and sent it to the cloud or stored it in local server for data analysis.
- ▶ Based on the data analysis, the control action would be taken.
- ▶ For example, switching off the water heater remotely when the water is heated as per requirement.

# Things in IoT

► Not just sensors, the following can also be called as things:

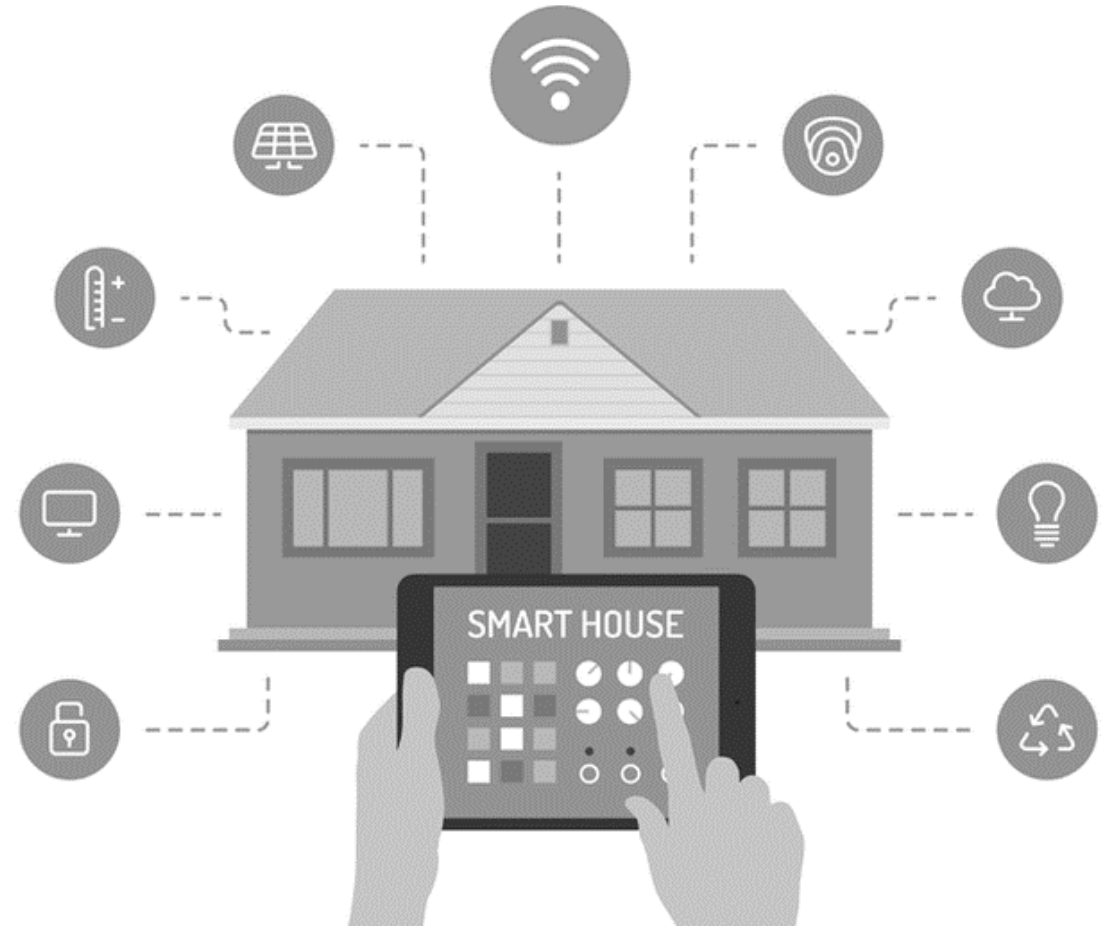


- ➔ Industrial motors
- ➔ Wearables (e.g., watch)
- ➔ Vehicles
- ➔ Shoes
- ➔ Heart monitoring implants (e.g., pacemaker, ECG real-time tracking)
- ➔ Biochip transponders (for animals in farms)
- ➔ Automobiles with built-in sensors (automobile feature real-time monitoring)
- ➔ Food/perishables quality measuring

# Things in IoT

► In IoT-based home automation, the “things” could be the following

- Lighting control and automation devices
- Ventilation devices
- Air conditioning [heating, ventilation and air conditioning (HVAC)] systems
- Appliances such as washer/dryer
- Air purifiers
- Ovens or refrigerators/freezers that use Wi-Fi for remote monitoring
- Security cameras
- Smart phones

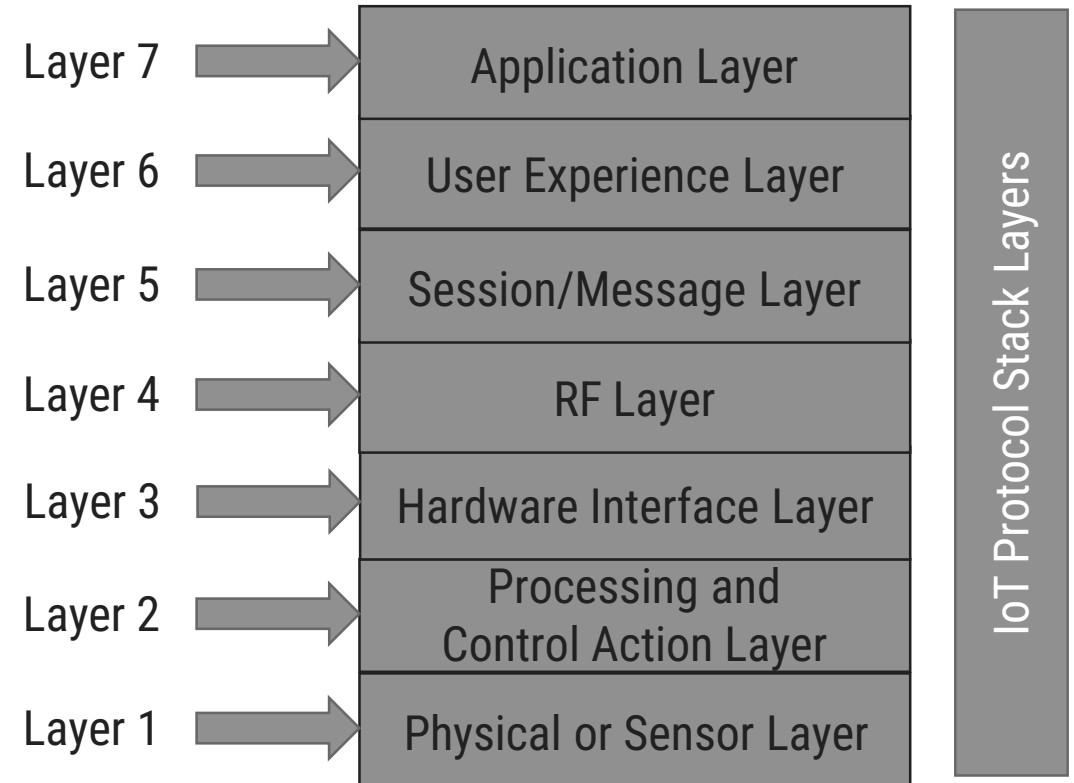


# IoT Stack

▶ Like other digital technology IoT has stack layers.

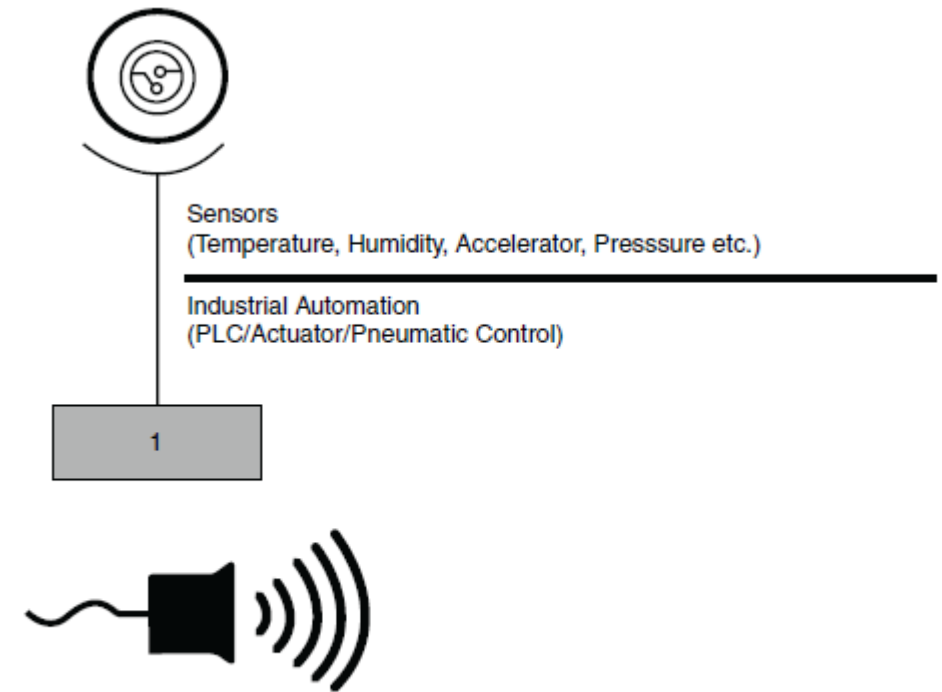
▶ Following are the identified seven layers in IoT stack.

- Layer 1 (Physical or Sensor Layer)
- Layer 2 (Processing and Control Action layer)
- Layer 3 (Hardware Interface Layer)
- Layer 4 (RF Layer)
- Layer 5 (Session/Message Layer)
- Layer 6 (User Experience Layer)
- Layer 7 (Application Layer)



# IoT Stack - Layer 1 (Physical or Sensor Layer)

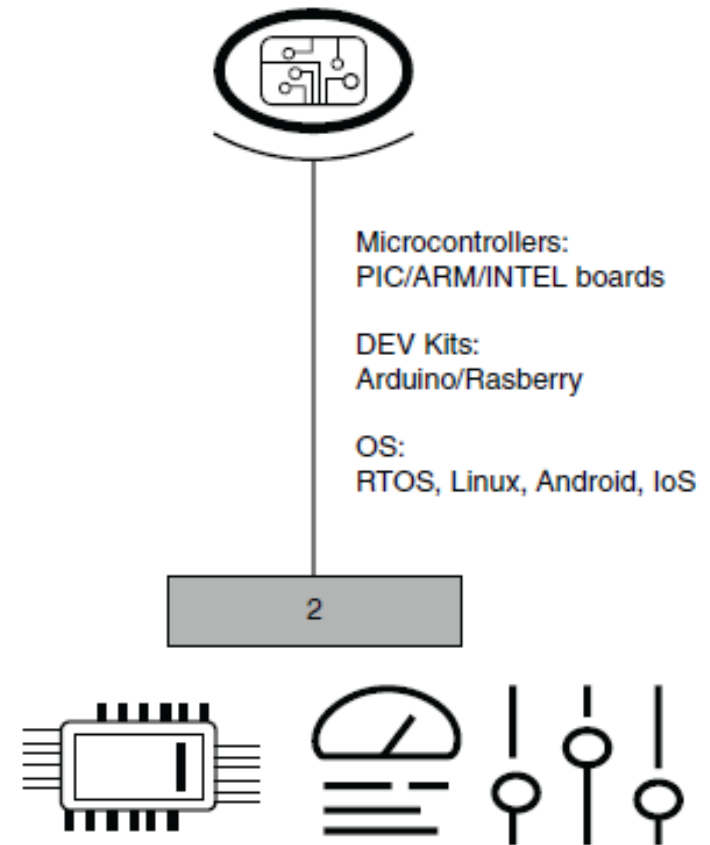
- ▶ This layer is concerned about the physical components, which mainly includes **sensors**.
- ▶ In this layer, the sensors are the core component.
- ▶ Temperature sensor, pressure sensor, humidity sensor, etc. can all be referred as physical layer components.
- ▶ In industrial automation, PLC, actuator, etc. are considered as physical layer components.
- ▶ This layer is responsible for data collection and action execution.
- ▶ Selection of sensors is important and choosing an appropriate sensor is the challenge in this layer.
- ▶ Action execution, sensing and data collection happens here.



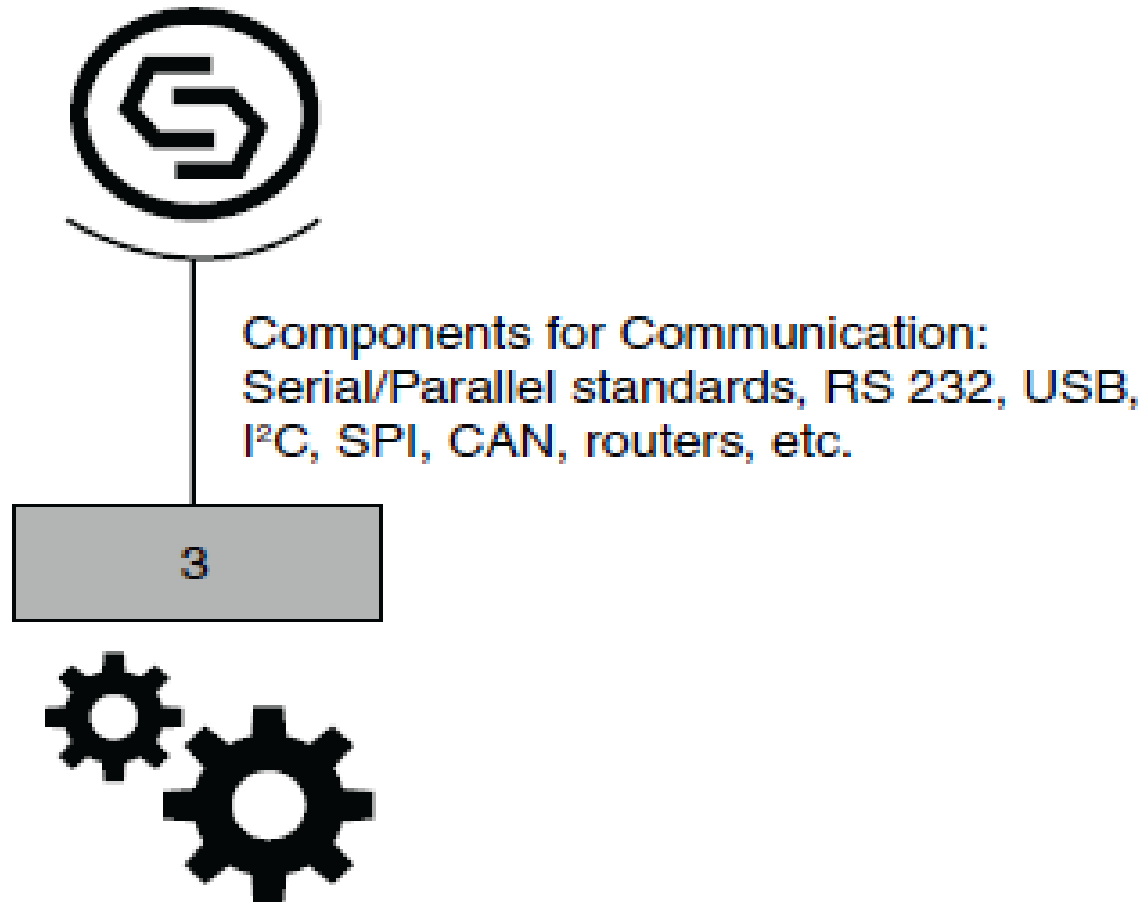
Layer 1: Sensor layer (physical layer); data collection happens here.

# IoT Stack - Layer 2 (Processing and Control Action Layer)

- ▶ This important layer contains core components of IoT system.
- ▶ The **microcontrollers or processors** are found in this layer.
- ▶ The data is received by the microcontrollers from the sensors.
- ▶ A variety of development kits are available in the market; like Arduino, Raspberry Pi, Node MCU, PIC, ARM development boards, etc.
- ▶ Microcontroller/Processor and operating system play vital role at this layer
- ▶ Data collected from the sensors is processed in this layer.

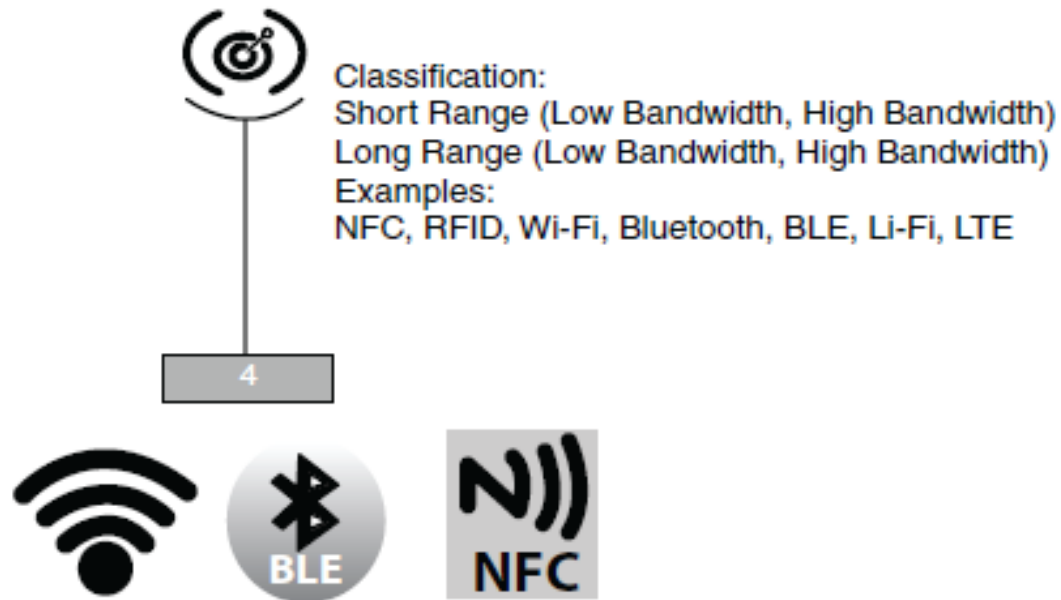


# IoT Stack - Layer 3 (Hardware Interface Layer)



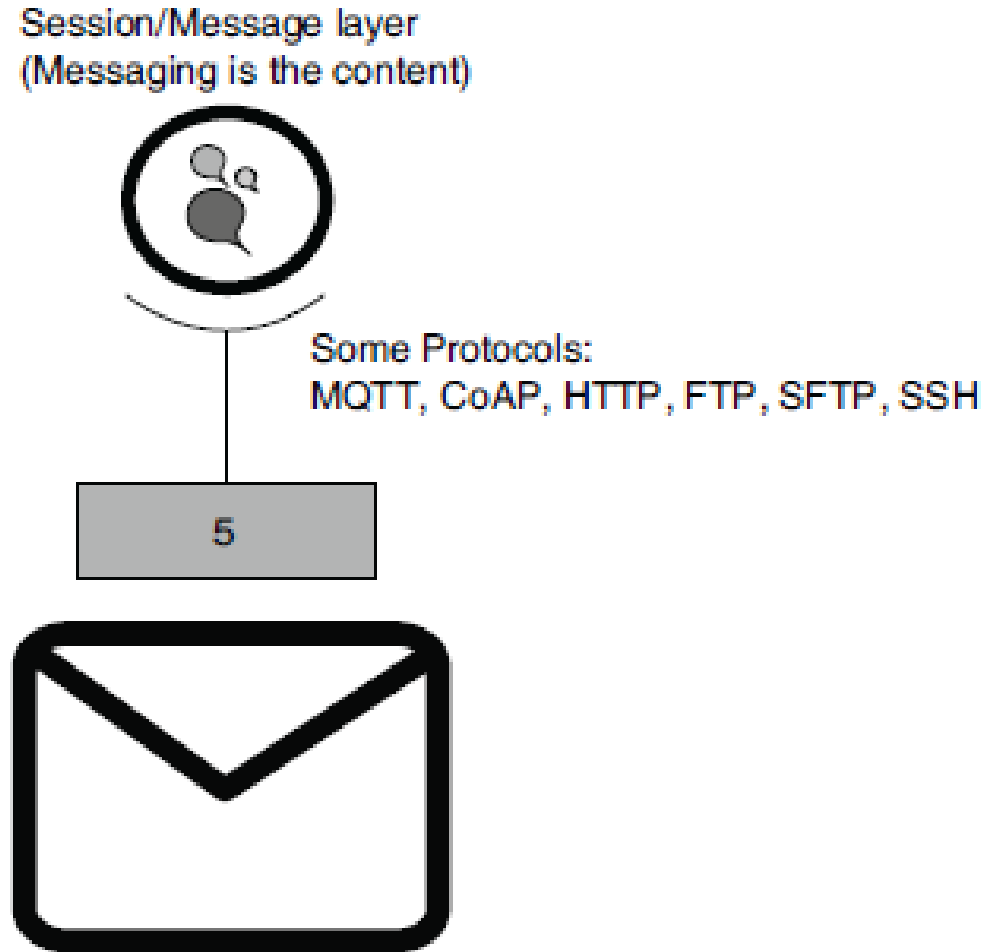
- ▶ The 3<sup>rd</sup> layer in the stack is the Hardware Interface Layer.
- ▶ **Hardware components and communication standards** such as RS232, CAN, SPI, SCI, I<sup>2</sup>C, etc. occupy this layer.
- ▶ All these components ensure flawless communication
- ▶ Handshake happens here.

# IoT Stack - Layer 4 (RF Layer)



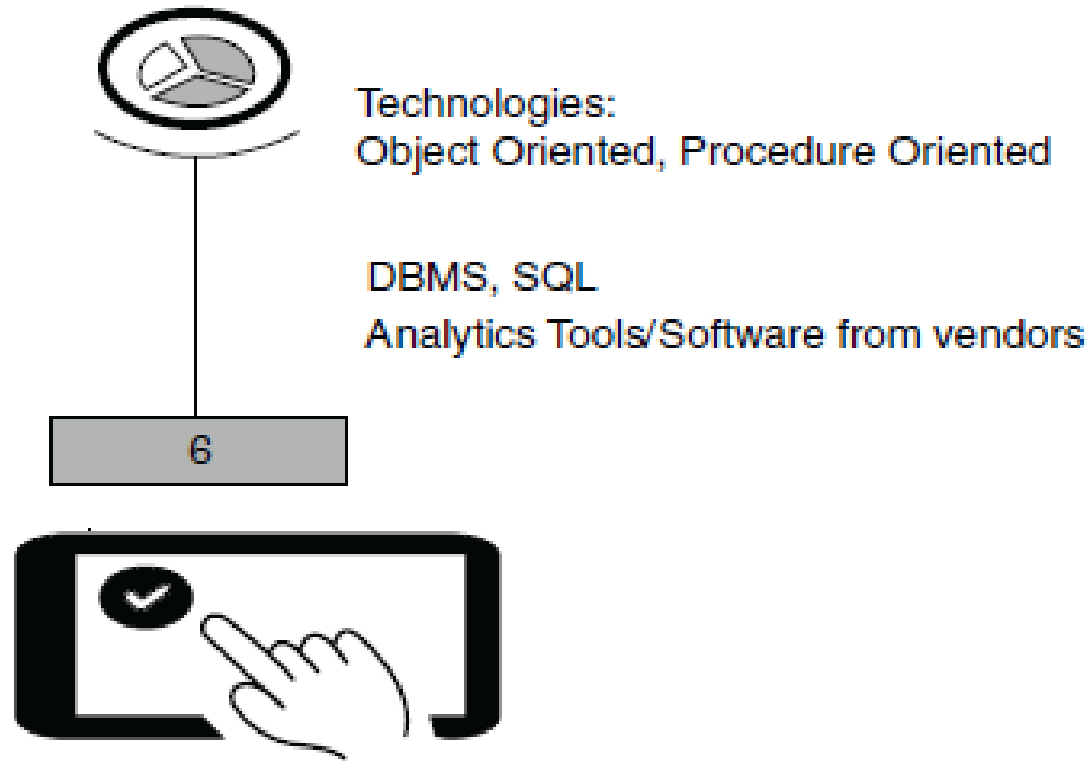
- ▶ Whenever one talks about IoT, RF is discussed and comes in picture.
- ▶ It plays a major role in the communication channel – whether it is short range or long range.
- ▶ Protocols used for communication and transport of data based on RF are listed in this layer.
- ▶ Some famous and common **protocols** are Wi-Fi, NFC, RFID, Bluetooth, Zigbee, etc.
- ▶ RF layer does communication of data using radio frequency based Electromagnetic (EM) waves
- ▶ This layer can also include Li-Fi; which are effective alternates for RF protocols.

# IoT Stack - Layer 5 (Session/Message Layer)



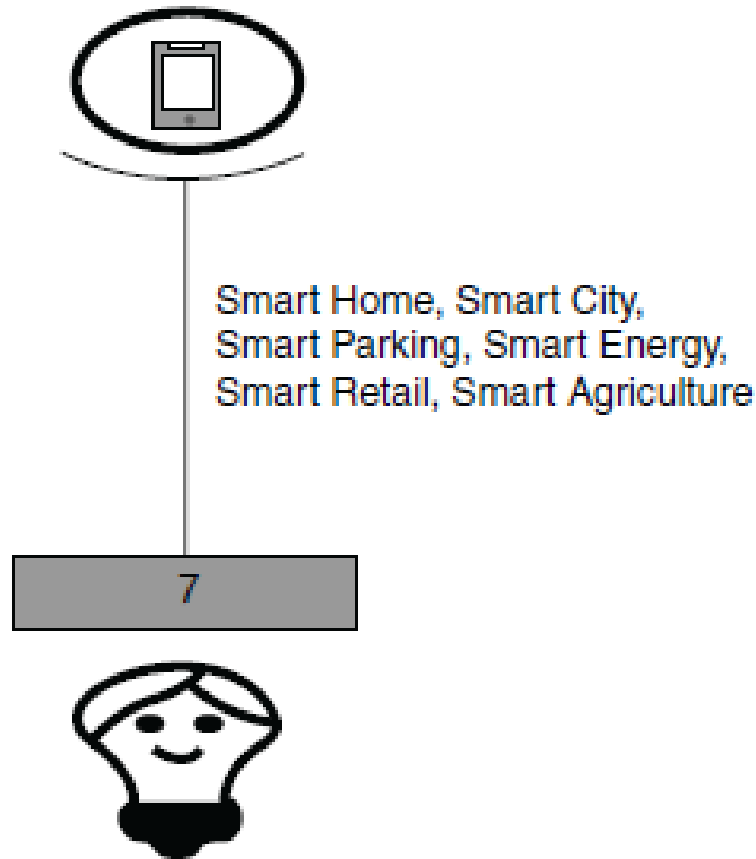
- ▶ Like computer network session management is also important in IoT.
- ▶ There are many protocols which manage how messages or data are broadcasted to the cloud.
- ▶ Layer 5 (session layer) deals with the various **messaging protocols** as MQTT, CoAP, etc. and also other protocols such as SSH and FTP.

# IoT Stack - Layer 6 (User Experience Layer)



- ▶ This layer deals with providing best experience to the end users of IoT products.
- ▶ The 6th layer takes care of rich UI designs with lots of features, which provide a pleasing experience while using the service/system or product.
- ▶ Object-oriented programming languages, scripting languages, analytics tools, etc. all should be included in this layer.
- ▶ This is also known as User Experience and Visualization Layer.

# IoT Stack - Layer 7 (Application Layer)



- ▶ Everything comes to perfection at this layer.
- ▶ This layer utilizes the rest six layers in order to develop desired application.
- ▶ It can range from a simple automation application to smart city application.
- ▶ After learning about the layers, it is now easier to relate them with an application, for example, vegetable quality monitoring during transport from source to the destination using IoT.

# Enabling Technologies

- ▶ IoT is a **collection or group of many technologies and devices.**
- ▶ The simplest of sensors, embedded systems, data analytics, communication protocols, security aspects and cloud computing with storage have all become enabling technologies.
- ▶ Enabling technologies/devices fall under one of the following **categories**:
  - ➔ Technologies that help in acquiring/sensing data.
  - ➔ Technologies that help in analyzing/processing data.
  - ➔ Technologies that help in taking control action.
  - ➔ Technologies that help in enhancing security/privacy.
- ▶ Let's discuss some of the enabling technologies.

## IoT Enabling Technologies

- **Wireless Sensor Network**



- **Cloud Computing**



- **Big Data Analytics**



- **Communication Protocols**



- **Embedded Systems**



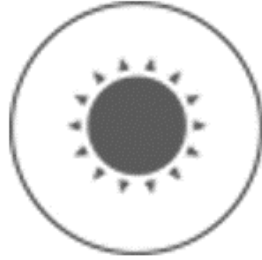
# Enabling Technologies - Sensors



**Temperature Sensor**  
Precision  $\pm 0.3^{\circ}\text{C}$   
Range - 40 to  $85^{\circ}\text{C}$   
- 40 to  $185^{\circ}\text{F}$



**Humidity Sensor**  
Precision  $\pm 3\% \text{RH}$   
Range 0 to 100%



**Ambient Light Sensor**  
Precision  $\pm 3\%$   
Range 0.01 to 83K lux



**Vibration Index Sensor**  
Precision 4mg  
Range - 16 to 16g



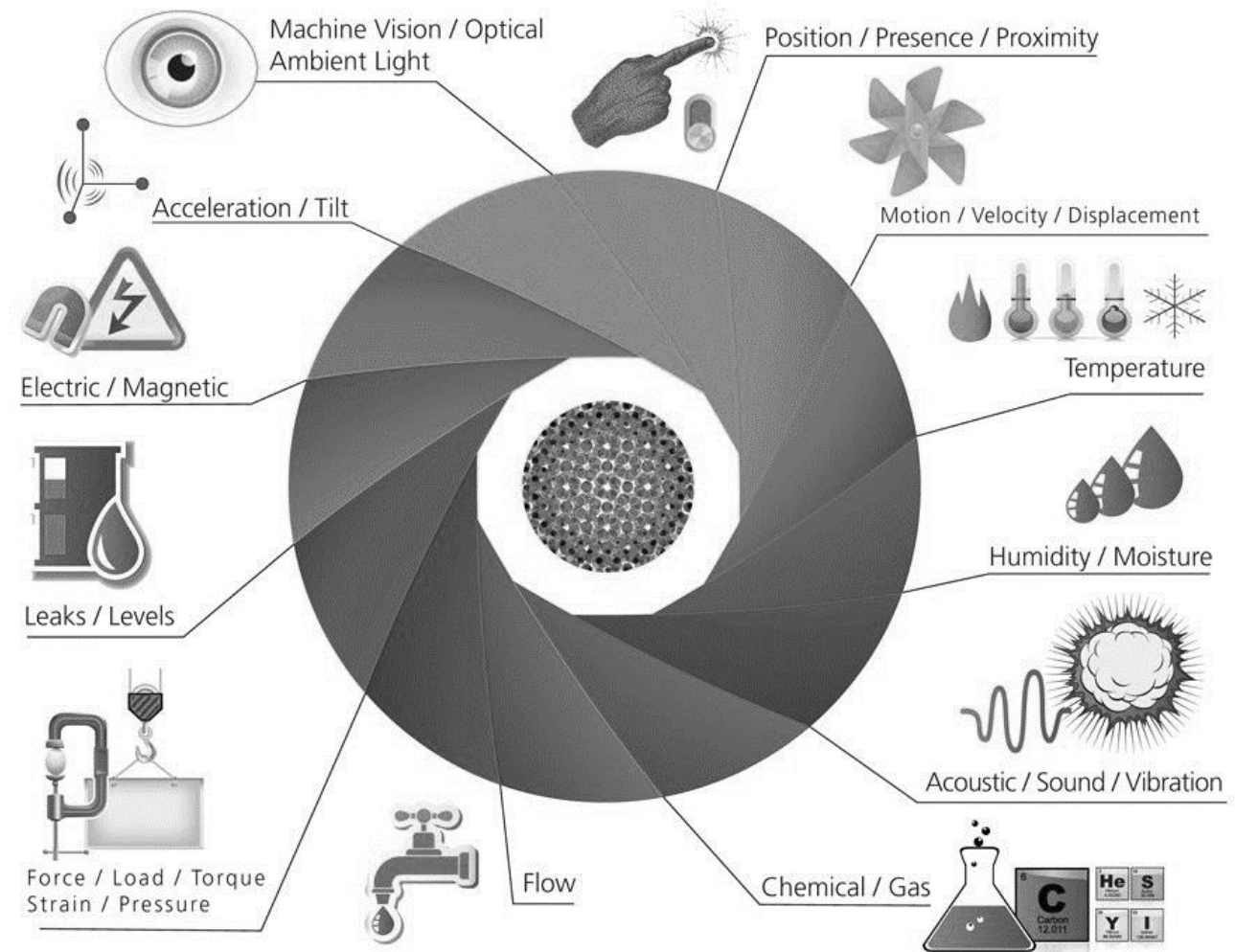
**External Temperature Sensor**  
Precision  $\pm 0.5^{\circ}\text{C}$   
Range - 55 to  $125^{\circ}\text{C}$   
- 67 to  $257^{\circ}\text{F}$

- ▶ Sensors are at the heart of any IoT application.
- ▶ As the name suggests, they sense the environment and retrieve data.
- ▶ Sensors are the starting point of any IoT application.
- ▶ It fetches data for us to operate on.
- ▶ Sensors could be analog or digital.
- ▶ Temperature sensor in a thermometer is an example of it. It is used to build temperature monitoring application.

# Enabling Technologies - Sensors

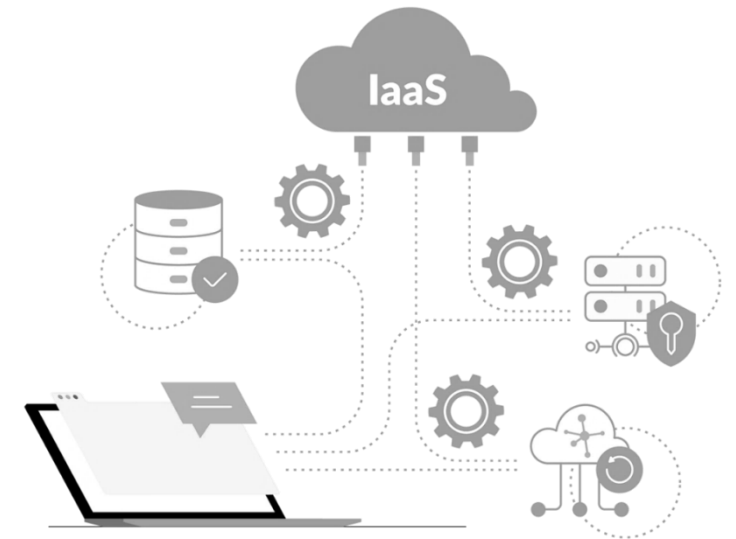
► Some examples of sensors that could be regarded as enabling technologies are as follows.

- ➔ Weather tracking system uses temperature/humidity/moisture sensors.
- ➔ Vehicle health monitoring sensors keep track of speed, tyre pressure, etc.
- ➔ On Board Diagnostics (OBDs) used for collecting all critical information from an automobile to detect error.
- ➔ Vibration sensors are used to track the quality of buildings/structures.
- ➔ Water quality is monitored through sensors that measure PH, chloride level, etc.
- ➔ PIR sensor is used in pedestrian signal operation with human presence detection.



# Enabling Technologies - Cloud Computing

- ▶ The next technology that is highly significant in IoT is cloud computing.
- ▶ Cloud has grown much more popular because it serves as an affordable, effective and efficient medium for data storage.
- ▶ Data storage plays a major role in IoT.
- ▶ Cloud services are categorized as follows:
  - ➔ **IaaS (Infrastructure-as-a-Service):**
    - In this cloud service, one can choose **virtual machines** over physical machines.
    - It is a form of cloud computing that provides virtualized computing resources over the Internet.
    - The users manage the machines, select the OS and underlying applications, and pay per their use.



# Enabling Technologies - Cloud Computing

## ➔ PaaS (Platform-as-a-Service):

- This is a cloud computing model in which the cloud service provider delivers **hardware and software tools needed for application development to users over the Internet.**
- A PaaS provider hosts the hardware and software on its own infrastructure. Users have to build, manage and maintain the applications as per their requirement.



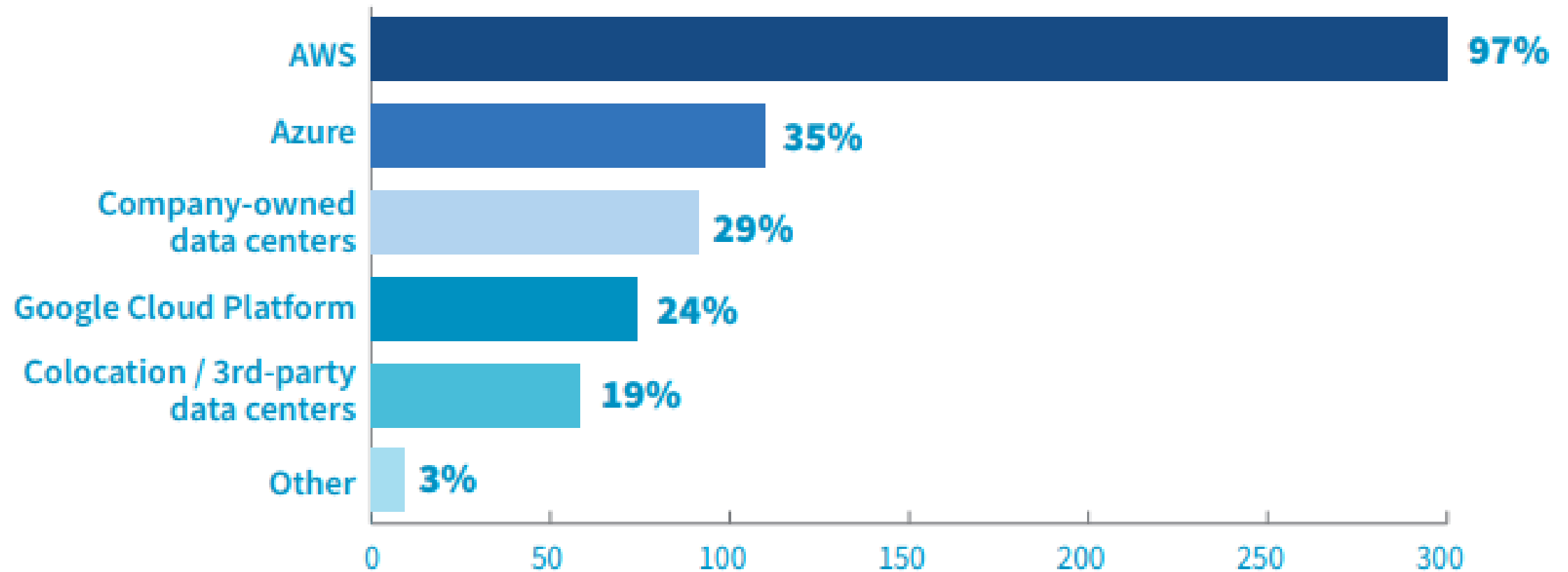
## ➔ SaaS (Software-as-a-Service):

- In this model, **a complete software application** is provided to the user.
- It can also be called application as a service. This service can be availed by paying a monthly, yearly, etc., subscription.
- Some well-known service providers in the market are Amazon web services, Azure and Adafruit.

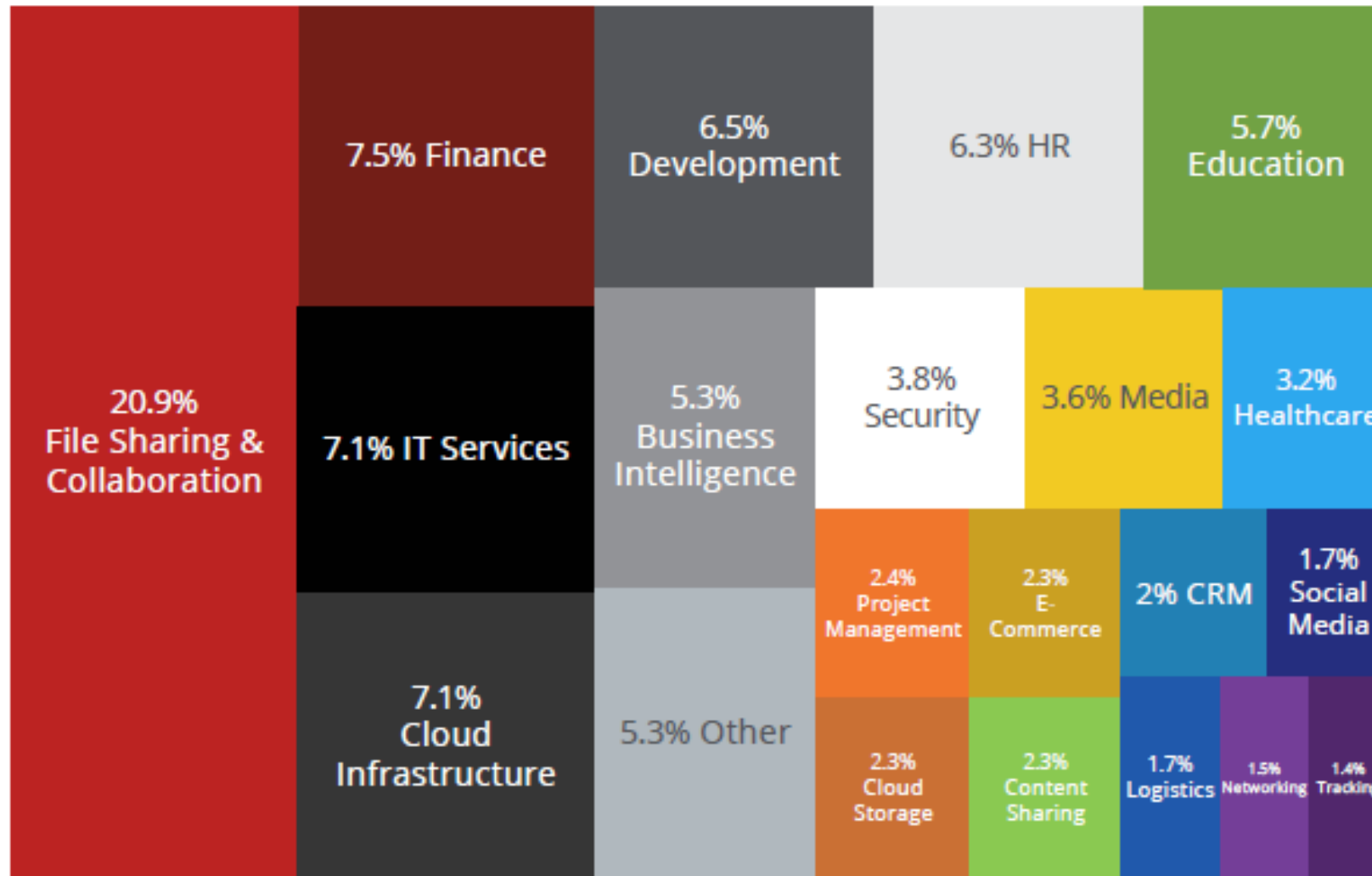
<https://www.zdnet.com/article/the-top-cloud-providers-of-2021-aws-microsoft-azure-google-cloud-hybrid-saas/>

# Enabling Technologies - Cloud Computing: Trends

WHICH DOES YOUR ORGANIZATION USE?



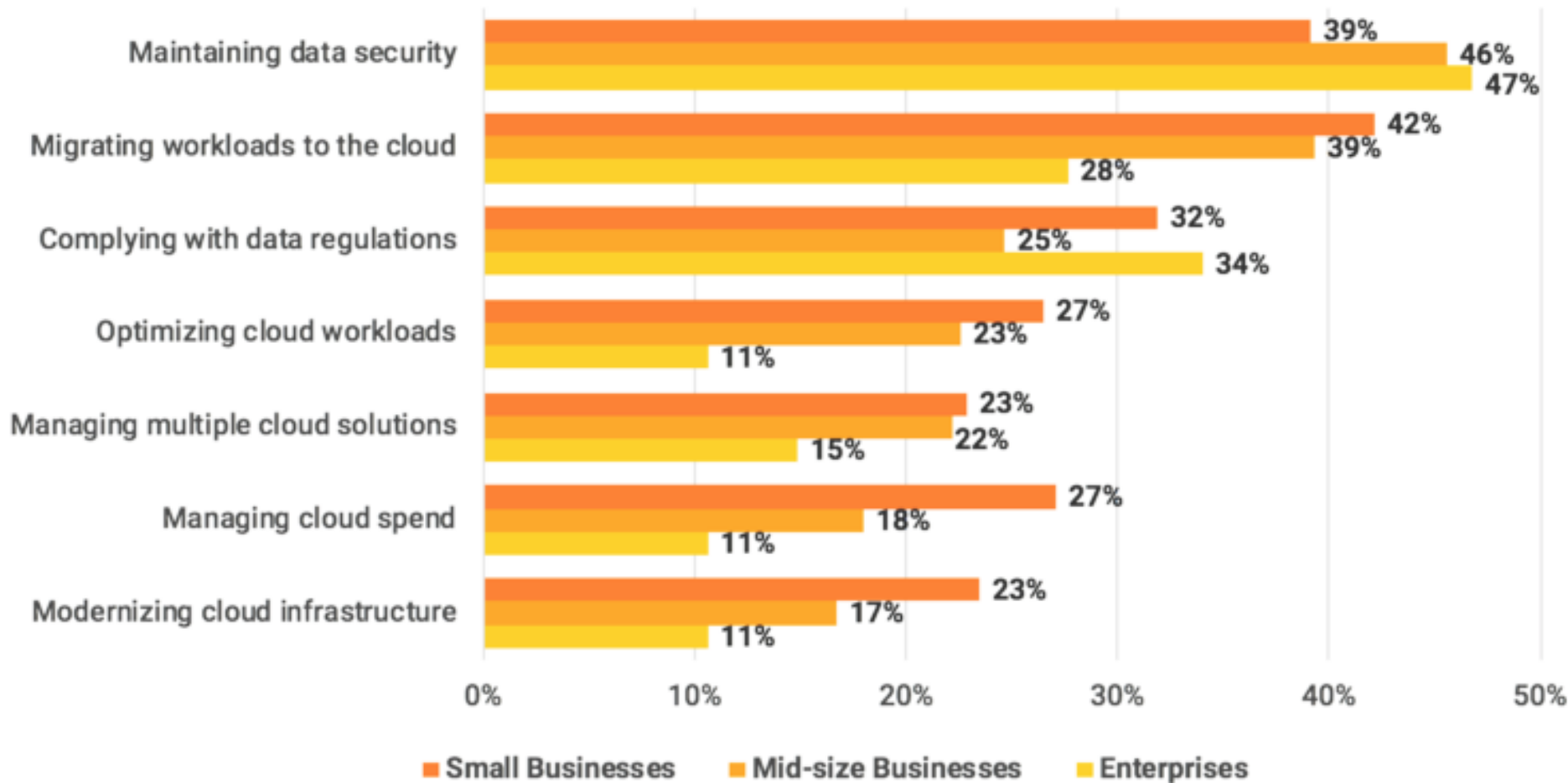
# Enabling Technologies - Cloud Computing : Application areas



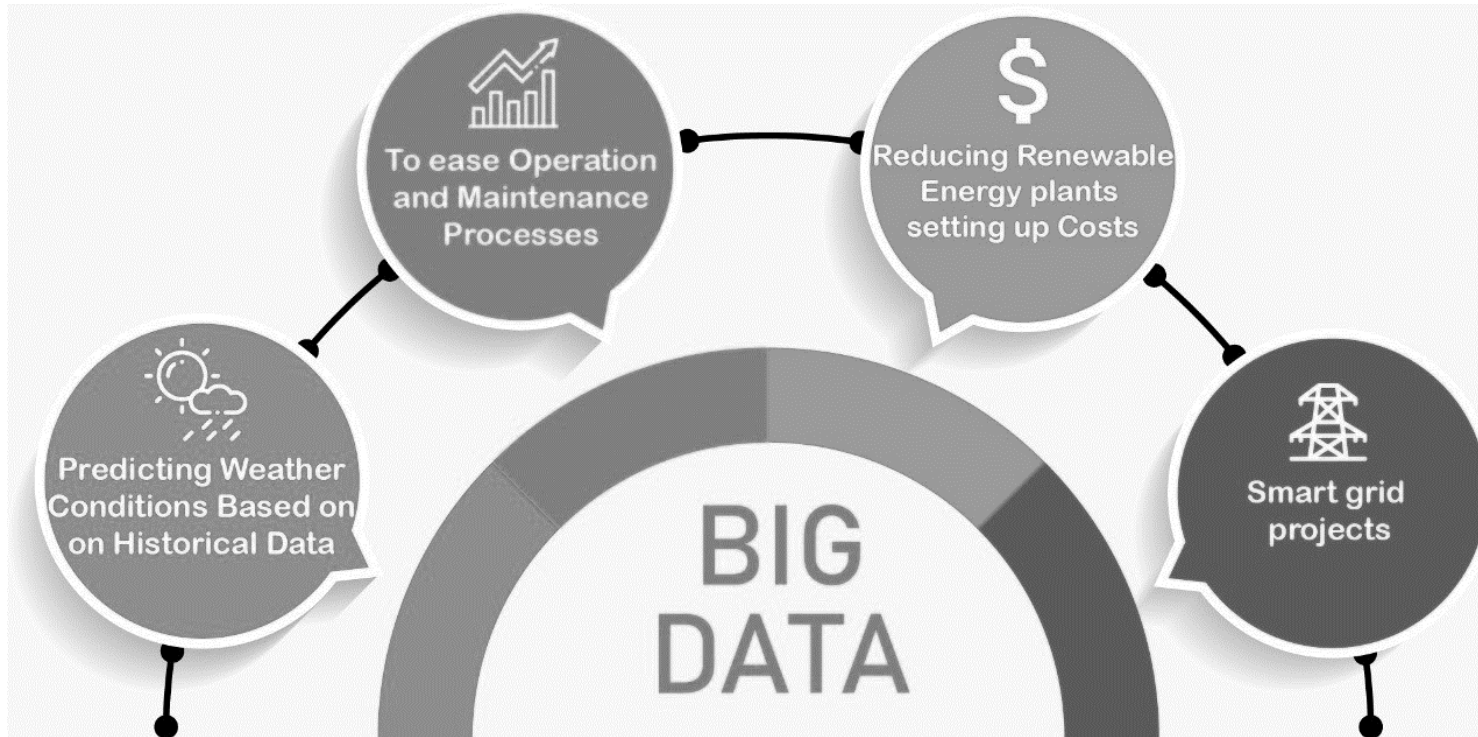
## Enabling Technologies - Cloud Computing

### Top Areas Where Businesses Need More Support From Cloud Vendors

*By Company Size*



# Enabling Technologies - Big Data Analytics



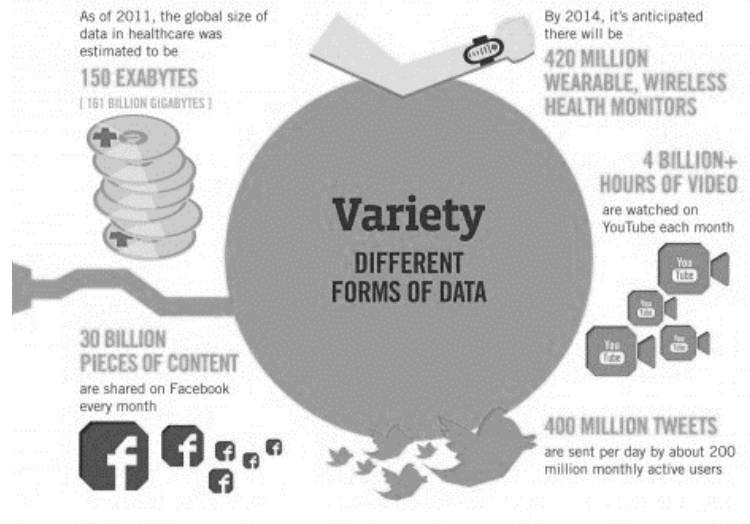
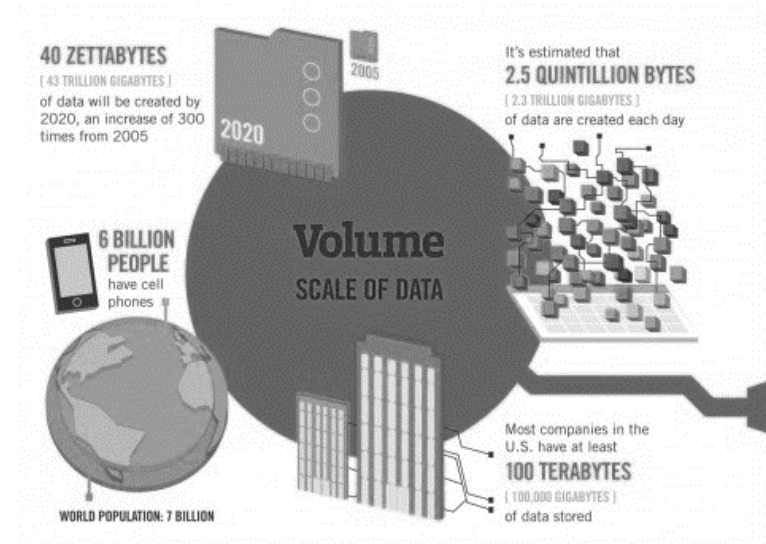
- ▶ Data is everywhere, and from every function or operation we get more data.
- ▶ IoT is all about collecting data from various sensory nodes.
- ▶ Handling the huge data is fundamental to make the application a success.
- ▶ The biggest challenge with big data is 4Vs. Volume, Variety, Speed (Velocity) at which it comes and its Veracity.

<https://www.datacenters.com/news/and-the-title-of-the-largest-data-center-in-the-world-and-largest-data-center-in#:~:text=But%20First%2C%20a%20Quick%20Journey,million%20square%20feet%20of%20space.>

# Enabling Technologies - Big Data Analytics

## ► Scale (Volume):

- ➔ Huge volume of data is generated every minute.
- ➔ Storage has become inexpensive and hence, cost-related challenges have reduced.
- ➔ Cloud storage and hardware storage both have become affordable because of the tremendous growth in the semiconductor industry.



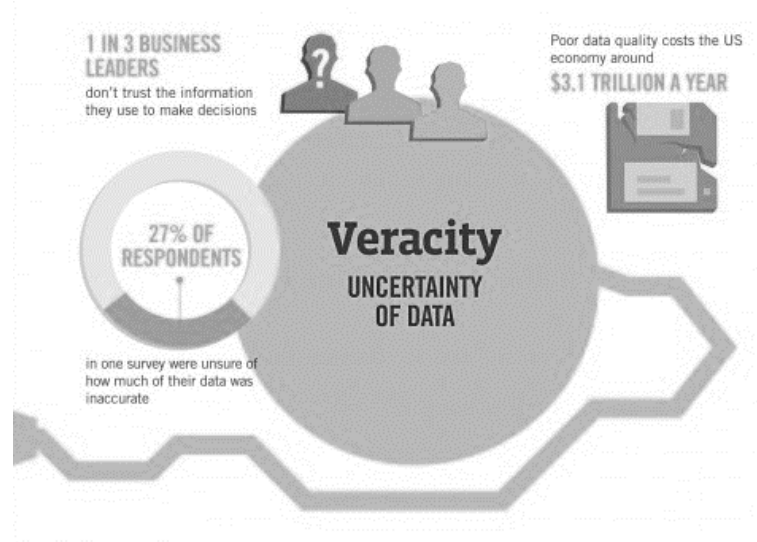
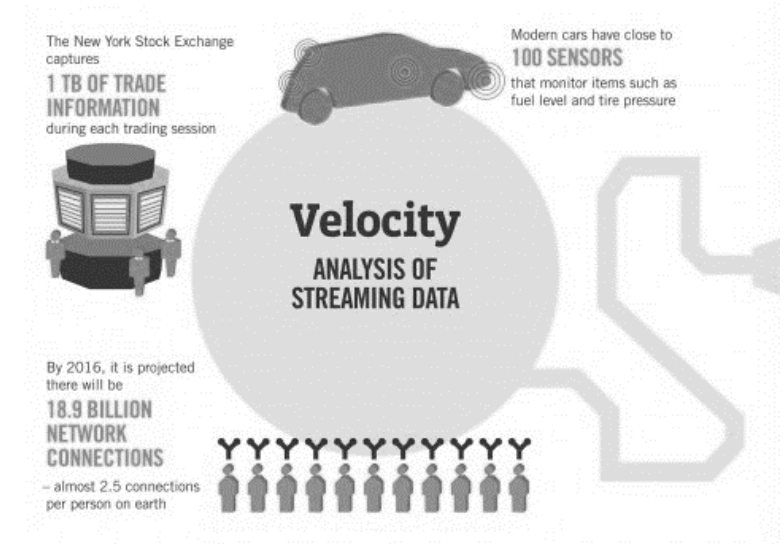
## ► Complexity (Variety):

- ➔ Data no longer comes from one single source.
- ➔ It also comes in different formats (e.g., audio, video, text and image) and has to be interpreted systematically.
- ➔ Varieties of data becomes a huge challenge.

# Enabling Technologies - Big Data Analytics

## ► Speed (Velocity):

- ➔ The rate at which data is generated very fast.
- ➔ Also, data dynamics changes very frequently.
- ➔ Nowadays, data comes from anywhere – from fit bit watches to refrigerators.
- ➔ All the data pours in at a very high speed, which makes it very challenging.

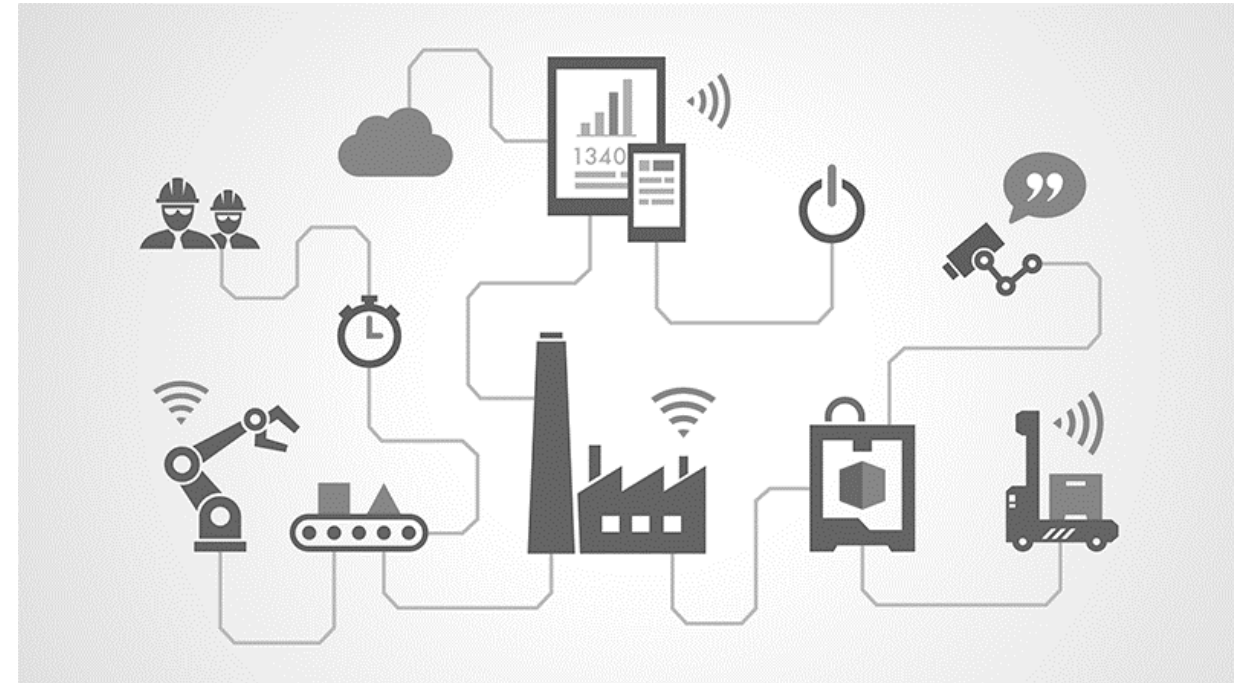


## ► Data in doubt (Veracity):

- ➔ How accurate is all this data anyway?
- ➔ Because we are now rely on it
- ➔ The data's nature alters dynamically and uncertainty is often seen.
- ➔ So, it would be challenging to process this unstable data.

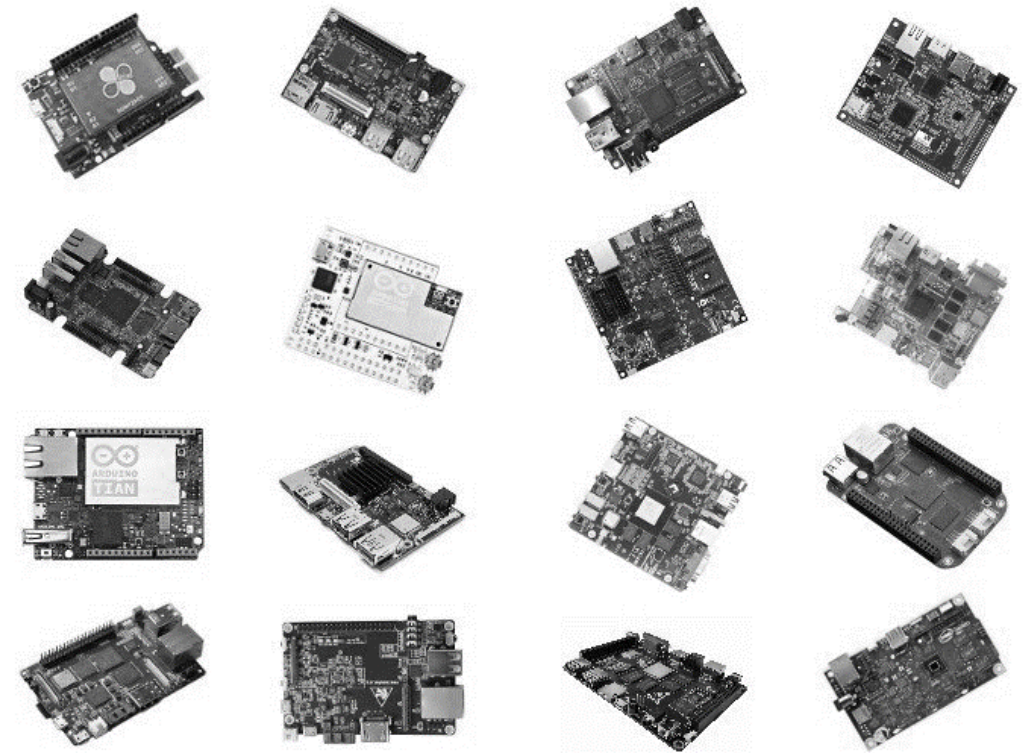
# Enabling Technologies - Big Data Analytics

- ▶ So the question is: “Who is generating all this data?” A partial list to answer this is as follows:
  - ➔ Sensors from security systems.
  - ➔ Sensors from weather monitoring systems.
  - ➔ Sensors from car/navigation systems.
  - ➔ Sensors from water quality monitoring systems.
  - ➔ Data from wearables (e.g., bands).
  - ➔ Data from industrial equipment (e.g., motor health).
  - ➔ Sensors from bridges/roads about traffic density and other factors.
  - ➔ Social media (e.g., tweets, photo uploads, etc.).
- ▶ In IoT data is everything. so, data analytics is one of the enabling technologies for building a complete and IoT application.



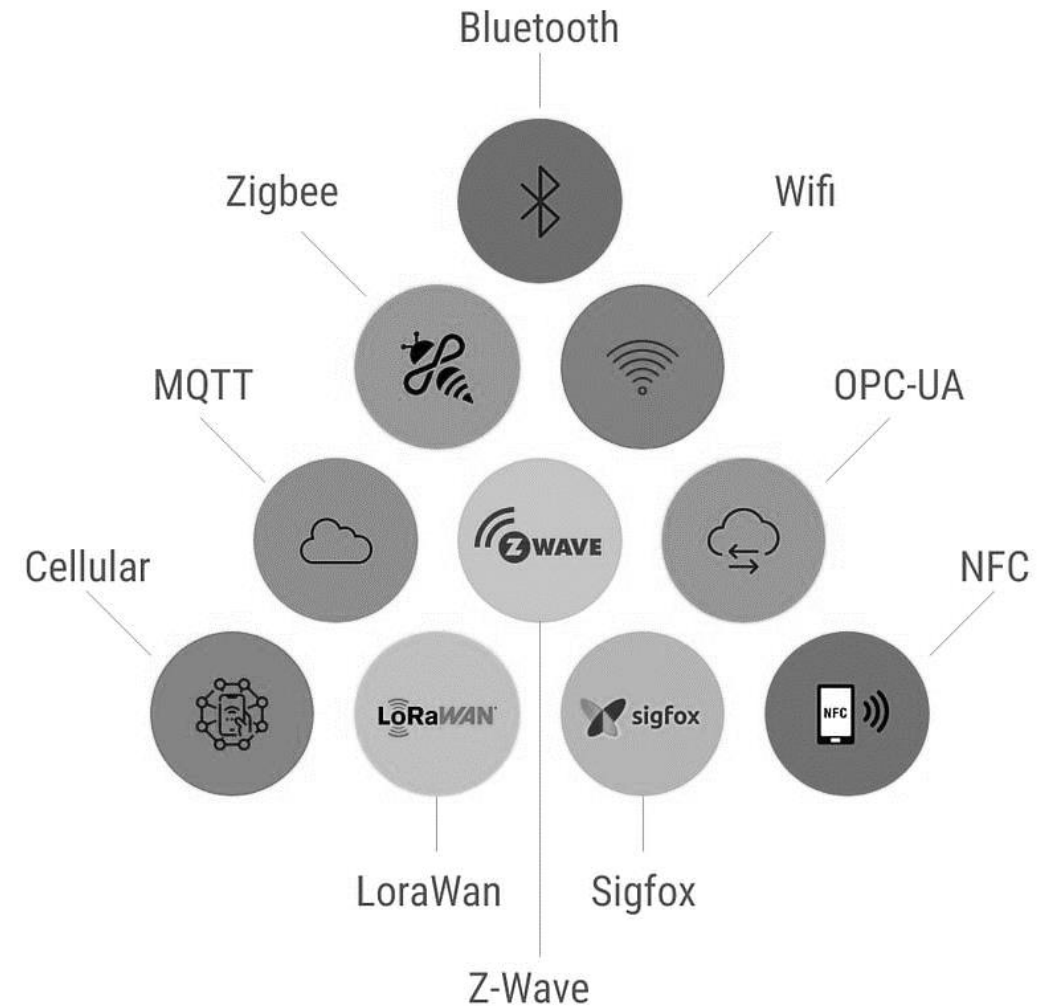
# Enabling Technologies - Embedded Computing Boards

- ▶ An embedded computing board a very important component to bring IoT design to reality.
- ▶ For making the prototype the computing boards play vital role.
- ▶ The computing boards available in the market are driven by microcontrollers or processors.
  - ➔ Some of the boards are as follows:
    - Raspberry Pi.
    - Arduino (many variants).
    - NodeMCU.
    - Intel Edison.
- ▶ All these boards are small, yet smart.
- ▶ Also, the cost involved is very minimal and one can get these boards at cheap rate.



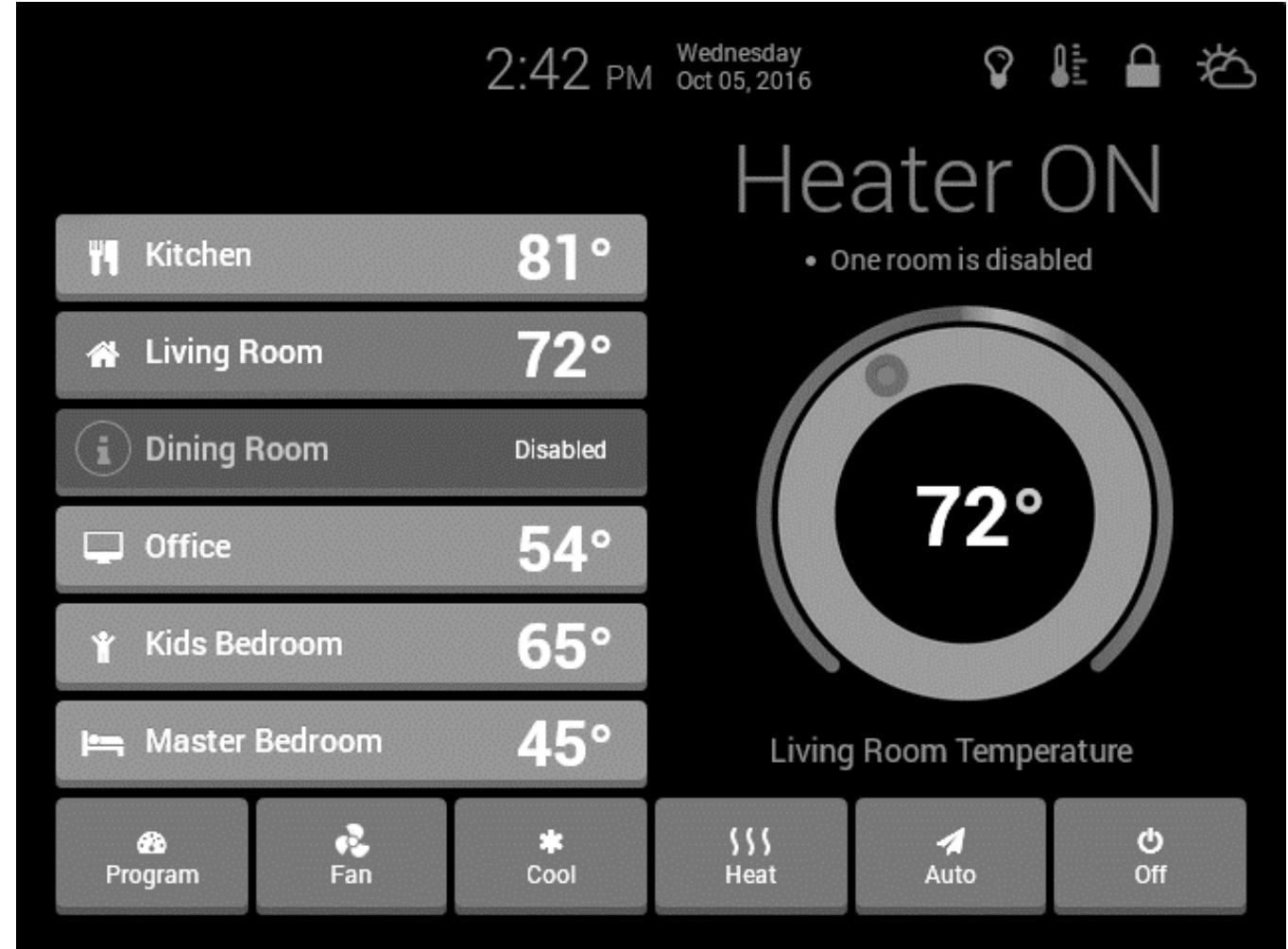
# Enabling Technologies - Communication Protocols

- ▶ Protocols are the pillars for good IoT infrastructure and hence are very important in communication.
- ▶ Data exchange happens through these protocols, which take care of the following:
  - ➔ Addressing.
  - ➔ Format of the messages.
  - ➔ Message security (encryption and decryption).
  - ➔ Routing.
  - ➔ Flow control.
  - ➔ Error monitoring.
  - ➔ Sequencing.
  - ➔ Retransmission guidelines.
  - ➔ Segmentation of the data packets.



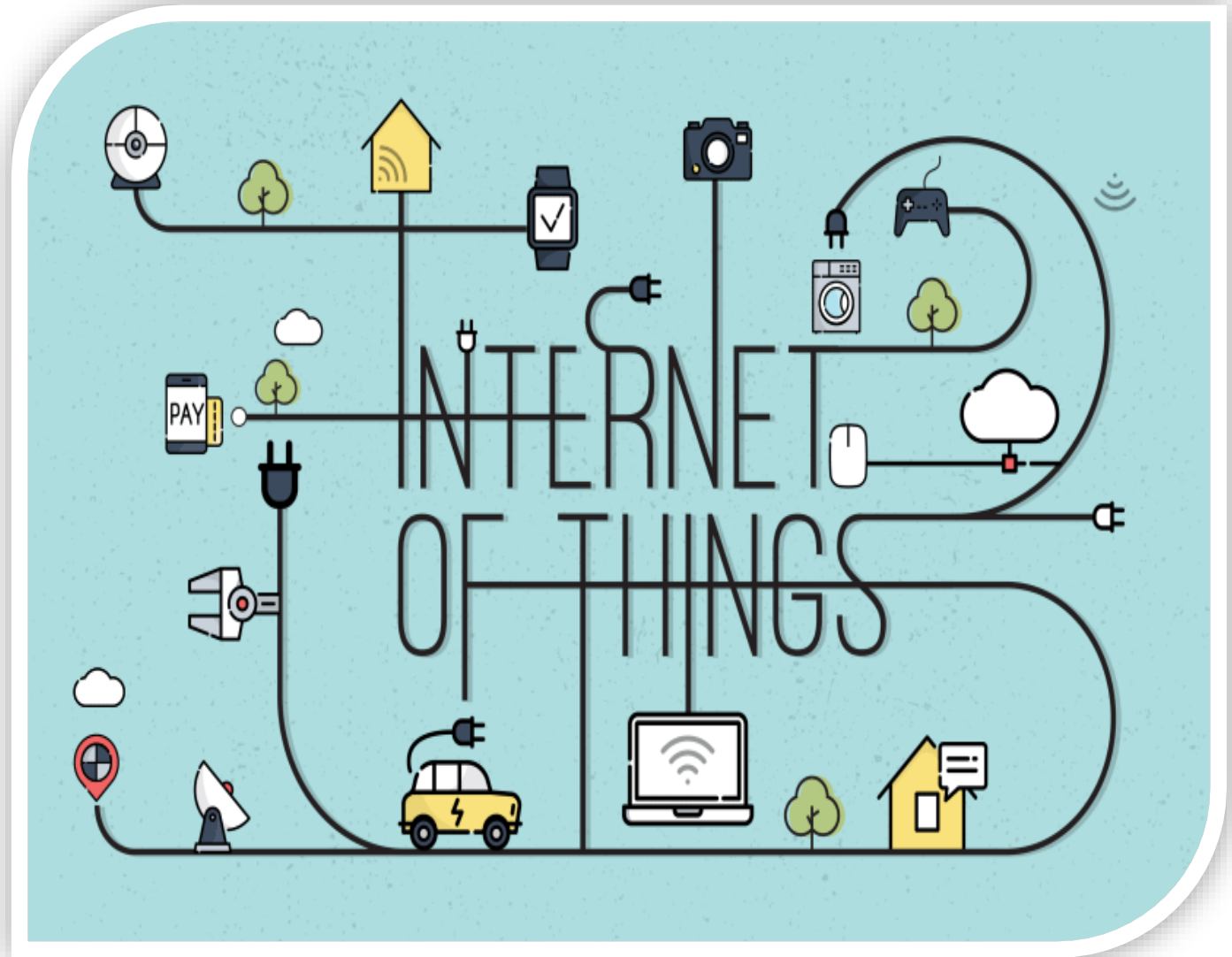
# Enabling Technologies - User Interfaces

- ▶ All devices should have an intuitive user interface.
- ▶ IoT devices/services should be designed in such a way that accessing and handling the services are easier and comfortable for the end user.
- ▶ Generally, the end user shall be provided “mobile application or web application”.
- ▶ The application should be stable and elegant.



# Unit Progress:

- ▶ Introduction to IoT
- ▶ Application areas of IoT
- ▶ Characteristics of IoT
- ▶ Things in IoT
- ▶ IoT Protocol Structures
- ▶ Enabling Technologies
- ▶ **IoT Challenges**
- ▶ **IoT Levels**
- ▶ **Physical Design of IoT**
- ▶ **Logical Design of IoT**



# IoT Challenges

- ▶ The following are some of the challenges - technical and non-technical during building an IoT application .

## 1. Security/Personnel safety:

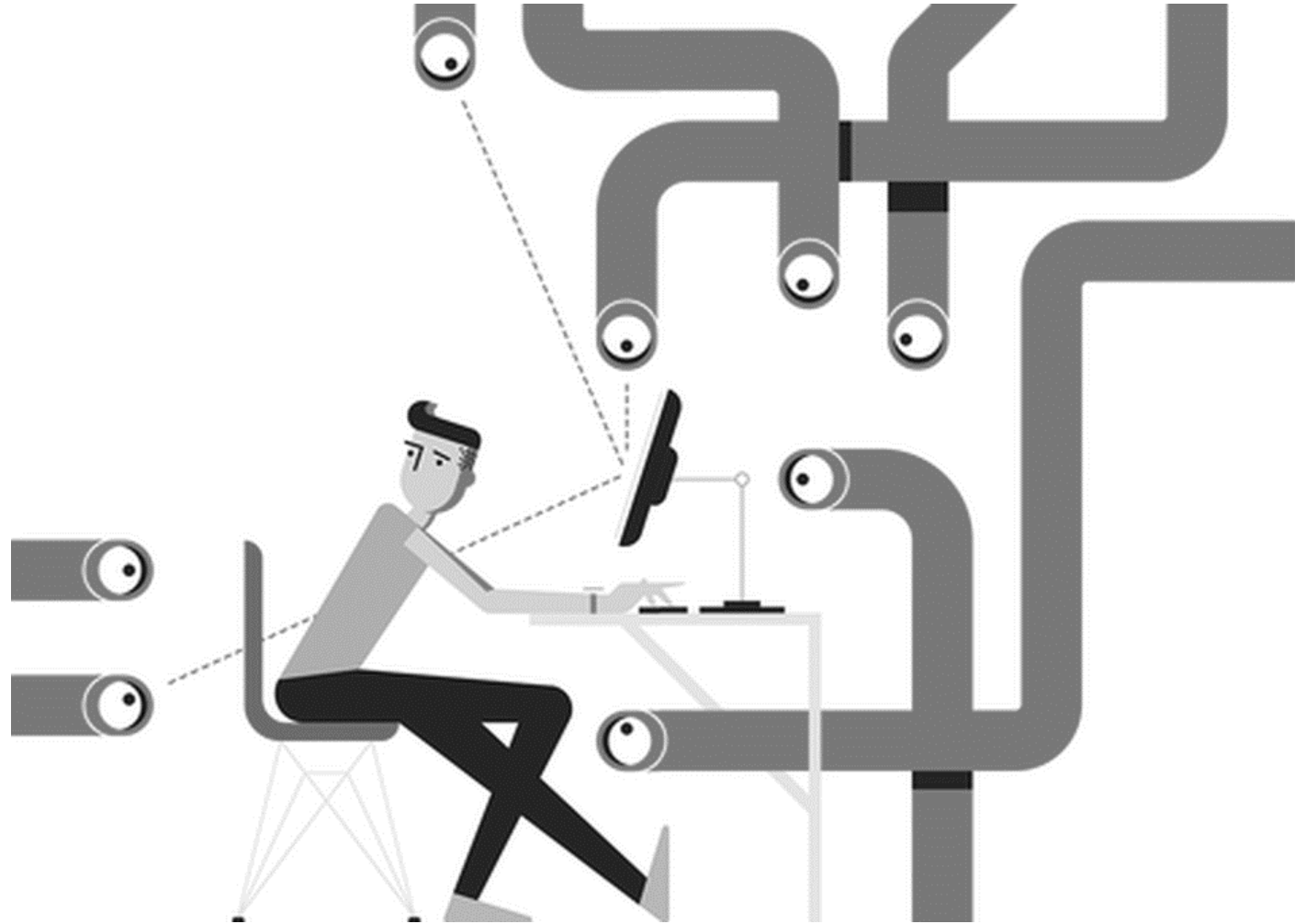
- ➔ Security is one of the most significant challenges.
- ➔ Number of IoT devices are gradually increasing, so user data becomes more vulnerable to theft.
- ➔ Poor security features can let attackers damage the whole network and the rest of the devices could also become vulnerable.
- ➔ People's personal safety is also a concern and challenge.
- ➔ The implants and wearable used by people should be safe even from physical harm.



# IoT Challenges

## 2. Privacy:

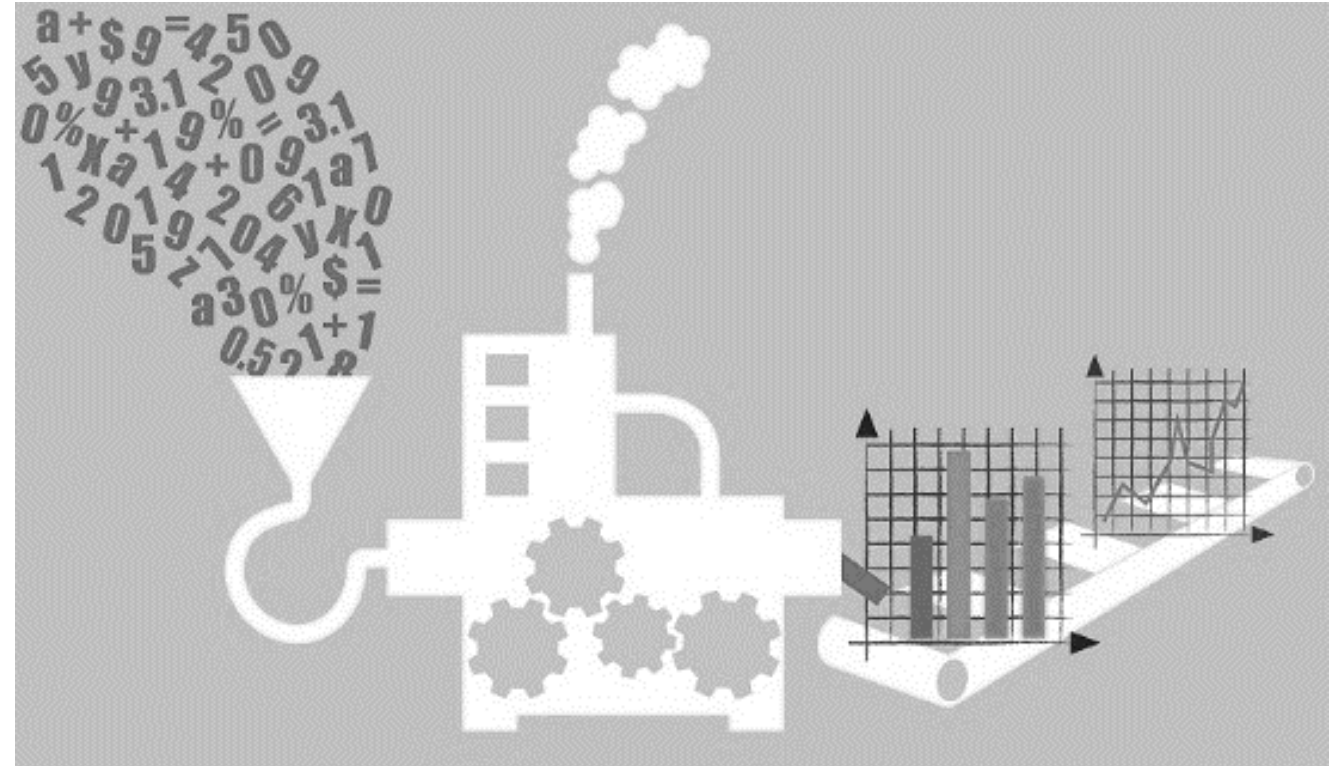
- One could be tracked/monitored by anyone, as we are connected 24 × 7 to the Internet.
- So, there is a threat on user data and raises a question on user privacy.
- “How do we ensure that the data that is sensed and collected from the user is with their permission?”



# IoT Challenges

## 3. Data extraction with consistency from complex environments:

- ➔ It is a huge challenge to sense and extract data from complex environments.
- ➔ For Example, variation in temperature could also damage the products being transported.
- ➔ Maintained the temperature is very critical and should be accurately monitored.
- ➔ If the temperature is about to change, then the corrective action has to be taken.
- ➔ Data extraction and storage in the cloud could be more challenging if the internet is not available.
- ➔ Extracting data inside a room is different from extracting data from an open environment.



# IoT Challenges

## 4. Connectivity:

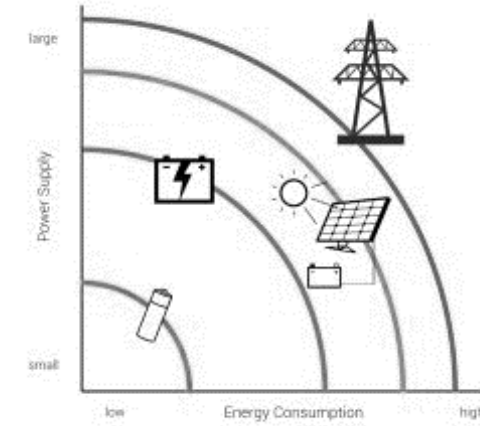
- ➔ This is a serious challenge that the IoT world must acknowledge.
- ➔ Since the Internet is itself a giant collection of networks and devices and IoT is a part of it. So requirement of wired and wireless connectivity is a necessity.
- ➔ The usage of frequency / spectrum is also to be noted.
- ➔ There are spectrum regulations to be followed based on the country for which the application is being developed.
- ➔ 2.4 GHz band is the ideal band everywhere.



# IoT Challenges

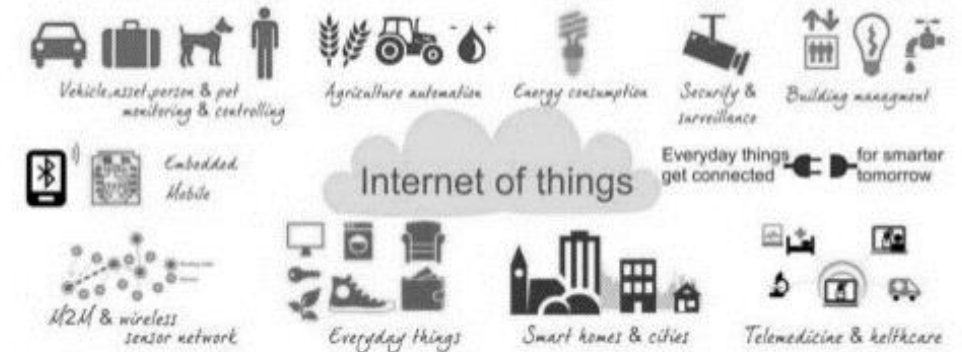
## 5. Power requirements:

- All the IoT devices require power and most of them are battery operated.
- Even though we now have long-lasting batteries that are economical, demand for power is on the rise.
- Usage of green power sources such as solar and wind should be motivated.
- If the power requirements are met appropriately, IoT can be even more powerful.



## 6. Complexity involved:

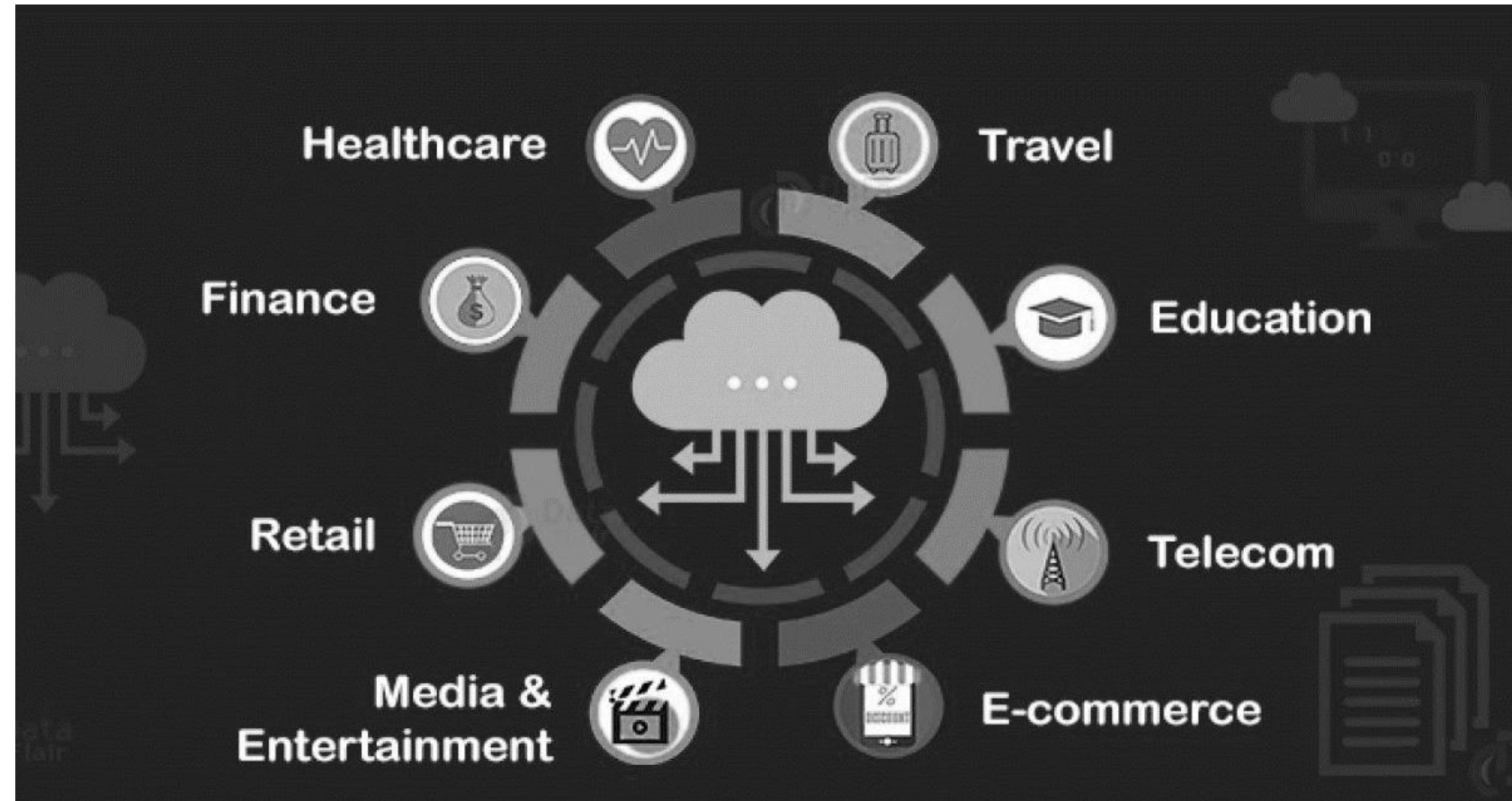
- IoT is not easy. It needs a lot of different domains to integrate into a cohesive system.
- There is very limited expertise available in the market, but the growth is very rapid.
- The toolkits, software and hardware are not abundant and real skill is required to build an application.



# IoT Challenges

## 7. Storage:

- Cloud is becoming mandatory for the data to be stored and analyzed.
- The challenge with respect to this aspect is connected to the following points:
  - Which cloud do we use (private, public, or hybrid)?
  - How do we identify the service provider?
  - How much does it cost?
  - Do we really need cloud?

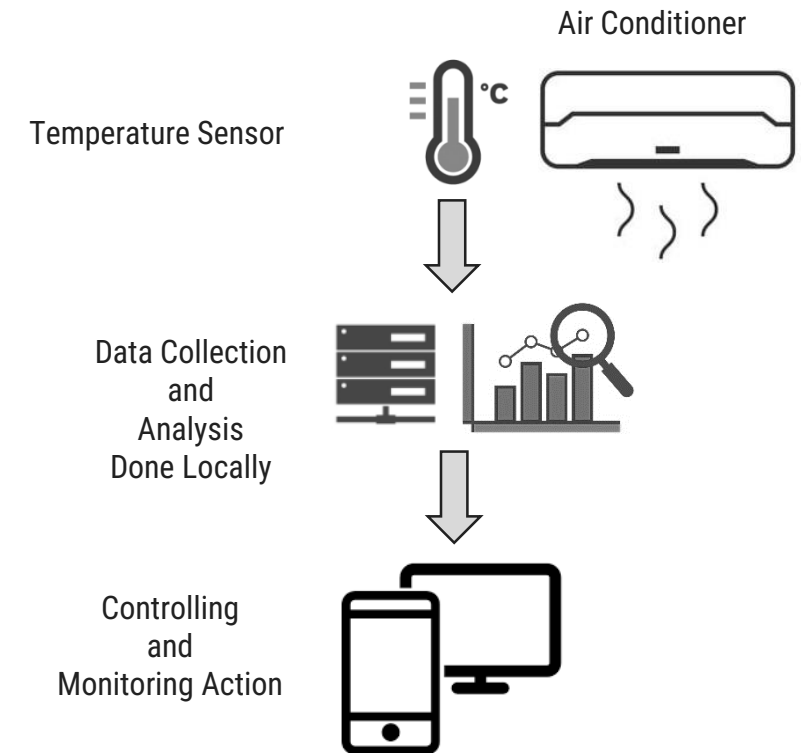


# IoT Levels

► Based on the architectural approach, IoT can be classified in five levels: Level 1 to Level 5.

## ► Level 1

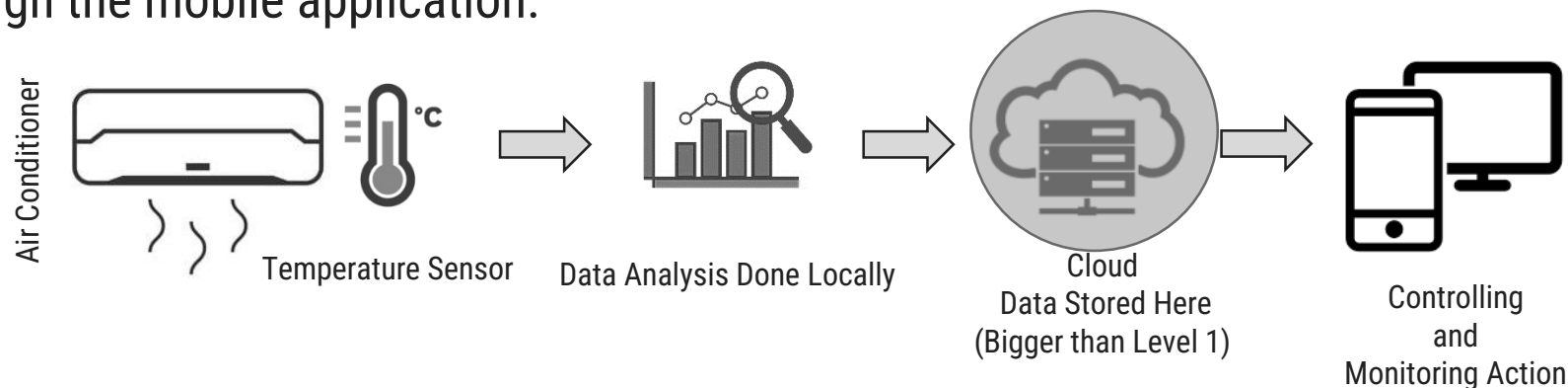
- It is of minimal complexity.
- The application has one sensor (temperature sensor, pressure sensor, etc).
- **The data sensed is stored locally and the data analysis is done locally.**
- Monitoring / control is done through an application.
- This is used for simple applications.
- Data generated in this level application is not huge.
- For example, a temperature sensor senses the room temperature and the data is stored and analyzed locally.
- Based on the analysis, the control action can be triggered through mobile application or it can help in monitoring the status.



# IoT Levels

## ► Level 2

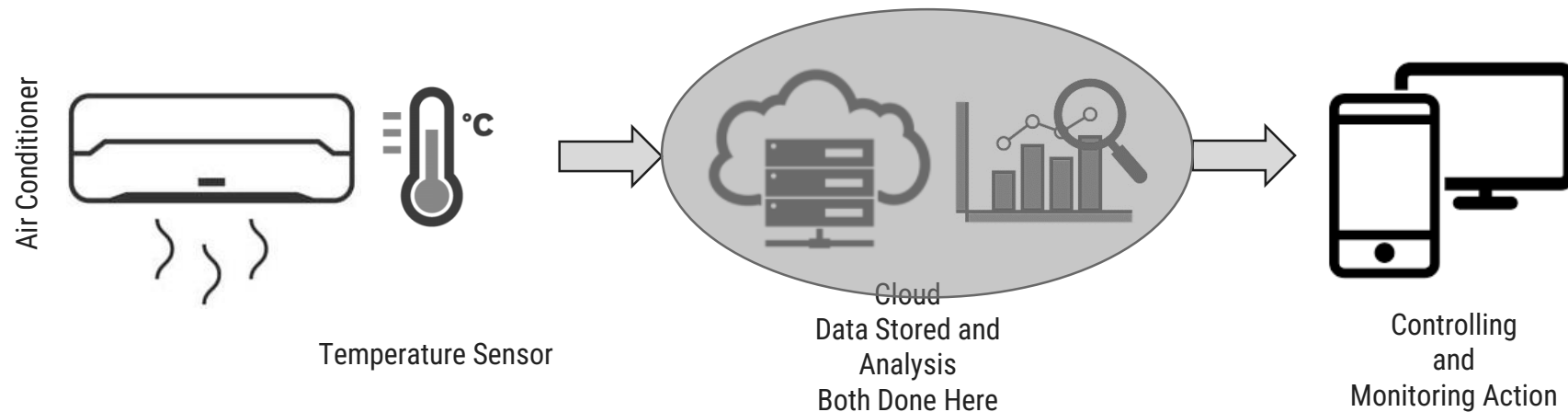
- The second level is slightly more complex than the previous level.
- **The data is more voluminous and hence, cloud storage is preferred.**
- The frequency of sensing done by the sensor is faster.
- The number of times sensing is done would be much more than Level 1.
- **The analysis is carried out locally, while cloud is meant for storage only.**
- Based on the data analysis, the control action can be triggered through the web application or mobile application.
- Some examples are agriculture applications, room freshening solutions based on odor, etc.
- IoT application of an air conditioner. The sensor reads the room temperature at a better pace and rate than Level 1; the data then goes on to the cloud for storage. Analysis is done locally and the action is triggered through the mobile application.



# IoT Levels

## ► Level 3

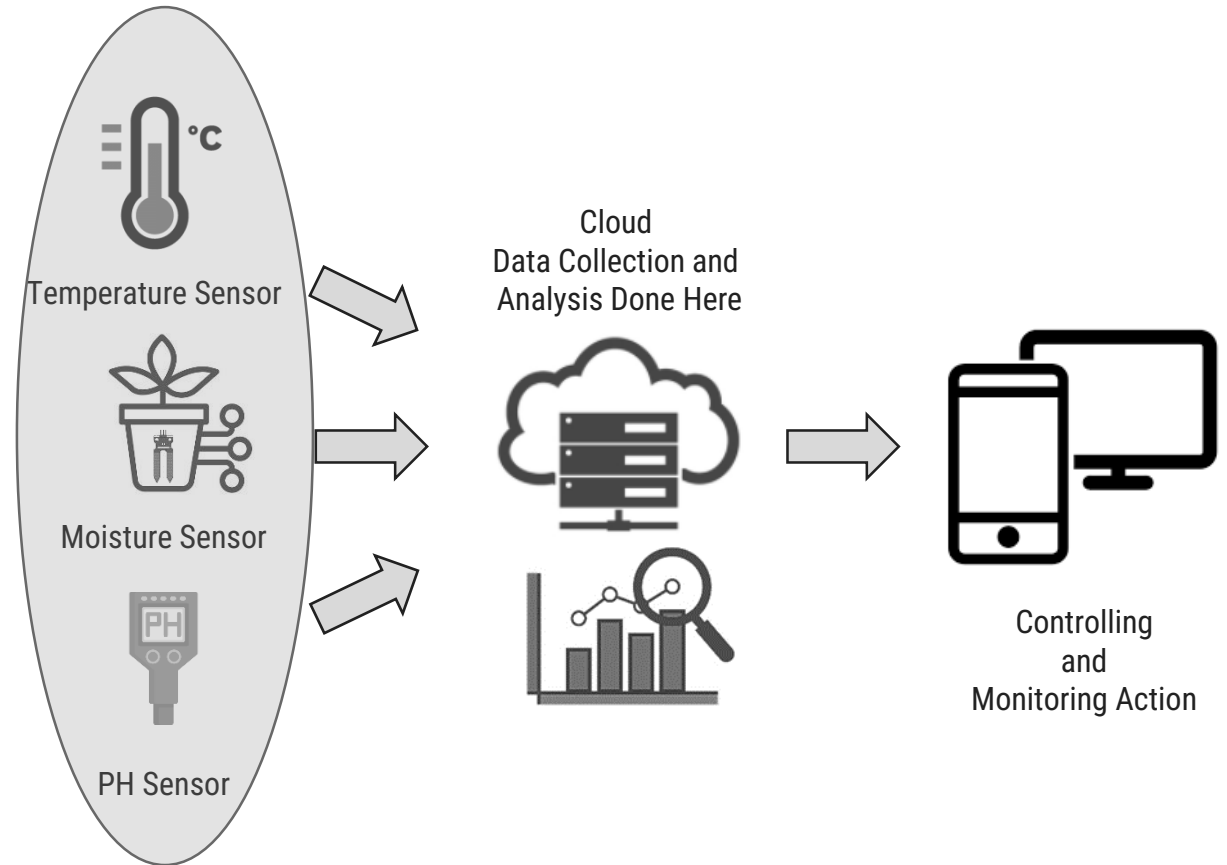
- The data is huge, frequency of sensing done by the sensor is faster and the data is stored on cloud.
- **The difference is that the analysis is also carried out on cloud.**
- Based on the data analysis, the control action can be triggered through the web application or mobile application.
- Some examples are agriculture applications, room freshening solutions based on odor, etc., where analysis of data occurs in the cloud.



# IoT Levels

## ► Level 4

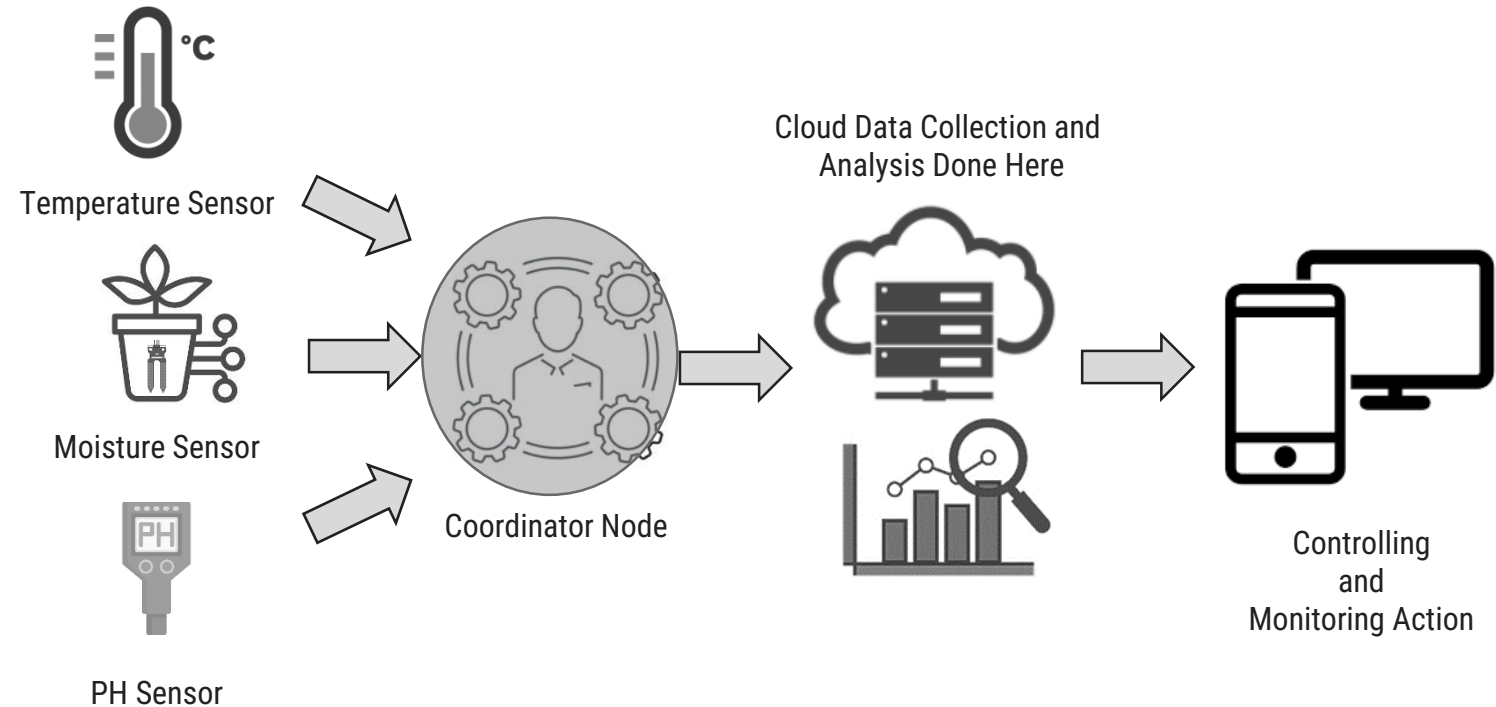
- ➔ With every passing level, the volume of data increases and hence the rate at which it is sensed also increases.
- ➔ At this level, **multiple nodes** are present which are independent of each other.
- ➔ These nodes upload data to the cloud.
- ➔ All the sensors upload the read sensory inputs on cloud storage.
- ➔ Analysis is also carried out on the cloud.
- ➔ Based on the analysis carried out, the control action shall be triggered through a web application or mobile application.



# IoT Levels

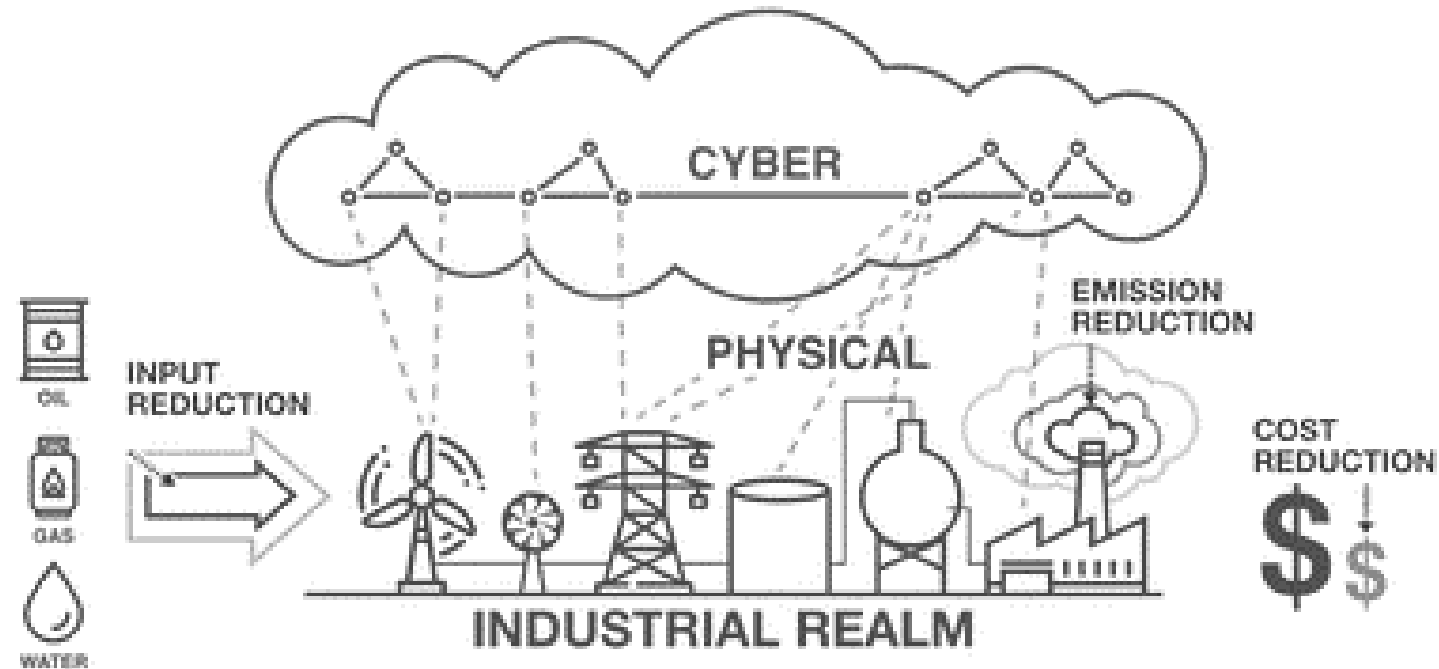
## ► Level 5

- ➔ At this level, the amount of data is extensive and is sensed much faster.
- ➔ Multiple nodes are involved in the applications categorized under Level 5 and these nodes are independent of each other.
- ➔ The sensing of data and its storage is the same as in all the previous levels.
- ➔ When an application is completely cloud oriented, it is computationally intensive in real time.
- ➔ Based on the data analysis, the control action can be triggered through web application or mobile application as in all other levels.



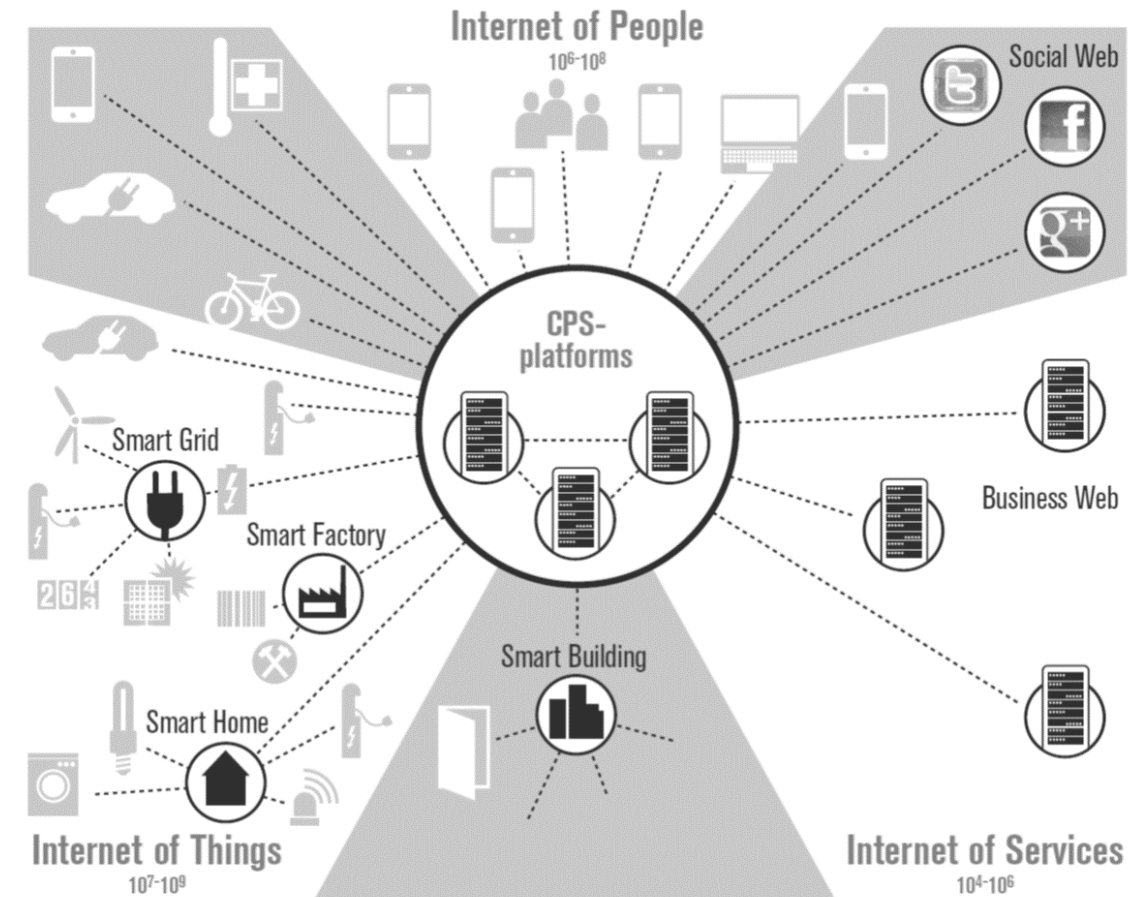
# Cyber Physical System versus IoT

- ▶ An important question is, Is IoT same as Cyber Physical System (CPS)?
  - ➔ There is a misconception that both the terms are the same.
  - ➔ We have learned the definition of IoT. The “thing” can also be accessed from anywhere, anytime by an authorized party.
  - ➔ The information or the sensed data of the things can be simple.
  - ➔ So complexity involved in the IoT applications is minimal.
  - ➔ For complex levels of operation and to address larger network of “things”, a new term called Cyber Physical System or CPS, has been introduced.



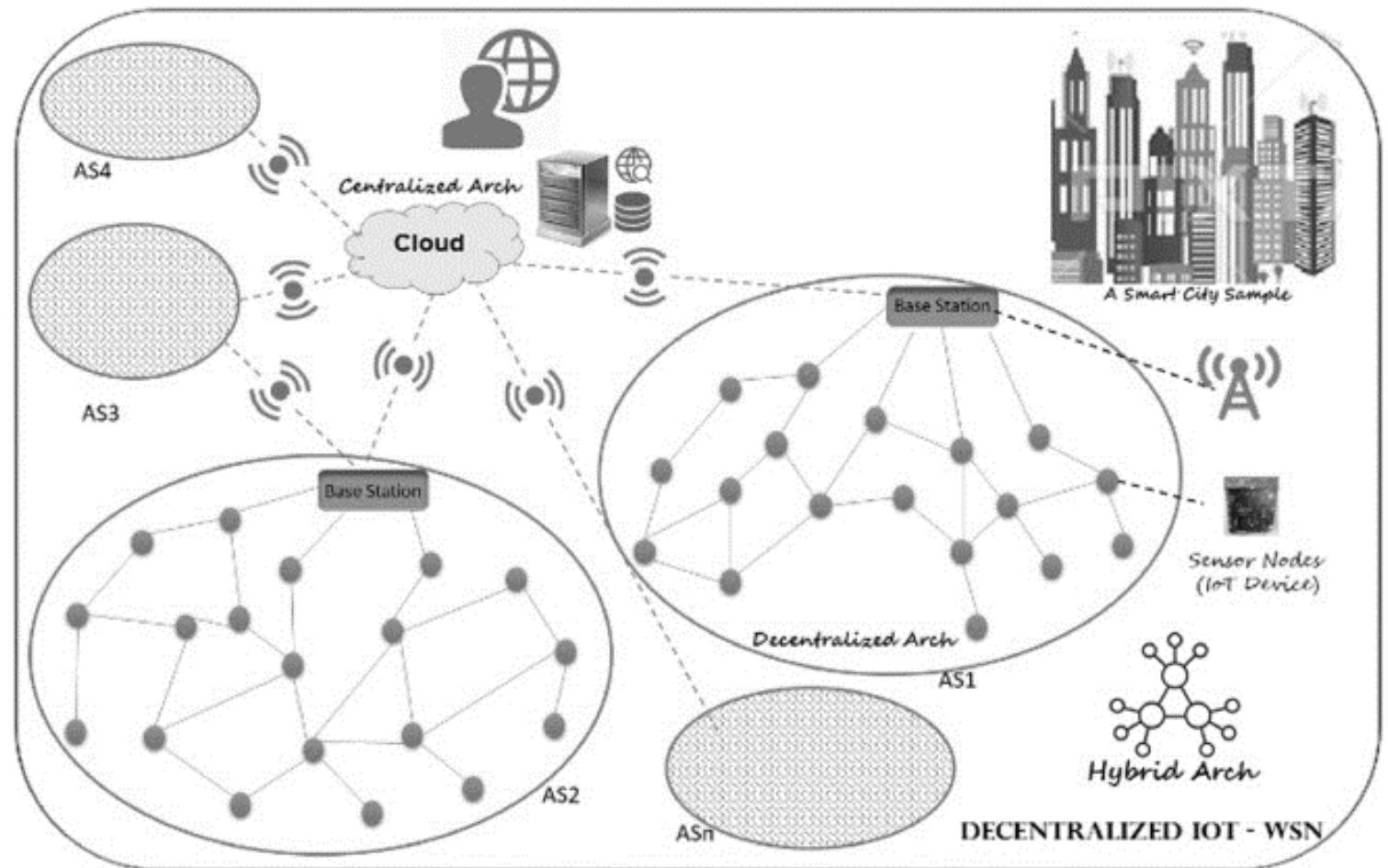
# Cyber Physical System versus IoT

- ➔ It is important to note that CPS is not IoT.
- ➔ CPS is more complex than IoT and is much more challenging.
- ➔ CPS has IoT as one of its components.
- ➔ It is a combination of multiple engineering domains coming together.
- ➔ The flight of an aero plane can be seen as a CPS which involves multiple domains of engineering.
- ➔ CPS is much more autonomous than IoT, taking appropriate decisions as and when needed.
- ➔ It is not just about identifying “things”; it is more about understanding and taking decisions in a more dynamic way.
- ➔ **CPS is mainly concerned about the collaborative activity of sensors or actuators to achieve a certain goal.**
- ➔ For that CPS uses an IoT system to achieve the collaborative work of the distributed systems.



# Wireless Sensor Network versus IoT

- ▶ WSN is a network of multiple autonomous sensors/nodes.
- ▶ Each node has one or more sensors.
- ▶ All the sensed data are passed to a centrally located server.
- ▶ The data passing happens in a coordinated pattern.
- ▶ We can say that WSN is all about coordinated data collection.
- ▶ On the other hand, IoT is much more than just data collection and the systems are more intelligent.

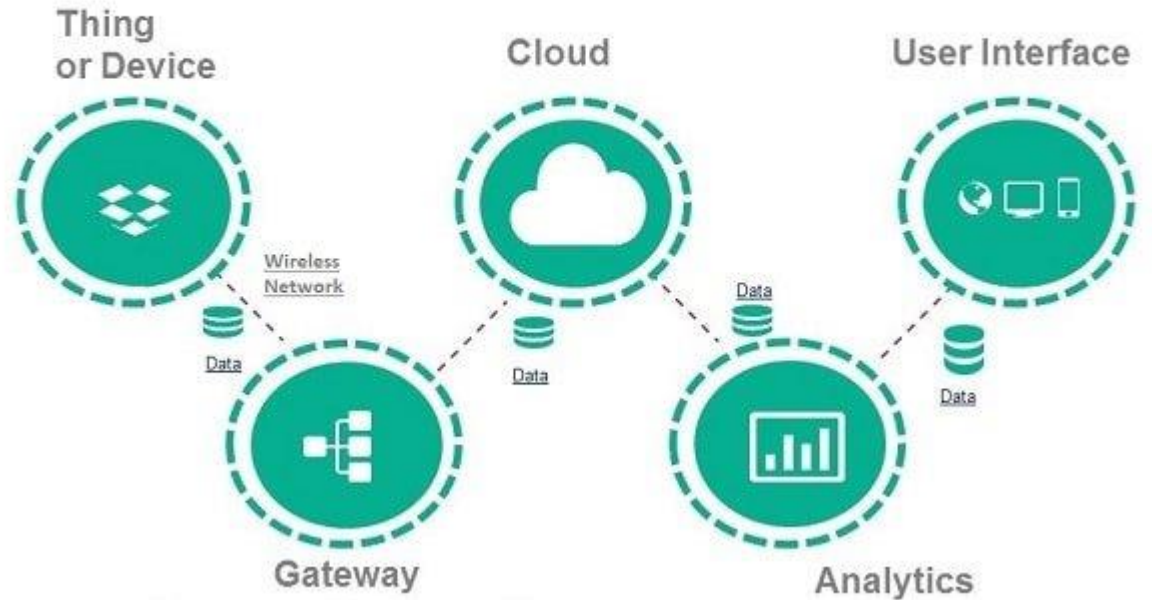


# Physical Design of IoT

There are three IoT components which enable seamless communication:

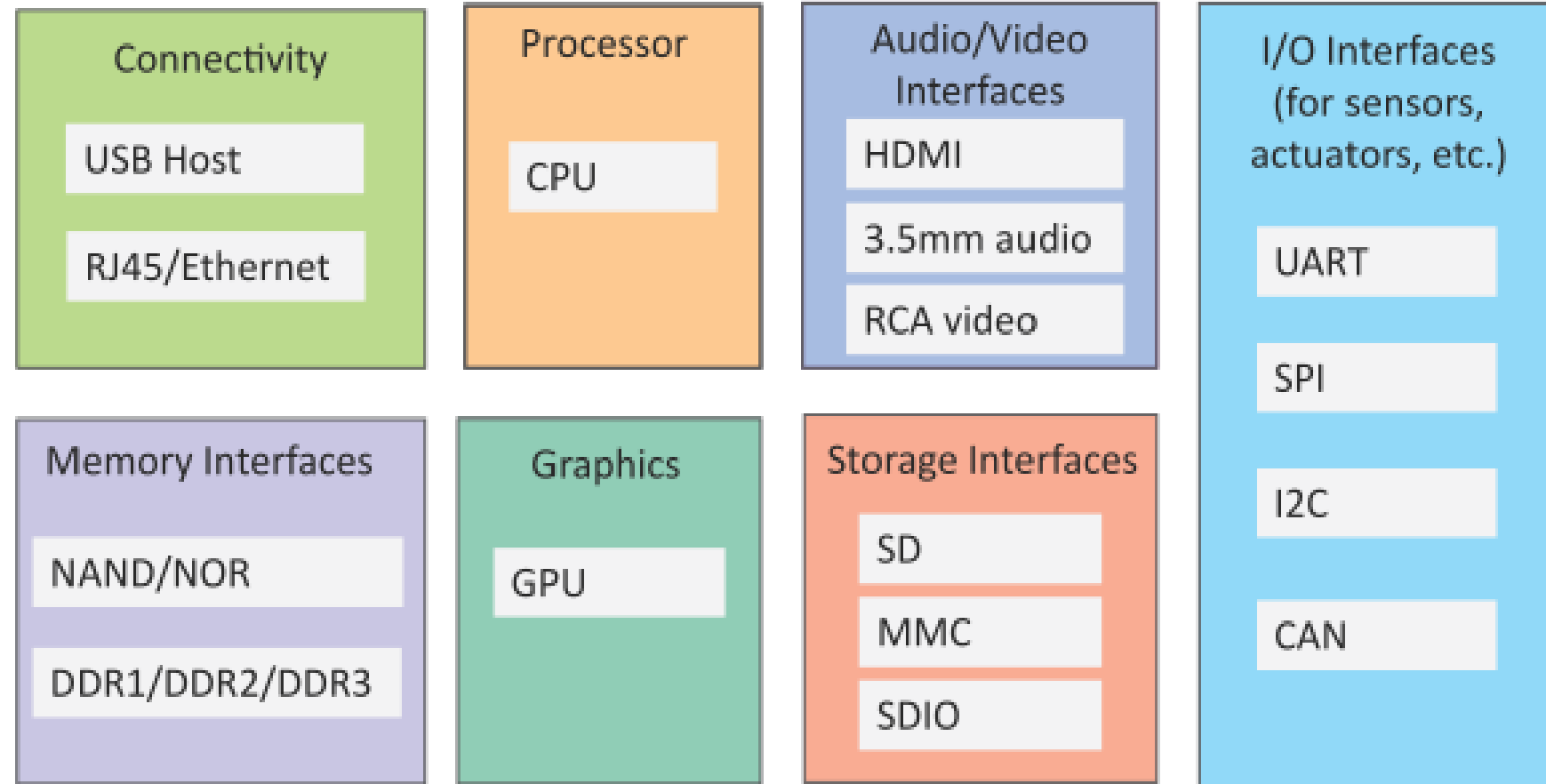
- a) Hardware - made up of sensors, actuators and embedded communication hardware
- b) Middleware - on demand storage and computing tools for data analytics
- c) Presentation - novel easy to understand visualization and interpretation tools which can be widely accessed on different platforms and which can be designed for different applications.

## Major Components of IoT



# Physical Design of IoT: Connectivity

- ▶ An IoT device may consist of several interfaces for connections to other devices, both wired and wireless.
- ▶ I/O interfaces for sensors
- ▶ Interfaces for internet connectivity
- ▶ Memory and storage interfaces
- ▶ Audio/video interfaces



# Physical Design of IoT: Communications

## ▶ Link Layer

- ➔ 802.3 – Ethernet
- ➔ 802.11 – WiFi
- ➔ 802.16 – WiMax
- ➔ 802.15.4 – LR-WPAN
- ➔ 2G/3G/4G

## ▶ Network Layer

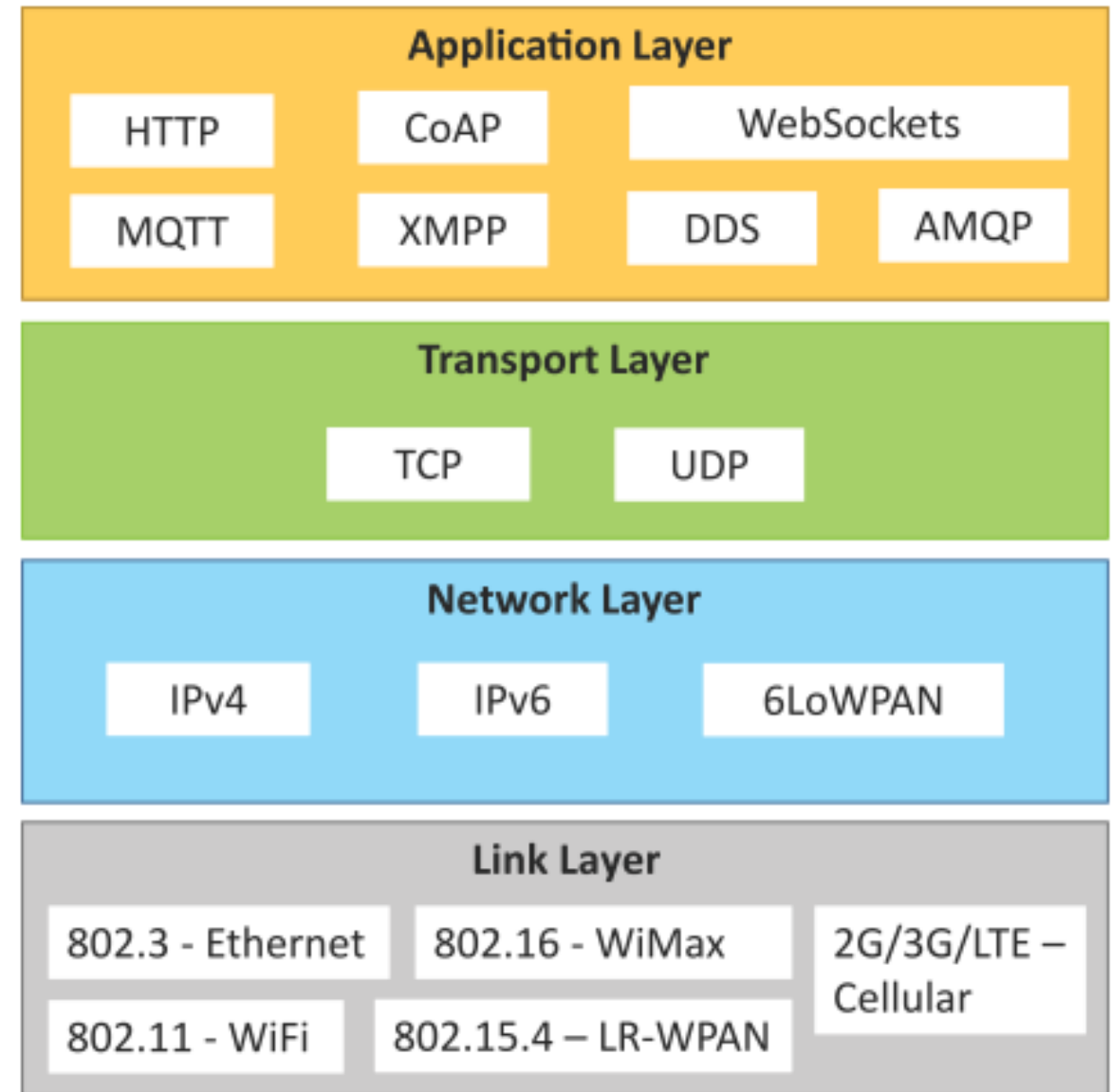
- ➔ IPv4
- ➔ IPv6
- ➔ 6LoWPAN

## ▶ Transport Layer

- ➔ TCP
- ➔ UDP

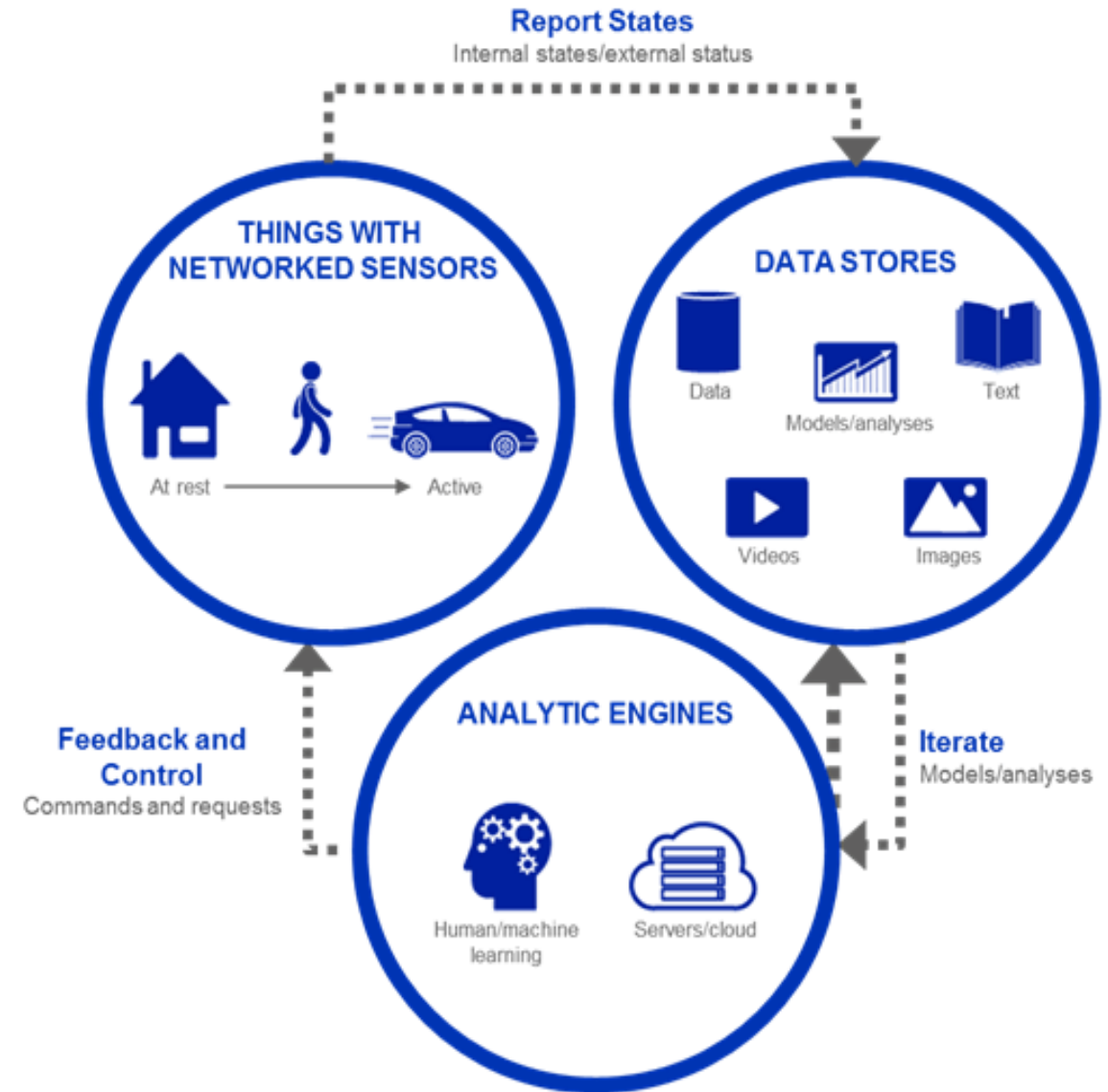
## ▶ Application Layer

- ➔ HTTP
- ➔ CoAP
- ➔ WebSocket
- ➔ MQTT
- ➔ XMPP
- ➔ DDS
- ➔ AMQP



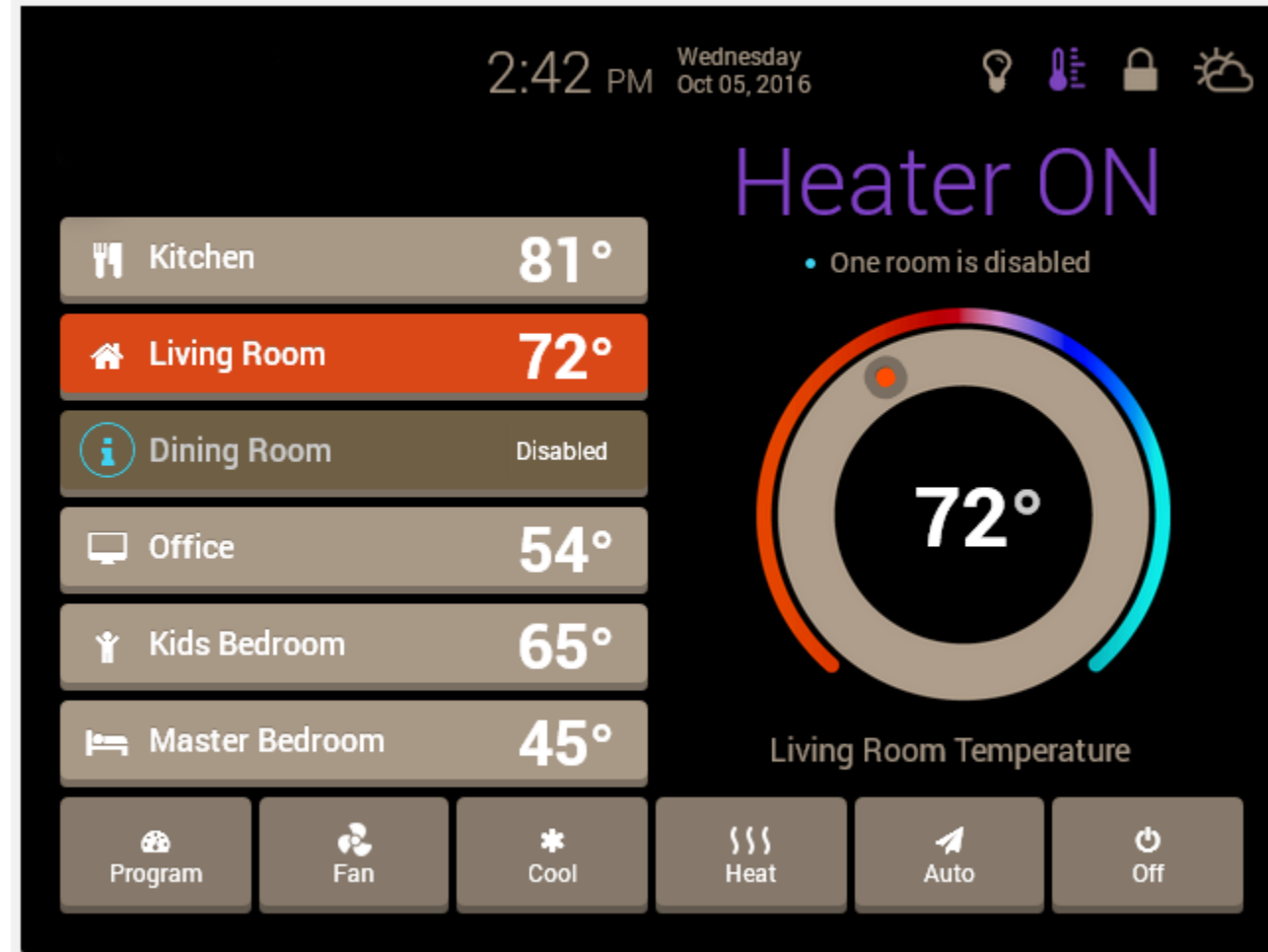
# Physical Design of IoT: Data Storage & Data Processing

- ▶ Once the data is collected and it gets to the cloud, the software performs processing on the acquired data.
- ▶ This can range from something very simple, such as checking that the temperature reading on devices such as AC or heaters is within an acceptable range.
- ▶ It can sometimes also be very complex, such as identifying objects (such as intruders in your house) using computer vision on video.
- ▶ But there might be a situation when a user interaction is required, example- what if when the temperature is too high or if there is an intruder in your house? That's where the user comes into the picture.



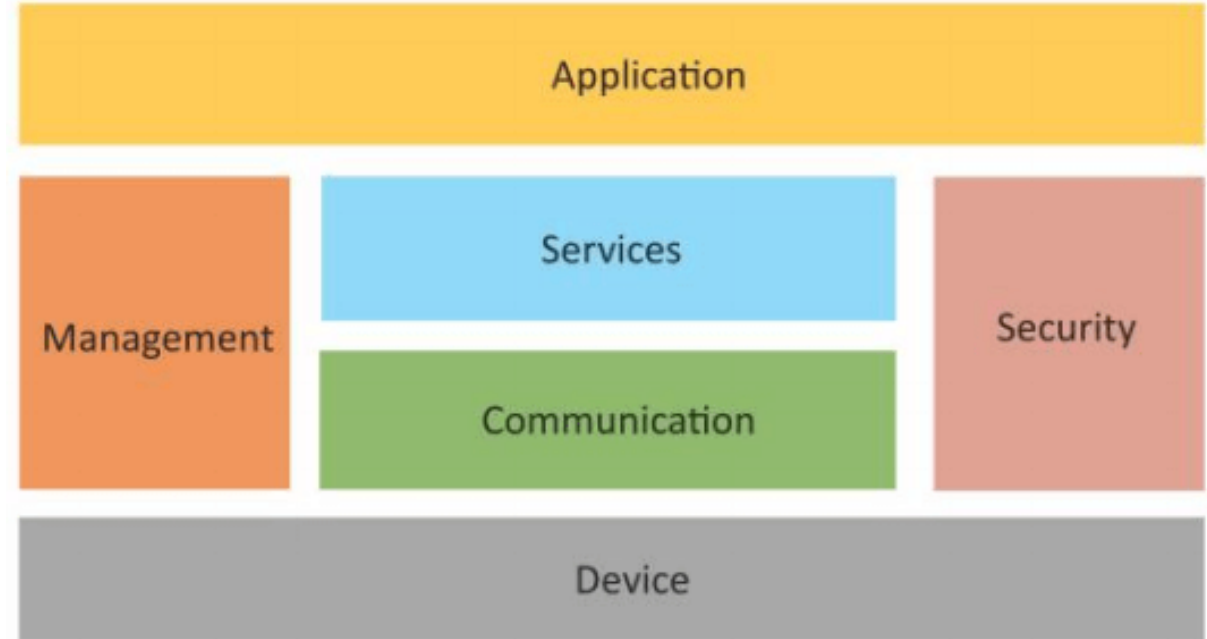
# Physical Design of IoT: User Interface

- ▶ Next, the information made available to the end-user in some way. This can be achieved by triggering alarms on their phones or notifying through texts or emails.
- ▶ Also, a user sometimes might also have an interface through which they can actively check in on their IOT system. For example, a user has a camera installed in his house, he might want to check the video recordings and all the feeds through a web server.
- ▶ User interface provides large visibility of IoT ecosystem.



# Logical Design of IoT:

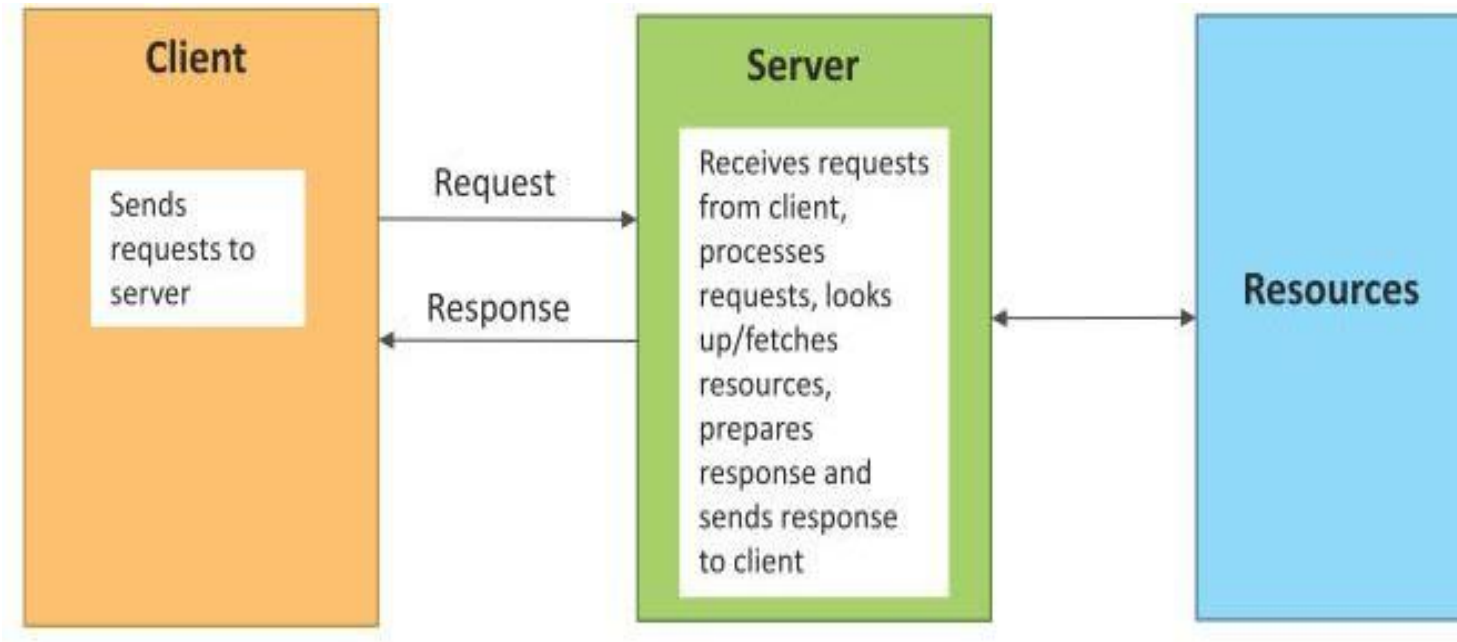
- ▶ Logical design of an IoT system refers to an **abstract representation** of the entities and processes without going into the low-level specifics of the implementation.
- ▶ An IoT system comprises of a number of functional blocks that provide the system the capabilities for identification, sensing, actuation, communication, and management.
- ▶ Depending on logical communication between source and destination of data, there are different kinds of logical communication model, which are:



1. Request-Response communication model
2. Publish-Subscribe communication model
3. Push-Pull communication model
4. Exclusive Pair communication model
5. REST-based Communication APIs
6. WebSocket-based Communication APIs

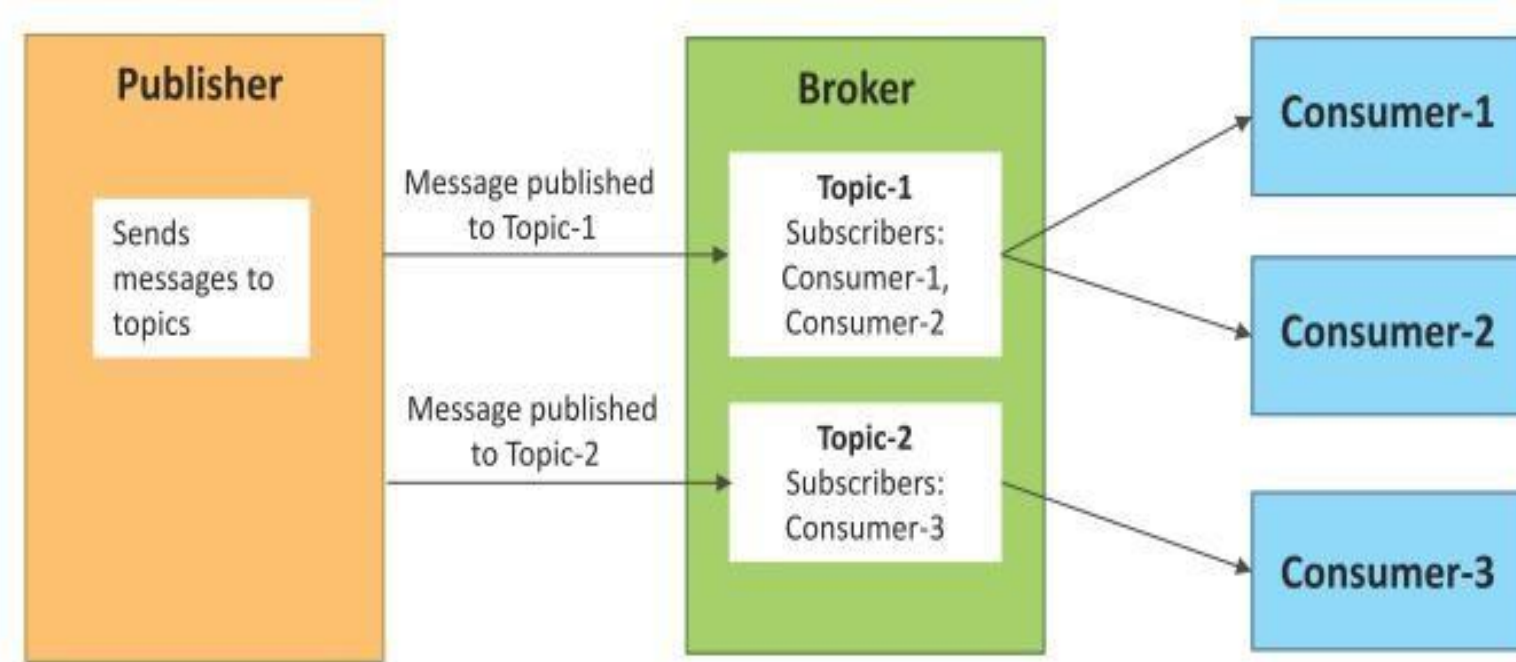
# Logical Design of IoT: Request-Response communication model

- ▶ Request-Response is a communication model in which the client sends requests to the server and the server responds to the requests.
- ▶ When the server receives a request, it decides how to respond, fetches the data, retrieves resource representations, prepares the response, and then sends the response to the client.



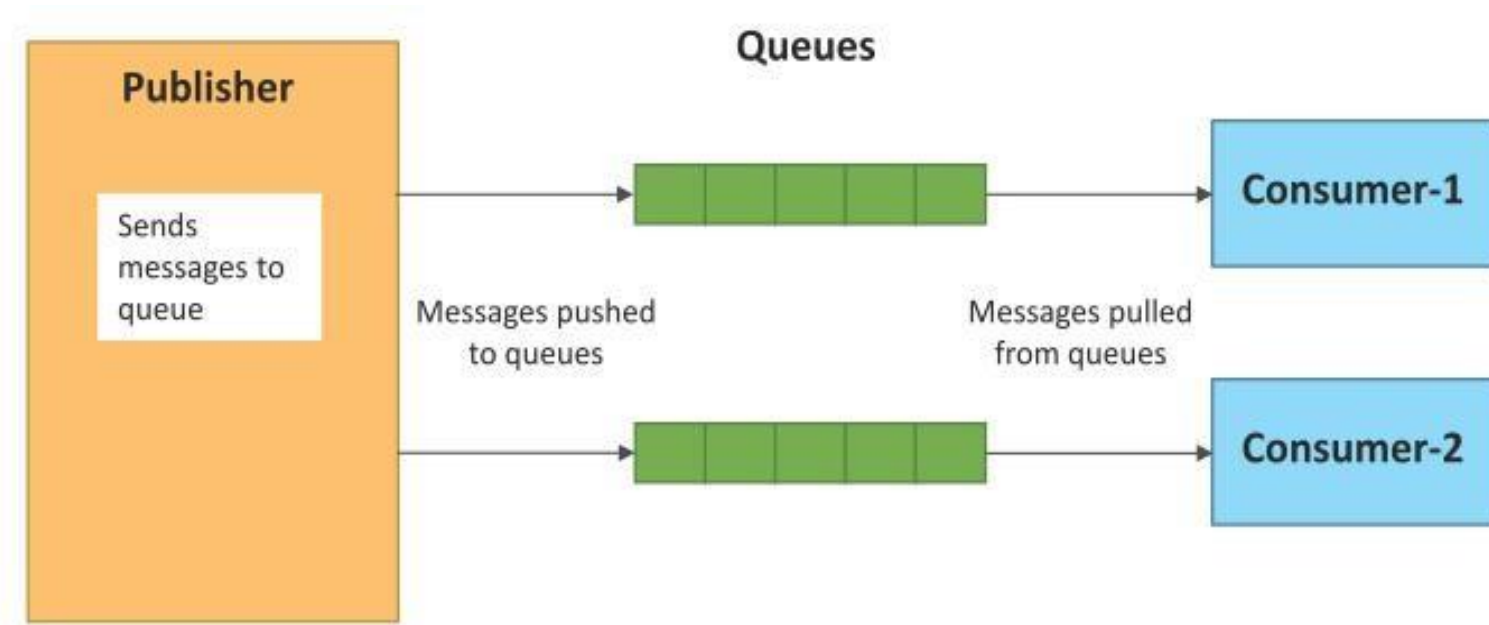
# Logical Design of IoT: **Publish-Subscribe communication model**

- ▶ Publish-Subscribe is a communication model that involves publishers, brokers and consumers.
- ▶ Publishers are the source of data. Publishers send the data to the topics which are managed by the broker. Publishers are not aware of the consumers.
- ▶ Consumers subscribe to the topics which are managed by the broker.
- ▶ When the broker receives data for a topic from the publisher, it sends the data to all the subscribed consumers.



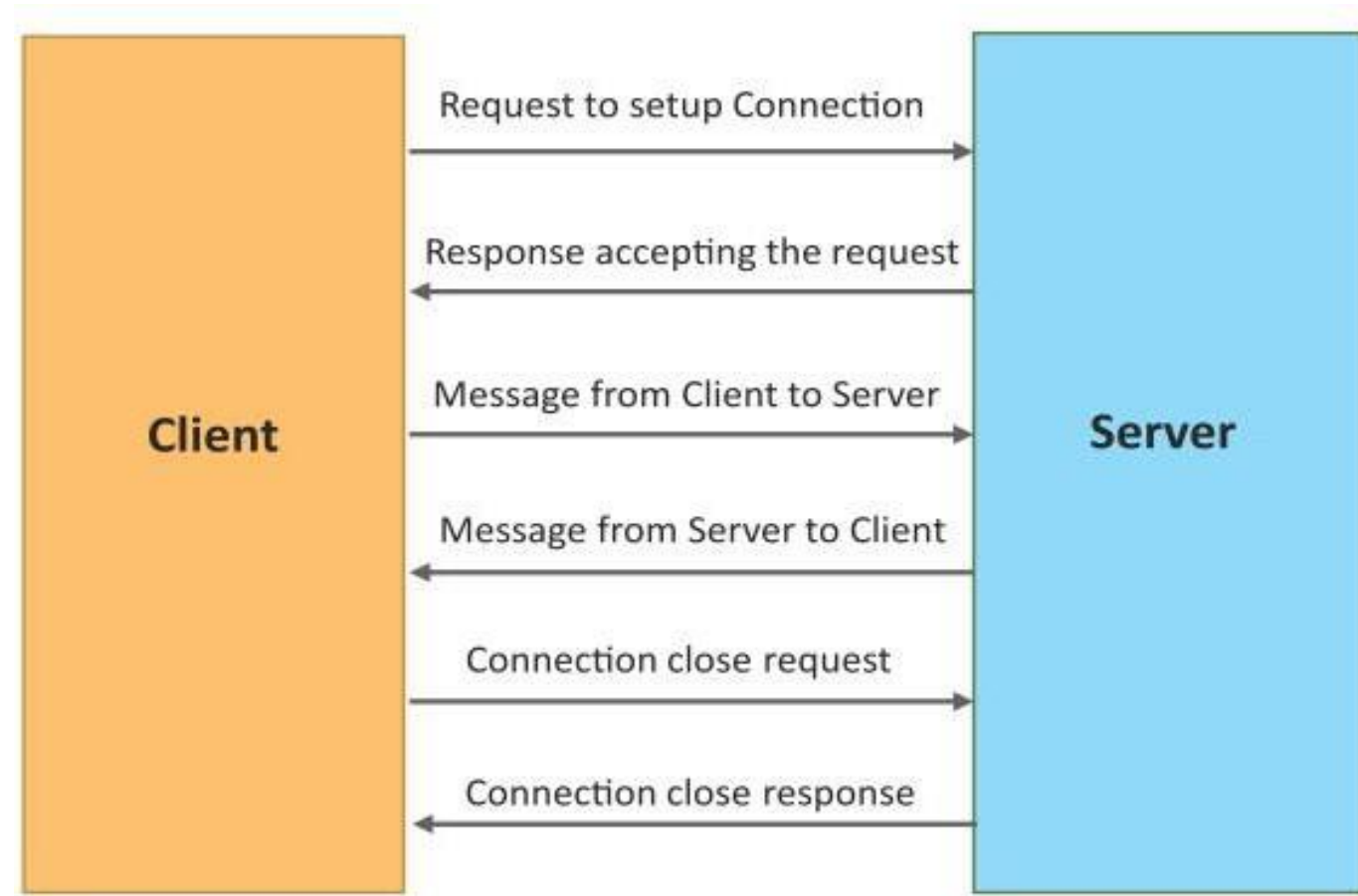
# Logical Design of IoT: **Push-Pull communication model**

- ▶ Push-Pull is a communication model in which the data producers push the data to queues and the consumers pull the data from the queues.
- ▶ Producers do not need to be aware of the consumers.
- ▶ Queues help in decoupling the messaging between the producers and consumers.
- ▶ Queues also act as a buffer which helps in situations when there is a mismatch between the rate at which the producers push data and the rate at which the consumers pull data.



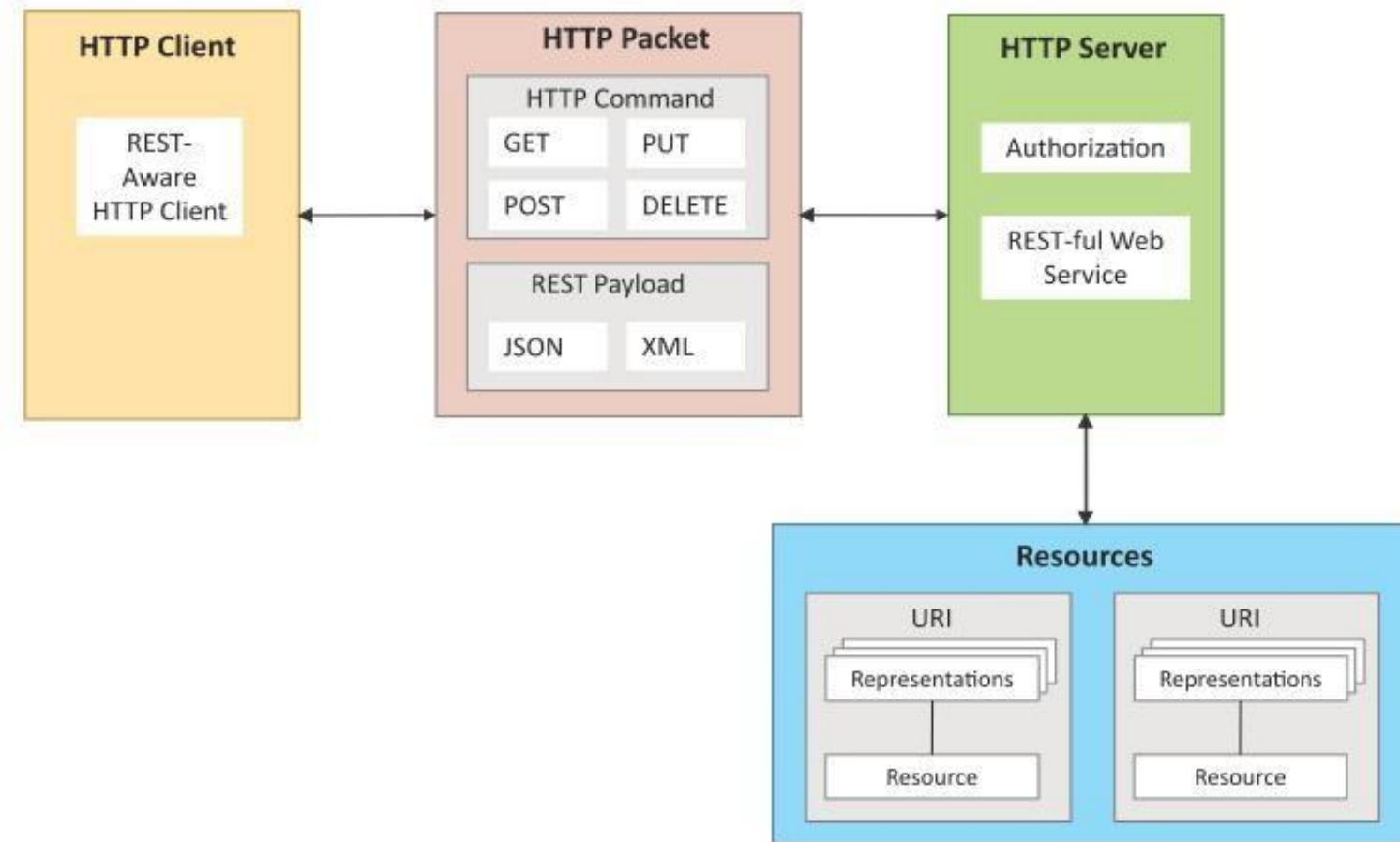
# Logical Design of IoT: Exclusive Pair communication model

- ▶ Exclusive Pair is a bidirectional, fully duplex communication model that uses a persistent connection between the client and server.
- ▶ Once the connection is setup it remains open until the client sends a request to close the connection.
- ▶ Client and server can send messages to each other after connection setup.



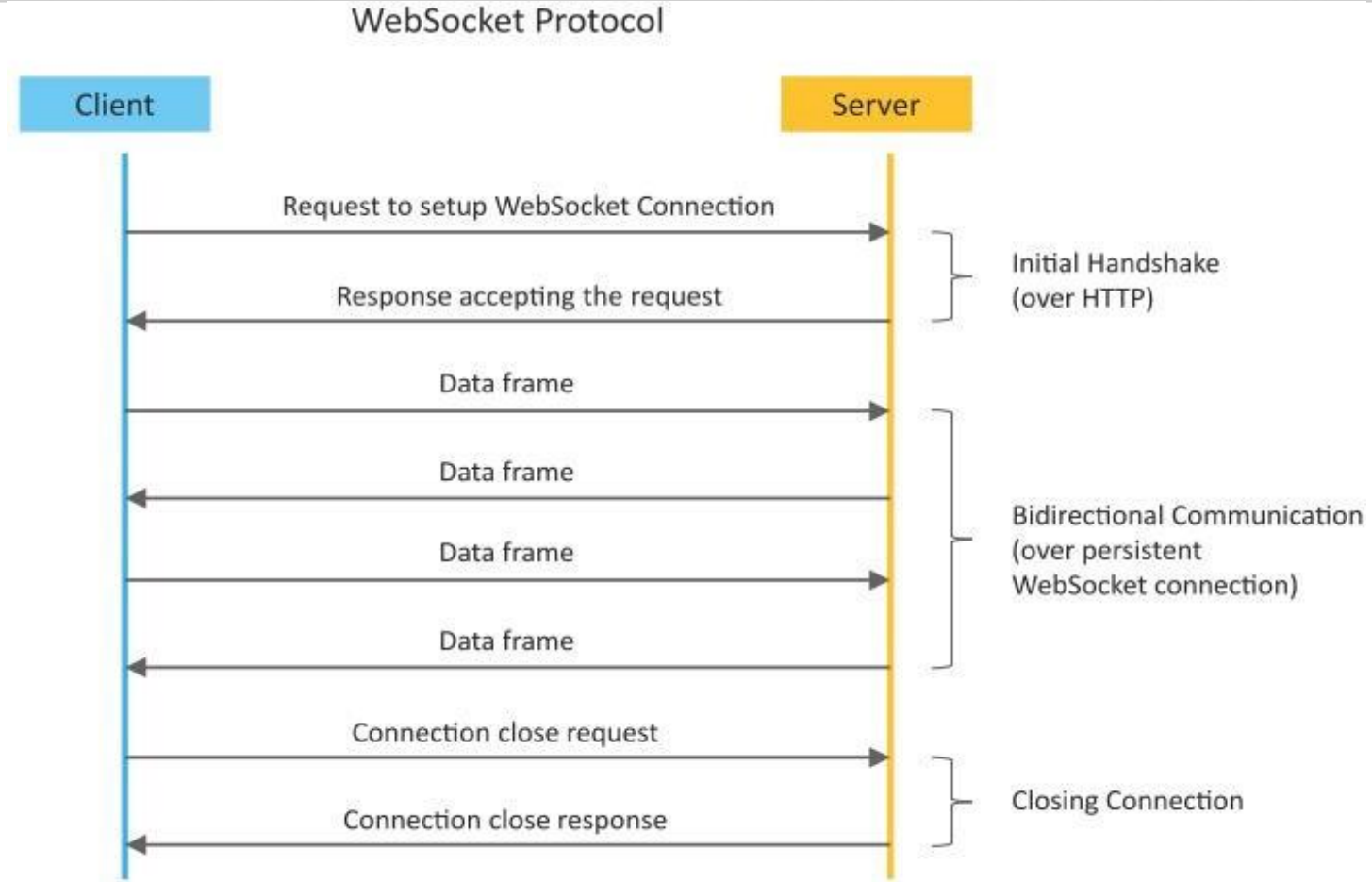
# Logical Design of IoT: **REST-based Communication APIs**

- ▶ Representational State Transfer (REST) is a set of architectural principles by which you can design web services and web APIs that focus on a system's resources and how resource states are addressed and transferred.
- ▶ REST APIs follow the request-response communication model.
- ▶ The REST architectural constraints apply to the components, connectors, and data elements, within a distributed hypermedia system.



# Logical Design of IoT: WebSocket-based Communication APIs

- ▶ WebSocket APIs allow bi-directional, full duplex communication between clients and servers.
- ▶ WebSocket APIs follow the exclusive pair communication model between client and server.



# Properties of IoT Ecosystem:

- ▶ Self-adaptation
- ▶ Self-organization
- ▶ Self-configuration
- ▶ Self-protection
- ▶ Self-healing
- ▶ Self-description
- ▶ Self-discovery
- ▶ Self-energy-supplying

# Summary

- ▶ IoT refers to the interconnection of computing devices embedded in everyday objects via the Internet, enabling them to send and receive data.
- ▶ IoT is not owned by any one engineering branch. It is a reality when multiple domains join forces and combine efforts.
- ▶ IoT is all about providing service to any device, anywhere, anybody, and any network.
- ▶ IoT has certain characteristics which are important: a. Connectivity. b. Intelligence and identity. c. Scalability. d. Dynamic and self-adapting (complexity). e. Architecture. f. Safety.
- ▶ “Things” refer to variety of devices. At times, even humans in the loop becomes a thing. For anything to qualify as a “thing”, it requires identity. The “thing” can monitor, measure, etc.; for example, a temperature sensor could be a “thing”.
- ▶ One should understand that “THINGS” = HARDWARE + SOFTWARE + DATA + SERVICE
- ▶ IoT stack has seven layers, starting with sensor layer and ending with application layer just as OSI.

# Summary

- ▶ Security/personnel safety, privacy, data extraction with consistency from complex environments, connectivity, power requirements, complexity involved, and storage are the major challenges we face while building an IoT application.
- ▶ IoT application can be classified as Level 1, 2, 3, 4 or 5 based on the complexity and architecture involved.
- ▶ IoT is all about sense, connect, store, analyze, control and sharing.
- ▶ Physical design indicates importance and role of each physical component in IoT Ecosystem.
- ▶ Logical design presents logical data flow between source and destination of data.
- ▶ At last, key properties are presented to showcase fundamental properties of a robust IoT system.